

Organization and Management

Complete Exam Notes

ME 708 · IV/I · BEL, BEX, BCT, BEI
Institute of Engineering, Tribhuvan University

Built from the syllabus, AJ Sir's lecture decks, Bikash Adhikari's notes, S.K. Joshi's textbook and **all 29 past papers (2065–2082 BS)**. Every topic carries a frequency tier counted from those papers, and `src/check.py` checks each tier chip against its own year list *and* against the sorted PYQ document, so no chapter can cite a paper that asks it nothing, and no paper that asks a chapter goes uncited.

No paper in 29 sets a numerical. What O&M asks instead is lists, comparisons and “in your opinion” questions; each chapter answers its opinion questions directly, and §6.3 collects them.

Short on time? Go straight to §6, Last-Minute Revision.

Ch	Topic	Hours	Papers	Marks	Share
1	Introduction	20	29/29	956	40.5%
2	Personnel Management	8	29/29	404	17.1%
3	Motivation, Leadership and Entrepreneurship	10	29/29	591	25.0%
4	Case Studies	2	14/29	92	3.9%
5	Management Information System	5	25/29	248	10.5%
–	Extra: 2065–2068 short notes, folded into the chapters		5/29	72	3.0%
Total		45		2363	100%

Marks track hours closely here. Chapter 1 is 44 % of the hours and 40 % of the marks; Chapter 3 is 22 % of the hours and 25 % of the marks, and like Chapters 1 and 2 it is asked in every paper. Chapter 4 is small but has been a near-certain 4-mark short note since 2079.

Shares are of the 2363 archive marks, which include 72 marks of 2065–2068 short notes that the Detailed PYQ keeps in a separate “Extra” section and these notes fold into the chapters they belong to. The archive averages 52 recorded marks per 80-mark paper because short-note blocks and the old any-five-of-six papers are counted low: read these as shares, not as an audit.

Contents

1	Introduction	3
2	Personnel Management	33
3	Motivation, Leadership and Entrepreneurship	49
4	Case Studies	65
5	Management Information System	82
6	Last-Minute Revision	94
	Abbreviations	98

1. Introduction

20 hrs · asked in **all 29 papers** · 6 syllabus units, 28 sub-topics · **956 marks, 40.5% of the archive** → the biggest chapter by far: about 21 of every 80 marks.

READ THIS FIRST

Chapter 1 is pure theory, and almost every question is **an 8-mark question split 4+4, 3+5 or 2+6**: two lists glued together. Learn the lists below as lists.

- Six lists carry the chapter**, each asked in 9 to 16 papers: importance of organization, principles of organization, formal vs informal organization, functions of management, levels + skills of management, and the three classical theories (Taylor, Fayol, Mayo).
- Forms of ownership and structures** are compare-and-choose questions: “JSC vs partnership”, “line & staff vs line”, “which structure for an engineering project”. The comparison tables in §1.4 and §1.5 answer them directly.
- Purchasing and marketing**: two questions set every year in some form (functions of purchasing; importance of marketing).

Tier chips count **papers out of 29** (2065–2082 BS): **TOP n/29** asked in 8+ papers **HOT n/29** 4–7 papers **PIN n/29** 1–3 papers inside an 8-mark question **LOW n/29** 1–3 papers, short note only. Bold year = Regular exam.

1.1 Organization

Define organization. Why is it necessary? Explain its importance and role in society **1 2 3 4 5 8**

TOP 16/29 82 Bh, 80 Bh, 78 Bh, 76 Ch, 76 Ash, 75 Ch, 75 Ash, 74 Ash, 73 Ch, 73 Shr, 72 Ch, 72 Ka, 71 Ch, 70 Ch, 69 Ch, 65 Shr · the opener to half the chapter’s questions

Definition

- **Mooney and Reiley**: “Organization is the form of every human association for the attainment of a common purpose.”
- Working definition: a **collection of people working together in a coordinated and structured way to achieve one or more goals**.
- As a **process**: identifying + grouping work → defining + delegating responsibility and authority → establishing relationships so people work effectively together.
- Three views of the same thing (AJ Sir): **goal-oriented system** (people work for specific goals), **social system** (people work in groups), **technological system** (people apply knowledge and techniques).

Necessity (why we need organization)

- **Complexity of industry**: large plants, complex methods need structure.
- **Growing competition**: required quantity + quality, at required time, at minimum cost.
- **Optimum use of resources**: men, machines, materials, money.
- **Fixes authority and responsibility** → no confusion, no conflict.
- **Coordination** of efforts of many people toward one goal.
- **Reduces labour problems; facilitates administration**.
- **Stimulates creativity** and specialization.
- People achieve together what no individual can alone; diverse skills, more resources.
- **Andrew Carnegie**: take away the money and the works but leave the organization, and he would rebuild within a few years.

Importance / role in society

- **Produces goods and services** society needs (food, power, transport).
- **Employment** and income → livelihood for millions.
- **Development**: hospitals, schools, universities, banks, telecom, roads.
- **Raises living standard**: better products at lower cost.
- **Innovation and technology**: research, new products.
- **Revenue to government** (taxes) → public services.
- **Social welfare**: CSR, cooperatives, NGOs serve the weak.
- **Civilization itself** is organized effort: without organization there is only self-sufficiency, no division of labour, no trade.
- **Can we sustain without organization?** No: every need beyond bare survival (education, health, security) is delivered by some organization.

Role of organization in an employee's professional growth 4

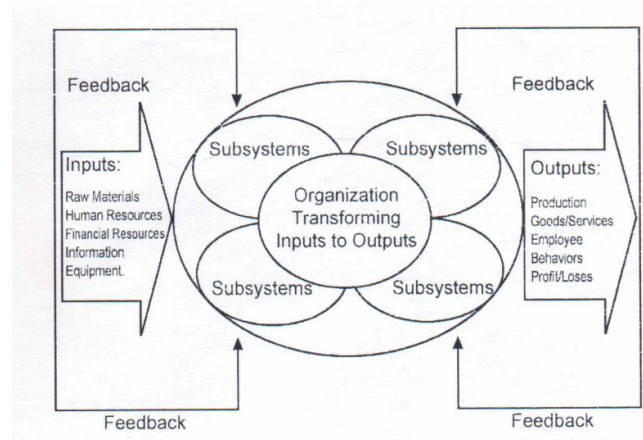
PIN 2/29 81 Bh, 75 Ch · always paired with “do we need informal organization?” (below)

- **Training and development:** on-the-job training, workshops, higher study.
- **Career path:** promotion through levels, job rotation, wider responsibility.
- **Performance appraisal / merit rating** → feedback, recognition.
- **Mentoring:** seniors and supervisors transfer knowledge (knowledge-oriented model).
- **Exposure:** teamwork, projects, clients, new technology build skills.
- **Security + remuneration:** stable income lets an employee invest in learning.

Organization as a system: open system, social system, system approach applied to organization 2 4

PIN 3/29 81 Bh, 79 Ba, 70 Ch

- **System** = set of interdependent parts working together toward a goal.
- Organization as a system:
 - **Inputs:** 5 M's, men, materials, machines, money, methods (+ information).
 - **Transformation process:** sub-systems (production, finance, marketing, personnel) convert inputs.
 - **Outputs:** products, services, profit, employee satisfaction.
 - **Feedback:** customer response, results vs plan → corrections.
 - **Environment:** external (social, political, economic, cultural, technology) and internal (board, employees, culture).
- **Open system:** continuously exchanges inputs and outputs with its environment and must adapt to changes in technology, market, customer needs, rules. A closed system ignores the environment and decays.
- **Social system:** people interact in groups; their attitudes, perceptions, norms shape how it works (Mayo, Barnard).
- **System approach applied:** every change (new technology, process) ripples to people, jobs, skills → plan for change, assess change forces, implement (Kaila). Also called a change-management plan.



Organization as an open system: inputs, transforming sub-systems, outputs, feedback from both ends. ioenotes Ch. 1, p1.

Organizations, people and management

- Organizations exist to accomplish work **people cannot do alone.**
- While their goals stay appropriate, **society lets them exist.**
- **People shape organizations:** their personality, attitudes and behaviour decide how it operates.
- Organizations **need management** to achieve their goals.

Principles of organization 4 5 6 8

TOP 9/29 82 Bh, 80 Bh, 76 Ash, 74 Ch, 72 Ch, 72 Ka, 71 Ch, 71 Shr, 70 Asa · “with relevant examples” in 82 Bh

1. **Unity of objectives:** clear goals for firm, each department, each post.
2. **Division of labour / specialization:** work split into jobs matched to skill.
3. **Departmentalization:** related activities grouped into units.
4. **Scalar chain / chain of command:** clear line of authority top to bottom.
5. **Unity of command:** one subordinate, one boss.
6. **Authority = responsibility:** authority given must match duty assigned.
7. **Span of control:** limited no. of subordinates per supervisor (5–6 at top, up to 20 at shop floor).
8. **Delegation:** authority pushed down to where decisions are made.
9. **Centralization vs decentralization** balanced to firm size.
10. **Coordination:** departments' work synchronized.
11. **Simplicity:** minimum levels → easy communication.
12. **Flexibility:** structure adapts to growth and change.
13. **Communication:** free two-way flow, with feedback.
14. **Continuity:** structure survives change of people.
15. **Efficiency:** goals at minimum cost and effort.

Examples (for 82 Bh)

- **Unity of command:** a hospital ward nurse reports to one ward in-charge only.
- **Span of control:** a factory foreman supervises 15 workers, a CEO only 5 managers.
- **Unity of objectives:** a bank splits its national target into branch targets.
- **Scalar chain:** a site engineer's leave request goes through the project manager, not straight to the MD.

How to write the 8-mark version

- Name 8 principles, one line each.
- Give each a one-line example.
- Close with one line on why they matter: clear roles, no duplication, faster decisions.

Two cautions worth a line

- **Over-specialization:** jobs split too finely become boring → performance falls. Balance specialization with job rotation or enlargement.
- **Span keeps widening:** with downsizing and IT, firms remove layers → each manager supervises more people.
- **Flexibility** means small alterations and expansions as circumstances change, not a new structure every year.

Importance of setting goals for an organization 3

PIN 1/29 82 Bh

- Gives **direction:** everyone knows what to achieve.
- **Basis for planning** and resource allocation.
- **Standard for control:** actual results measured against goals.
- **Motivates:** clear, achievable targets (see MBO, §1.3).
- **Coordinates** departments toward one end; resolves conflicts.

Formal and informal organization: definitions, differences, need for informal organization 2 3 4 5

TOP 11/29 82 Ba, 81 Bh, 79 Bh, 79 Ba, 76 Ch, 75 Ch, 74 Ch, 73 Shr, 71 Shr, 70 Asa, 65 Shr · short note in 65 Shr

- **Formal organization:** structure **deliberately created by management** for enterprise objectives; official network of authority, responsibility and communication. **Chester Barnard:** “a system of consciously coordinated activities of two or more persons towards a common objective”; exists when people (a) can communicate, (b) are willing to act, (c) share a purpose.
- **Informal organization:** network of personal and social relations that **emerges spontaneously** from friendship, language, culture, common interests. Barnard: “joint personal activity without conscious common purpose, though contributing to joint results”. Has its own informal leader, norms and grapevine.

Basis	Formal	Informal
Origin	Deliberately created by management	Spontaneous, social and psychological forces
Purpose	Organization's legitimate goals	Members' social satisfaction
Size	Can be very large	Small groups
Stability	Stable, long-lived	Unstable
Authority	Hierarchy, flows top down	All equal; power flows from the group
Behaviour	Rules and regulations	Group norms, beliefs, values
Communication	Official chain of command	Grapevine, informal channels
Leadership	Vested in the manager	Chosen by the group, may out-power the manager
Abolition	Management can abolish it	Management cannot abolish it
Chart	Shown on organization chart	Never shown

Need for informal organization ("do we need it indeed?")

- Yes: it **fills gaps** the formal structure leaves.
- **Social satisfaction**, sense of belonging → morale.
- **Fast communication** (grapevine) across departments.
- **Cooperation and teamwork** beyond job descriptions.
- **Safety valve** for frustration; eases work pressure.
- **Feedback to management** on real employee mood.
- Risks: rumours, resistance to change, group conflict.
Management should **understand and use it**, not suppress it (Hawthorne, §1.3).

Informal organization within the same family? (79 Ba)

- **Yes**. A family is itself an informal organization (no written hierarchy), yet sub-groups form inside it: siblings, cousins of the same age, members sharing a hobby or a family business.
- In a **family business** (single owner or partnership) the formal roles (owner, manager, accountant) sit alongside informal family loyalties, which often decide more than the formal roles do.

Historical development of organization 3

PIN 1/29 79 Bh

- Concept of organization grew with **economic activity**: each stage needed a larger, more structured way of working together.

1. Pre-industrial era

- Self-sufficiency (village economy).
- Commercial stage with marketable surplus: barter → money system.
- Means of transport (ships, railways) → wider trade.

2. Modern industrial era

- Industrial revolution (James Watt's steam engine, machine tools) → factories.
- Corporate form: joint stock company with limited liability.
- Managerial revolution: scientific management (Taylor, Fayol).
- Mechanization, automation; economic liberalization.
- Globalization: multinationals; GATT (1947), ISO (1947), WTO (1995), SAFTA.

Characteristics / elements of an organization 4

PIN 2/29 73 Ch, 68 Ch

- **Group of people** (two or more) + **common goal**.
- **Division of work** and specialization.
- **Authority-responsibility structure** (hierarchy) and **coordination**.
- **Communication network**; **rules and procedures**.
- **Resources**: men, money, materials, machines, methods.
- **Environment** it serves and adapts to; **continuity** beyond members.
- For business operations add: profit motive, risk bearing, legal identity, customers.

Organizational behaviour: a multidisciplinary field; how management affects it 4

HOT 4/29 68 Ch, 68 Ba, 67 Asa, 66 Bh · 2065–2068 papers only **verified/added**

- **OB** = study of how individuals and groups behave in organizations, to improve performance and satisfaction.
- **Multidisciplinary**, it borrows from:
 - **Psychology**: individual learning, motivation, perception, personality.
 - **Sociology**: groups, roles, status, culture.
 - **Social psychology**: attitude change, group decisions, communication.
 - **Anthropology**: values, cultures across societies.
 - **Political science**: power, conflict, politics in organizations.

How management affects OB

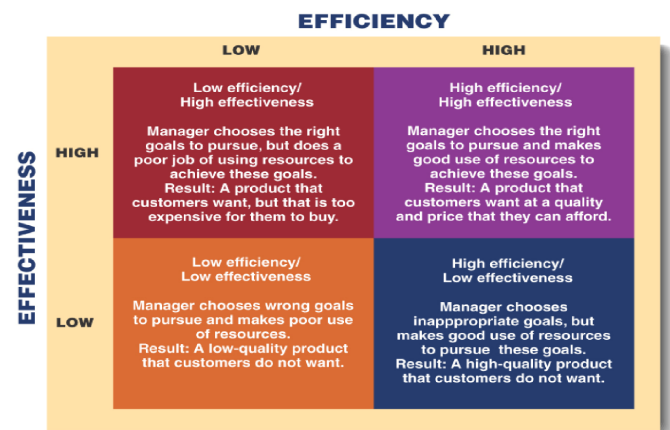
- **Structure**: tall, rigid hierarchy → caution and passing the buck; flat structure → initiative.
- **Leadership style**: autocratic control breeds compliance; participation breeds commitment.
- **Reward system**: what is rewarded gets repeated.
- **Job design**: enriched jobs raise interest; monotonous jobs raise absenteeism.
- **Communication and culture**: open channels build trust; rumours fill silence.

1.2 Management

Define management. Explain the functions of management 2 3 4 5 8

TOP 15/29 82 Bh, 81 Ba, 80 Ba, 79 Ba, 78 Bh, 76 Ch, 76 Ash, 75 Ch, 75 Ash, 73 Shr, 72 Ka, 70 Asa, 69 Ch, 68 Ch, 66 Bh · the most asked single list in the subject

- **Simple**: getting things done through other people.
- **Full**: “a social process involving coordination of human and material resources through planning, organizing, staffing, leading and controlling, to accomplish stated objectives” **effectively** (right goals) and **efficiently** (least resources).
- Resources it coordinates: people, machinery, raw materials, computers and IT, patents, finance, loyal customers.
- The **brain** of an organization; as old as civilization (Sun Tzu, Chanakya’s Arthashastra).
- **Efficiency** = doing things right (least input per output). **Effectiveness** = doing the right things (the goal customers want). High performers are both (figure).

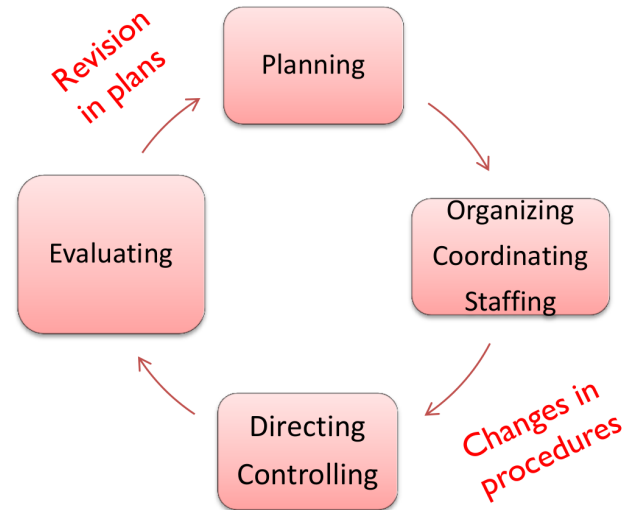


High-performing organizations are efficient *and* effective.

Efficiency (use of resources) across, effectiveness (choice of goal) down.

- Management is an **ongoing activity** made of eight functions (Adhikari), detailed below.
- They form a **process**, not a sequence done once: evaluation feeds revised plans and changed procedures (right).
- **Fayol's five** (AJ Sir): planning, organizing, directing (commanding), coordinating, controlling. Write these five if the question says "basic" functions.

Management Process



The management process as a loop.

Functions (POSDCoRB, extended)

- 1. Planning:** "thinking before doing".
 - Deciding in advance what, how, when and by whom (Koontz & O'Donnell).
 - Sets objectives and the path to them; basis of every other function.
 - Eg a hydropower firm fixes its commissioning date and budget.
- 2. Organizing:** the backbone of management.
 - Identify and group activities into departments and jobs.
 - Assign them to people; set authority-responsibility relations.
- 3. Staffing:** filling the posts of the structure.
 - Recruit, select, place, train, appraise, pay (Ch. 2).
 - Right person in the right job at the right time.
- 4. Directing:** putting the plan into action.
 - Orders, guidance, supervision of subordinates.
 - Needs leadership and communication, not just commands.
- 5. Motivating:** stimulating willingness to work.
 - Uses internal (interest, pride) and external (pay, praise) motivators.
 - Satisfied needs → effort and loyalty (Ch. 3).
- 6. Coordinating:** "orderly arrangement of group effort to provide unity of action" (Mooney & Reiley).
 - Synchronizes departments so no one works against another.
 - Internal, external, vertical, horizontal.
- 7. Controlling:** checking results against plans.
 - Set standards, measure performance, correct deviations.
 - Physical, financial and budgetary control.
- 8. Communicating:** the link through which all others work.
 - Sender, message, receiver, feedback.
 - Orders flow down, reports and suggestions flow up.

Planning: meaning, objectives, steps, advantages, limitations, components 3 5

the first function; expand it when a question says “explain the functions” at 8 marks

- **Meaning:** most basic function; **thinking before doing;** preparation for action.
- Includes forecasting and framing objectives, policies, programs, schedules, procedures, budgets.
- **Koontz and O'Donnell:** “deciding in advance what to do, how to do it, when to do it and who is to do it. Planning **bridges the gap** from where we are to where we want to go.”

Objectives of planning

1. **Effective forecasting:** anticipate demand, costs, risks.
2. **Certainty in activities:** everyone knows what comes next.
3. **Performance standards:** targets that control can measure against.
4. **Specific direction:** all departments pull toward one goal.
5. **Tune the organization to its environment:** market, policy, technology.
6. **Economy in management:** no idle resources, no duplicated effort.

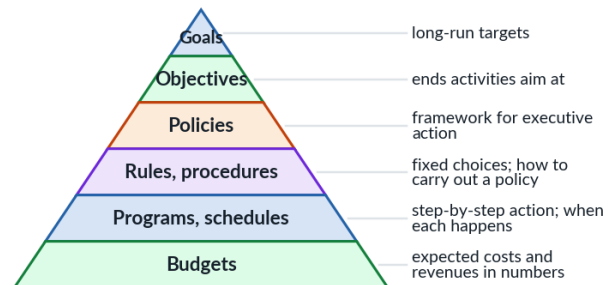
Advantages

- Maximum use of resources; reduces uncertainty.
- Basis for managerial action and for control.
- Encourages innovation and creativity.

Limitations

- Forecasts can be wrong.
- Costly and time consuming.
- External factors (policy, market, disaster) upset plans.
- Resistance to change; rigidity and inflexibility.

Steps in planning (Adhikari)



Components of planning from the broadest (goals) to the most numerical (budgets).

Organizing and staffing: meaning and steps

- **Organizing:** identify and group activities, divide them among people, create authority-responsibility relationships among them.
- The **backbone of management:** links people, work and resources.
- Avoids duplication and gaps: every task has an owner.

Steps in organizing



- **Staffing:** fill the posts of the structure with **competent, qualified** people (detail in Ch. 2).
- Continuous: people retire, leave or are promoted, so posts keep reopening.

Steps in staffing



Directing and motivating

Directing

- Motivating, guiding and supervising subordinates toward company objectives.
- **Only giving orders is not directing:** plans succeed only when diverse human efforts are coordinated, guided and directed.
- It **actuates** members to work effectively and efficiently.
- Steps:
 1. issue orders and instructions;
 2. guide and train subordinates;
 3. supervise their work.
- Elements: supervision, communication, leadership, motivation.

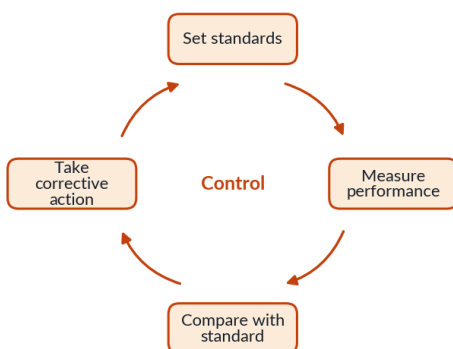
Motivating: fundamentals

- Inspiring people to **intensify their desire and willingness** to perform and co-operate; “push the right button to get the desired action”.
- First fundamental: people want to **exist and survive**: food, clothing, shelter, education, medical care.
- Second: the desire to **achieve a goal**: psychological, social, security, ego (self-respect) needs. People work sincerely when sure these will be met.

Controlling and co-ordinating

Controlling

- Setting standards, measuring actual performance, taking corrective action.
- **Koontz and O'Donnell:** “measurement of accomplishment against the standard and correction of deviations to ensure attainment of objectives according to plans.”
- **Types**, by objective:
 - **physical:** quantity and quality of products;
 - **financial:** unit cost, material cost, labour expense;
 - **budgetary:** future physical and financial standards set, results compared.



Internal (intrinsic)	External (extrinsic)
Interests	Attractive salary, bonus
Emotional attachment	Praise, incentive
Burning desire	Punishment
Fighting spirit for a noble cause	Fear of losing the job

Essentials of a sound motivation system

1. **Good wages:** enough for a decent living.
2. **Financial incentives:** bonus, profit share linked to output.
3. **Achievement and growth:** training, promotion paths.
4. **Human relations:** respect, recognition, fair treatment.
5. **Good working conditions:** safety, light, rest.
6. **Job satisfaction:** interesting, meaningful work.
7. **Job security:** no fear of arbitrary dismissal.

Co-ordinating

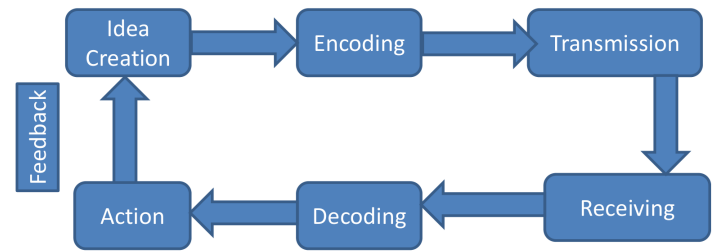
- Ongoing process by which a manager develops an **integrated, orderly, synchronized** pattern of group effort, for **unity of effort** toward a common purpose.
- **Mooney and Reiley:** “the orderly arrangement of group effort, to provide unity of action in the pursuit of common purpose.”
- Achieved through clear objectives, sound structure, communication, committees, leadership.

Types of co-ordination

Where	Internal among departments, levels and people inside the firm	External with customers, suppliers, government, public
Direction	Vertical between levels: top, middle, lower	Horizontal same level: production, sales, finance

Communicating: components and process

- Process by which instructions, ideas, thoughts or information are **transmitted, received and understood** by people in the organization.
- **Components:** sender, message, receiver, feedback.
- Sender prepares the message properly; receiver must understand it clearly.
- **Never complete** until the sender knows it was received and understood, through feedback or the receiver's behaviour.



Process of effective communication.

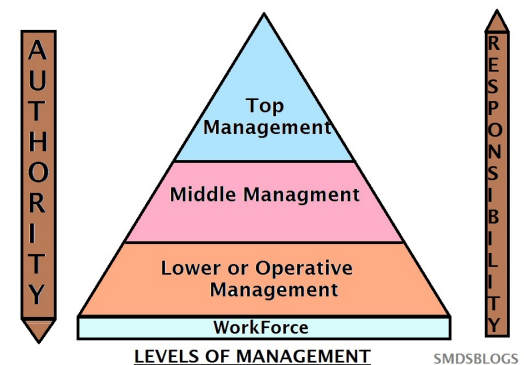
Two principal functions and why they matter most (79 Ba) 4+2

- Choose **planning** and **controlling**: the two ends of the same loop.
- **Planning:**
 - Sets the target, the method and the timetable.
 - Every other function starts from it: organizing groups the work the plan needs, staffing fills it, directing executes it.
 - Reduces uncertainty and waste.
- **Controlling:**
 - Checks the result against the plan.
 - Catches deviations early, while they are cheap to fix.
 - Feeds lessons into the next plan.
- **Why these two:** no plan → nothing to control; no control → plans are only wishes.

Levels of management: why different levels are needed, and functions at each level 2 4 5 6

TOP 9/29 82 Bh, 82 Ba, 80 Bh, 78 Bh, 75 Ash, 74 Ch, 73 Ch, 72 Ch, 69 Ch

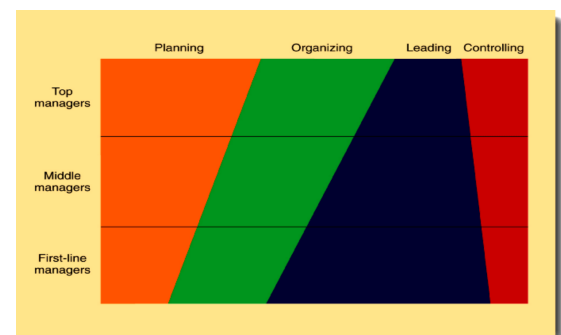
- **Top level** (board of directors, CEO, MD, chairman):
 - Sets goals, objectives, policies, budget.
 - Main function: **decision making, planning**; cross-department responsibility.
 - Answerable for success or failure of the whole firm.
- **Middle level** (department heads, section officers):
 - Implements top's policies; **organizes** resources.
 - Link between top and lower; plans for own department.
 - Selects, trains, motivates lower managers.
- **Lower / operative level** (foreman, supervisor, inspector):
 - Day-to-day supervision of workers; **directing** and on-the-spot control.
 - Passes workers' problems upward.



Authority rises and responsibility widens toward the top. AJ Sir, Ch. 1 deck, slide 15.

Why levels are needed

- **Span of control:** one person cannot direct hundreds.
- **Division of work:** strategic vs tactical vs operational decisions.
- **Clear authority and responsibility;** accountability at each step.
- **Communication:** orders down, reports up.
- **Specialization and career growth:** a ladder for managers.



Share of time on each function by level: top managers plan and organize most; first-line managers lead (direct) most.

Managerial skills and the degree needed at each level; qualities of a good manager 1 3 4 5

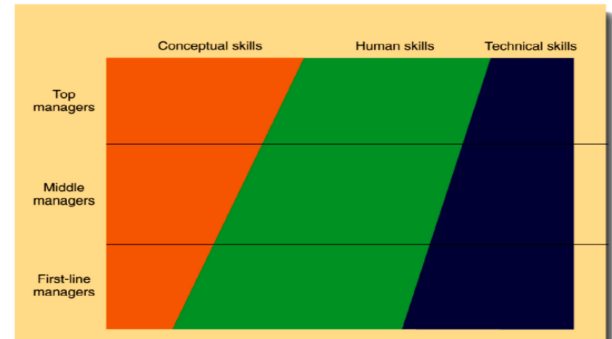
TOP 10/29 82 Ba, 81 Bh, 81 Ba, 78 Bh, 76 Ch, 75 Ash, 74 Ch, 73 Ch, 72 Ch, 71 Shr · short note in 81 Bh

Katz's three skills

- **Technical skill:** job-specific methods, processes, tools, equipment of a specialized field. **Most needed at lower level.**
- **Human (interpersonal) skill:** understand, communicate with, motivate and lead people and groups. **Needed equally at all levels.**
- **Conceptual skill:** see the organization as a whole, analyse and diagnose situations, cause vs effect, long-range decisions. **Most needed at top.**

Modern manager also needs

- **Decision-making, communication, IT / digital skills.**
- **Time management, diagnostic skill, design skill** (solving problems, not just spotting them).



Degree of each skill by level: conceptual shrinks and technical grows going down; human skill stays wide throughout.

Qualities of a good manager

- **Leadership and vision:** sees where the unit must go, carries people along.
- **Decisiveness:** decides on time with incomplete information.
- **Communication:** clear orders, and listens to subordinates.
- **Integrity and fairness:** treats all alike; earns trust.
- **Technical knowledge** of the field, and the ability to keep learning.
- **Emotional stability:** calm under pressure; patient.
- **Initiative:** acts on problems before being told.

Roles of a manager (Mintzberg) 4

PIN 1/29 76 Ash verified/added

- **Interpersonal roles:**
 - **Figurehead:** ceremonial head; signs documents, receives visitors.
 - **Leader:** hires, trains, motivates subordinates.
 - **Liaison:** builds a network outside the unit.
- **Informational roles:**
 - **Monitor:** gathers news about the unit and its environment.
 - **Disseminator:** passes that information inside.
 - **Spokesperson:** speaks for the unit to outsiders.
- **Decisional roles:**
 - **Entrepreneur:** spots opportunities, starts change.
 - **Disturbance handler:** fixes crises, strikes, breakdowns.
 - **Resource allocator:** decides who gets what money, people, time.
 - **Negotiator:** bargains with unions, suppliers, clients.
- AJ Sir's "interpersonal, informational, decisional skills" are these three role groups.

Importance of management 3 4

PIN 2/29 71 Shr, 70 Asa

- **Achieves group goals:** directs everyone to common objectives.
- **Optimum use of resources:** efficient + effective; less waste. The better a society uses scarce resources, the greater its well-being.
- **Reduces costs** → competitive prices.
- **Establishes sound organization:** right person, right job.
- **Adapts to change:** new technology, markets.
- **For individuals:** helps people deal with bosses and co-workers; opens a path to a well-paying job and a satisfying career.
- **Development of society:** "no underdeveloped countries, only undermanaged ones"; jobs, better living standard.
- **Breaks the poverty cycle:** low productivity → low income → low saving → low investment; good management lifts productivity and turns the loop upward (ioenotes p5).

Models of management: name them, explain any two or three with examples 3 4 5

HOT 4/29 81 Bh, 80 Bh, 76 Ash, 70 Ch · team-effort model as a short note in 81 Bh

1. Hierarchical:

- Authority comes from the superior; responsibility is owed to the superior.
- Eg army, civil service.
- **Misses:** leadership, charisma, informal communication.

2. Task-oriented:

- Manager performs five tasks: plan, organize, direct, coordinate, control.
- The tasks **cycle**, they are not a line (figure below).
- Eg a production plant. **Misses:** can create artificial manager-worker links.

3. Allocational:

- Manager obtains, allocates, shares, monitors and returns resources (money, manpower, materials): "keeping the cup topped up".
- Eg finance or project resource manager.
- **Misses:** no true **process** view of managing.

4. Transactional:

- Management as a stage in the value-added chain; adds value to raw resources.
- Shows **how** management adds value, not just control.
- Eg supply chain firms.

5. Team effort:

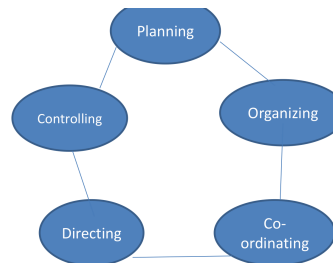
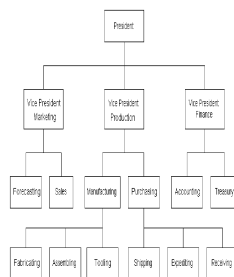
- Interlocking teams; manager as team leader stressing cooperation and coordination.
- Team output exceeds the sum of individuals.
- Leader embodies the group's traits to motivate it.
- Eg an agile software team, a surgical team.

6. Knowledge-oriented:

- Manager builds knowledge by experience, diffuses it by mentoring, teaching, coaching, training.
- The firm learns to adapt and wins in change.
- Eg consulting, R&D.

7. Goal-concentrated:

- Management as control toward goals, including resource management.
- Information from the environment triggers adjustments so the plan holds.
- Eg a sales team with targets.
- A real firm usually mixes several models.



Hierarchical model (left) and task-oriented model (right).

Management: science or art? 3

PIN 1/29 78 Bh

- **Science:** systematic body of knowledge; principles (Fayol, Taylor) tested by observation; cause-effect. Eg time study sets a standard output.
- Inexact science: deals with people, so results are not perfectly repeatable.

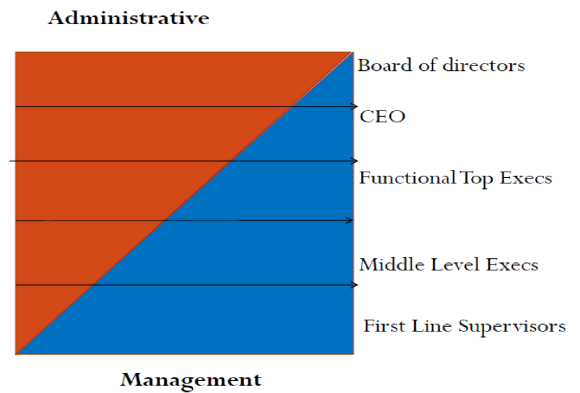
- **Art:** personal skill, creativity, judgement in applying knowledge; improves with practice. Eg two managers apply the same rule with different success.
- **Both:** science gives the principles, art applies them. Complementary.

Organization vs management (vs administration) 2

PIN 3/29 79 Ba, 76 Ash, 71 Ch

Basis	Organization	Management
Meaning	Structure: people + jobs + relations	Process: planning, organizing, directing, controlling
Nature	A tool / framework (the body)	The force that runs it (the brain)
Relation	One function of management	Uses organization to reach goals
Output	Defined roles, authority	Goals achieved efficiently

- **Administration:** lays down corporate policy, obtains finance, sets the limits within which the firm works: top level, **thinking**.
- **Management:** actual execution of that policy within those limits: lower levels, **doing**.
- Both exist at every level in different shares (figure): the board is mostly administration, first-line supervisors mostly management.



Administration (orange) shrinks and management (blue) grows going down the levels.

Which management theory is most practical today / best for organizations in Nepal? 4 8

PIN 2/29 80 Ba, 74 Ash · an opinion question: take a side and justify

- Argue for the **contingency / system approach**: no single best way; mix theories to fit the situation.
- **Why, for Nepal:**
 - Mixed sector: family firms, cooperatives, public corporations, MNCs, IT startups → each needs a different style.
 - Labour-intensive manufacturing (garments, cement) → Taylor's time study and incentives still raise output.
 - Public enterprises (NEA, NOC) → Fayol's clear hierarchy, unity of command, discipline cure overlapping authority.
 - Service and IT firms, remittance-driven youth workforce → behavioural approach: motivation, participation, recognition to retain staff.
 - Political instability, frequent policy change → open-system thinking, watching the environment.
- Conclude: **contingency approach**, drawing on all the classical tools.

1.3 Theories of Management

Scientific management (Taylor): principles, experiments, features, applicability today 3 4 5 8

TOP 11/29 82 Bh, 81 Ba, 80 Bh, 79 Ba, 75 Ch, 75 Ash, 72 Ch, 71 Ch, 68 Ch, 68 Ba, 65 Shr · short note in 68 Ch

- **F.W. Taylor** (1856–1915), US engineer at Midvale and Bethlehem Steel; “Father of Scientific Management”. Book: *Principles of Scientific Management* (1911). Term coined by Louis Brandeis (1910); Charles Babbage anticipated it (1832).
- **Definition:** “the art of knowing exactly what is to be done and the best way of doing it”; replacing rule of thumb with scientific study of work.
- Applies scientific knowledge and method to every aspect of management → best use of men, materials, machines, capital.
- **Historical development:**
 - Before: rule-of-thumb factories, soldiering (deliberate slow work), no standards.
 - Taylor's shop studies (1880s).
 - Gilbreths' motion study; Gantt chart.
 - Mass production (Ford's assembly line, 1913).

Taylor's own principles

1. **Science, not rule of thumb**, for each element of work.
2. **Scientific selection, training and development** of workers.
3. **Cooperation**, not individualism, between management and labour.
4. **Equal division of responsibility**: management plans, workers do.
5. Harmony, not discord; maximum output, not restricted output.

Adhikari's list of principles (write it if asked "principles" at 8 marks)

1. Science for each element of work.
2. Scientific selection, placement and training.
3. Division of labour.
4. Standardization of methods and tools.
5. Time and motion study.
6. Differential wages.
7. Labour-management co-operation.
8. Management by exception.

Techniques (features)

- **Time study**: standard time → fair day's work.
- **Motion study**: remove wasted motions (Gilbreth).
- **Fatigue study**: rest pauses.
- **Standardization** of tools, methods, conditions.
- **Differential piece-rate** wages: higher rate above standard.
- **Functional foremanship**: 8 specialist foremen.
- **Management by exception**: only deviations go to top.

Scientific management vs management science 3

PIN 1/29 72 Ch

- **Scientific management** (Taylor, 1900s): improves **worker productivity** at shop level by time and motion study, standardization, incentive wages.
- **Management science** (quantitative approach, 1940s on): uses **mathematical models**, operations research, statistics, simulation, linear programming for **managerial decisions** at every level.
- **Scope**: the first studies how a job is done; the second studies what decision to take (how much to stock, which route, which project).
- **Tools**: stopwatch, motion charts, piece rates vs computers, mathematical models, queuing, inventory models.
- **Link**: management science grew out of Taylor's idea of measuring instead of guessing.

Experiments that support it

- **Pig-iron handling** (Bethlehem Steel, 1898): 75 labourers loading 42 kg pigs; by method study, rest periods and incentive pay, output rose **12.5 → 47.5 tons/day** (AJ Sir's figures: 12.7 → 48.8), wages **\$1.15 → \$1.85**.
- **Shovelling**: shovel sized to a 21.5 lb load for each material; 400–600 men cut to 140, cost per ton halved.
- **Metal cutting**: 26 years, 30,000 tests on speeds, feeds → high-speed steel.

Still applicable today? Yes, largely

- Assembly lines, fast food, BPO call centres use standard times and methods.
- Lean manufacturing, Six Sigma, industrial engineering are its heirs.
- Incentive pay, work study, ERP-based control.
- But: human needs ignored, monotony, worker resentment → combine with behavioural approach.

Role of HR manager; how HR manager implements scientific management 4+4

PIN 1/29 75 Ch

Role / significance of the HR manager (4)

- **Manpower planning:** forecasts how many people, of what skill, when.
- **Recruitment and selection:** attracts and picks the right candidates.
- **Training and development:** builds skills; plans careers.
- **Performance appraisal:** measures work, feeds promotion and pay.
- **Wage and salary administration:** fair, competitive pay; incentives.
- **Welfare and safety:** working conditions, health, canteen, housing.
- **Industrial relations:** handles grievances, unions, bargaining.
- **Retention:** keeps good staff; records and HR information (see Ch. 2).

How the HR manager implements scientific management (4)

1. **Job analysis:** studies each job; writes job description and specification.
2. **Time and motion study:** fixes the standard time and the one best method.
3. **Scientific selection:** tests and interviews match people to jobs.
4. **Training** in the standard method, not rule of thumb.
5. **Standardization:** tools, working conditions, rest pauses.
6. **Differential incentive wages:** higher rate for output above standard.
7. **Worker-management cooperation:** explains the system, shares gains, handles grievances so workers do not resist.
8. **Management by exception:** reports only deviations upward.

Administrative management (Fayol): 14 principles, and are they applicable today? 4 8

TOP 9/29 82 Bh, 80 Ba, 79 Bh, 79 Ba, 75 Ash, 72 Ka, 71 Shr, 70 Ch, 70 Asa

- **Henri Fayol** (1841–1925), French mining engineer; “Father of Modern Management”. *General and Industrial Management* (1916).
- Studied management **from the top down** (Taylor: from the shop floor up); set the pattern of management and the **pyramidal** organization.
- Technical ability dominates at lower levels; managerial ability at higher levels.

The 14 principles

1. **Division of work:** each person does a specialized part → skill, speed, output rise.
2. **Authority and responsibility:** go together; whoever has the right to order answers for the result.
3. **Discipline:** obedience and respect for agreements; fair penalties; needs good superiors at every level.
4. **Unity of command:** orders from one boss only; two bosses → confusion.
5. **Unity of direction:** one plan and one head for each group of activities with the same objective.
6. **Subordination of individual interest** to the general interest of the firm.
7. **Remuneration:** fair to both employee and employer; satisfies both.
- **Six activities** of an industrial undertaking:
 1. **Technical:** production, manufacture, adaptation.
 2. **Commercial:** purchasing, selling, exchange.
 3. **Financial:** optimum use of capital.
 4. **Security:** protection of property and persons.
 5. **Accounting:** stock taking, balance sheet, costing, statistics.
 6. **Managerial:** planning, organizing, commanding, co-ordinating, controlling.
8. **Centralization:** how far decisions stay at the top; the degree depends on size and situation.
9. **Scalar chain:** unbroken line of authority from top to bottom; a “gang plank” lets peers talk directly when speed matters.
10. **Order:** a right place for everything and everyone (material and social order).
11. **Equity:** kindness plus justice toward all employees → loyalty.
12. **Stability of tenure:** low turnover; people need time to learn a job.
13. **Initiative:** let subordinates think out and carry out plans.
14. **Esprit de corps:** “union is strength”; team spirit and harmony.

Applicable today? Yes

- Every organization chart uses scalar chain, unity of command, division of work.
- Fayol's functions (POCCC) are still the standard management process.
- Matrix structures bend unity of command; startups bend centralization. Principles are **flexible guidelines**, applied by situation (Fayol said so himself).

Experiments / contributions around it

- Fayol: 30 years running Commentry-Fourchambault, turned a near-bankrupt mining firm profitable by applying these principles.
- **Frank Gilbreth** (father of motion study): process chart; fatigue cut by rest periods, seating layout, better conditions, principles of motion economy; therbligs.
- **Gantt**: Gantt chart, a scheduling tool comparing **planned vs actual** progress at a glance → corrective action on a significant deviation (see the plan chart in §4.4).
- **Weber**: bureaucracy (rules, hierarchy, impersonality).

Behavioural management approach (Mayo, Hawthorne): and how it compares with scientific and administrative approaches 3 4 5 8

TOP 11/29 82 Ba, 81 Bh, 81 Ba, 80 Ba, 79 Ba, 78 Bh, 76 Ch, 75 Ash, 73 Shr, 71 Shr, 66 Bh

- Scientific management concentrated on **physical** resources, not human ones.
- **Elton Mayo** and associates (1920s–30s): organization is a **social system**, not only a techno-economic one. Productivity depends on human behaviour and social environment, not only money.
- Good human relations → harmony, higher satisfaction → higher efficiency and productivity.
- Also: **Maslow** (need hierarchy), **McGregor** (X-Y), **Herzberg**, Follett, Barnard. Human relations movement.
- **Motivators** are the job and its conditions: recognition, belonging, challenging work, independence, participation, achievement, **job enlargement and enrichment**.

Elements of behavioural theory

1. **The individual**: recognize individual differences; each person has feelings, emotions, perceptions, attitudes, a changing psychology.
2. **The work group**: informal organization matters; integrate it intelligently with the formal one; group psychology drives motivation and productivity.
3. **Participative management**: workers share decision making and problem solving, mainly to raise productivity.

Hawthorne experiments (Western Electric, 1924–32)

1. **Illumination** (1924–27): output rose whether light went up *or down* → physical factors were not the cause.
2. **Relay assembly test room**: 6 women; rest, hours, pay varied; output rose almost every time → attention, group morale matter.
3. **Interviewing programme**: 21,000 workers → feelings, informal groups shape performance.
4. **Bank wiring room**: 14 men; group set its own output norm and enforced it → informal group controls output.

Conclusions

- Attention from managers raises output (Hawthorne effect).
- Workers are social beings; money is not the only motivator.
- Informal groups set norms; must be integrated with the formal organization.
- Satisfaction → productivity; participation and communication matter.
- Managers need **social skills**, not only technical ones.

Limitations

- Over-stresses the social side; satisfied workers are not always productive.
- Hawthorne method criticized (small samples).

Basis	Scientific (Taylor)	Administrative (Fayol)	Behavioural (Mayo)
Focus	Worker, shop floor	Manager, whole firm	People and groups
View of person	Economic man, money motivates	Part of a structure	Social man, needs motivate
Aim	Max output per worker	Sound structure, principles	Satisfaction → productivity
Method	Time and motion study	Principles from experience	Experiments, observation
Informal group	Ignored	Ignored	Central
Contribution	Standards, incentives, efficiency	Management process, 14 principles	Motivation, participation, human relations
Limitation	Mechanical, monotony	Rigid, universal claims	Neglects structure, economics

Advantages of behavioural over scientific (78 Bh, 73 Shr)

- Treats workers as humans, not machines.
- Uses informal groups instead of ignoring them.
- Participation raises commitment.
- Lower turnover, absenteeism and conflict.
- Quality improves with morale.

Advantages over modern theory (82 Ba)

- Simpler to apply; needs no models or data.
- Directly improves morale and motivation, where quantitative models ignore people.
- Still the base of HR practice.
- Admit that modern theory integrates it.

Modern management theories: system approach and contingency approach 4 5 7

HOT 4/29 82 Ba, 68 Ch, 67 Asa, 66 Bh · short note in 66 Bh

- After 1950; four streams:

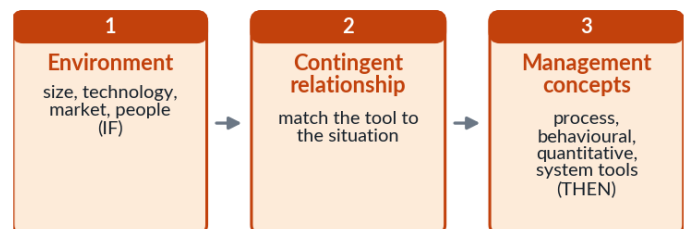
1. **Quantitative:** build a **model** expressing the system's effectiveness as a function of variables, at least one controllable. Tools: OR, maths, simulation, model building. Taylor's method was its early form. Systematizes managerial thinking.
2. **System:** organization = open system of interrelated sub-systems; inputs (information, energy, materials) → conversion (men-machine system) → outputs + feedback, all steered by a plan (Boulding, Johnson, Rosenzweig, Churchman, late 1960s). System = "an established arrangement of components which leads to accomplishment of particular objectives as per plan".
3. **Contingency:** "it depends".
4. **Operational:** Koontz, O'Donnell, Weihrich: management is a **universal body of knowledge** for every level and enterprise, applied with judgement to each firm's real problems.

Characteristics of modern thought

- System approach; dynamic; adaptive.
- Multidisciplinary; multivariable in thought.
- Multi-motivated in action.
- Multi-level and multi-dimensional.
- Probabilistic, not deterministic.

Contingency approach

- Woodward, Fiedler, Lawrence and Lorsch.
- **No one best way:** the right structure, style, technique depends on the situation (size, technology, environment, people).
- Eg a routine garment line suits tight rules; an R&D lab suits loose, participative control.



Importance

- Practical and flexible: fits the answer to the problem.
- Integrates classical, behavioural and quantitative tools.
- Fits fast-changing environments.
- Trains managers to diagnose before acting.
- Weakness: no firm principles, hard to teach.

Management by objectives (MBO) 4

LOW 1/29 68 Ch · short note verified/added

- **Peter Drucker (1954)**: superior and subordinate **jointly set objectives**, then review performance against them.
- Objectives are specific, measurable and time-bound.
- Process is a cycle (right).

Merits

- Clear goals: everyone knows what is expected.
- Commitment: people own targets they helped set.
- Self-control replaces close supervision.
- Objective appraisal against agreed results.

Demerits

- Time consuming; heavy paperwork.
- Favours short-term, measurable goals over quality.
- Fails if top management is not committed.



1.4 Forms of Ownership

Forms of ownership: list with brief explanation; choosing one; how an organization changes its form

4 8

HOT 4/29 82 Ba, 74 Ch, 73 Shr, 71 Shr

- **Firm** = ownership unit combining men, materials, machines to produce and sell at a profit.
- **Five forms**:
 1. **Single ownership**: one owner, unlimited liability.
 2. **Partnership**: 2+ partners by a deed, share profit and loss.
 3. **Joint stock company**: capital from shareholders, limited liability, separate legal entity.
 4. **Cooperative society**: voluntary, one member one vote, service motive.
 5. **Public corporation**: created by special act, owned by government.

Choosing a form: evaluation criteria

- **Size and nature** of the business.
- **Capital**: start-up and future needs.
- **Liability exposure**: can the owners risk personal property?
- **Control** the owners want to keep.
- **Tax** consideration.
- **Managerial ability** available.
- **Business goals** and **succession plans** (continuity).
- **Cost of formation** and government rules.

Changing form of ownership (73 Shr)

- Growth needs capital and skills one owner lacks → owner takes partners.
- Partnership's unlimited liability and instability → convert to a **private limited company**.
- More capital needed → **public limited** via IPO.
- Reverse path: loss-making public enterprise → privatized.

Nepali examples verified/added

- **Chaudhary Group**: family trading shop → group of companies → listed CG / Nabil holdings.
- **Nepal Telecommunications Corporation** → Nepal Doorsanchar Company Ltd (2004) → listed public company.
- **Bhrikuti Paper, Bansbari Leather**: public enterprises privatized in the 1990s.

Basis	Single	Partnership	JSC	Co-operative	Public corp.
Owners	One	2+, by deed	Shareholders	Members, 1 vote each	Government
Liability	Unlimited	Unlimited	Limited	Limited	Limited
Legal entity	No	No	Separate	Separate	Separate
Life	Ends with owner	Ends with a partner	Perpetual	Perpetual	Perpetual
Motive	Profit	Profit	Profit	Service	Public service

Single ownership (sole proprietorship): features, advantages, limitations 4 8

PIN 3/29 80 Bh, 71 Shr, 65 Shr · short note in 65 Shr; 80 Bh compares it with a JSC (table below)

Features

- Oldest, simplest; one person owns, finances, manages, bears all risk.
- Owner and business are the same in law.
- **Unlimited liability**: personal property liable for business debts.
- Suits small shops, cottage industry, handicrafts, farming.

Advantages

- Easy and cheap to form and dissolve; few legal formalities.
- Full control, prompt decisions, flexibility.
- Direct motivation: all profit to owner.
- Business secrecy.
- Personal contact with customers and employees.
- Little government control.

Limitations

- Limited capital.
- Limited managerial skill (one person cannot master production, marketing, finance, HR).
- Unlimited liability.
- Uncertain life: ends with owner's death or illness.
- Cannot compete with large firms; no economies of scale.
- Heavy workload, stress; isolation in decisions.

Partnership: features, types of partners, advantages and difficulties 3 4 5

PIN 3/29 81 Bh, 74 Ash, 67 Asa · short note in 67 Asa

Features

- Association of 2+ persons to run a business and share profit.
- Formed by agreement: **partnership deed** (capital, profit ratio, duties, dissolution).
- Unlimited, joint and several liability (except limited partners).
- Mutual agency: each partner binds the firm.
- No separate legal entity; restricted transfer of interest.

Types of partners

- **General / active**: manage, unlimited liability.
- **Limited**: liability up to capital; no say in management.
- **Sleeping / silent**: capital only, no management, still liable.
- **Nominal**: lends name only; no capital.
- **Minor**: under 18, benefits only.
- **Partner by estoppel**: implies being a partner, liable to third parties.

Advantages

- Easy to form; little regulation.
- More capital than sole owner.
- Complementary skills; division of labour.
- Shared risk; prompt decisions (few people).
- Flexibility; personal interest of partners.

Difficulties / disadvantages

- Unlimited liability of at least one partner.
- Conflict, misunderstanding between partners → instability.
- Limited life: ends on death, insolvency, retirement of a partner.
- Mutual agency: one partner's mistake binds all.
- Cannot easily transfer share; capital still limited.
- Unequal effort vs equal sharing breeds resentment.
- Less secrecy than a sole owner; hard to dispose of one's interest.

Joint stock company: meaning, characteristics, types, formation in Nepal, advantages and disadvantages 2 3 4 5 6 8

TOP 10/29 81 Ba, 80 Bh, 80 Ba, 79 Ba, 76 Ch, 76 Ash, 75 Ash, 70 Ch, 69 Ch, 67 Asa · **most asked form of ownership**

- **Meaning**: voluntary association of persons, registered under the Companies Act, whose capital is divided into transferable shares. Shareholders own it; an elected **board of directors** manages it (elected at the AGM).

Characteristics

- Created by law (registration); **separate legal entity**.
- **Perpetual succession**: unaffected by death of shareholders.
- **Limited liability**: up to share value.
- **Transferable shares** (freely in public company).
- Common seal; can sue and be sued.
- Ownership separate from management.
- Large membership, large capital.

Types (Companies Act 2063, Nepal) verified/added

- **Private limited**: 1 to 101 shareholders; cannot invite the public to buy shares; share transfer restricted; name ends "Pvt. Ltd."
- **Public limited**: min 7 promoters, no upper limit; min paid-up capital Rs 1 crore; shares offered to public and listed; "Ltd."
- Adhikari's "max 50" for private is the old Indian figure; Nepal's is 101.

Advantages

- Huge capital from many investors; fresh issue possible.
- Limited liability → attracts investors.
- Transferable shares → liquidity.
- Stability, continuity.
- Professional, efficient management.
- Economies of large-scale production.
- Public confidence (audited, published accounts); easy loans.
- Social benefit: jobs, investment opportunity.

Formation / registration in Nepal verified/added

- Promotion**: promoters identify the idea, feasibility, name.
- Name approval** from the **Office of the Company Registrar (OCR)** (online, ocr.gov.np).
- Documents** filed:
 - Application form.
 - **Memorandum of Association**: name, objectives, registered office, authorized / issued / paid-up capital.
 - **Articles of Association**: internal rules, board, meetings, share transfer.
 - Promoters' citizenship (passport for foreigners); agreement among promoters; consent of directors.
 - Registration fee (scaled to authorized capital).
- OCR scrutinizes → **certificate of incorporation**.
- PAN / VAT at Inland Revenue; local ward registration.
- Public company only: prospectus approved by SEBON, IPO, **certificate of commencement of business**, listing at NEPSE.

Disadvantages

- Difficult, costly, lengthy formation; legal formalities.
- Lack of personal interest (salaried managers, scattered owners).
- Delay in decisions (board meetings).
- Lack of secrecy.
- Excessive government regulation.
- Directors may exploit shareholders; speculation in shares; favouritism.
- Impersonal labour relations.

Basis	Single ownership	Partnership	Joint stock company
Owners	One	2 or more, by deed	Many shareholders
Formation	Easiest, cheapest	Easy, by agreement	Registration, costly
Legal entity	None	None	Separate
Liability	Unlimited	Unlimited (joint)	Limited
Capital	Small	Moderate	Large
Management	Owner	Partners	Board of directors, professional
Continuity	Ends with owner	Ends with a partner	Perpetual
Transfer of interest	At will	Needs all partners' consent	Free (public)
Secrecy	Full	Fair	Little
Regulation	Minimum	Low	Heavy

Why JSC is better than partnership (79 Ba, 76 Ch, 76 Ash)

- **Limited liability**: shareholders risk only their shares.
- **Larger capital** from many investors.
- **Perpetual life**: not ended by a death or quarrel.
- **Transferable shares**: an owner can exit without closing the firm.
- **Professional management** by an elected board.
- **Public confidence**, audited accounts → easier loans.

Private limited vs public limited company 5

PIN 1/29 67 Asa

Basis	Private limited	Public limited
Members (Nepal)	1 to 101	Min 7 promoters, no max
Shares to public	Not allowed	Offered to public, listed
Transfer of shares	Restricted	Free
Min capital	No statutory minimum	Paid-up Rs 1 crore
Start of business	On incorporation	Needs certificate of commencement
Disclosure	Less	Full, audited, published

Co-operative societies: meaning, features, types, merits and demerits 5 8

HOT 4/29 82 Bh, 81 Bh, 79 Bh, 72 Ka

- **Meaning:** voluntary association of persons on equal footing for common economic interest; **service motive**, not profit. Motto: “each for all and all for each”.
- **Features:**
 - Voluntary, open membership.
 - Democratic management: **one member one vote**, regardless of shares.
 - Limited dividend; surplus shared as patronage bonus.
 - Self-help and equality.
 - Separate legal entity after registration.
- **Nepal:** registered under the Cooperative Act 2074; a primary cooperative needs at least 30 members verified/added.

Merits

- Easy formation; limited liability; continuity.
- Democratic; protects weak members.
- Eliminates middlemen → cheaper goods, better prices.
- Government support, tax concessions; reduces inequality.

Types

1. **Producers’:** small producers pool raw materials, tools, sell output (dairy, handicraft, cottage industry).
2. **Consumers’:** buy wholesale, sell daily goods to members cheaply.
3. **Marketing:** farmers, artisans sell produce jointly (Eg milk cooperatives).
4. **Housing:** members get land, build houses at low cost.
5. **Credit / savings and credit:** pool savings, lend to members at fair rates (most common in Nepal).
6. **Multipurpose:** several services together.
7. **Farming:** small farmers pool land for mechanized farming.

Demerits

- Limited capital; inefficient, amateur management.
- Lack of motivation (fixed dividend); lack of secrecy.
- Lack of co-ordination; members lose the **spirit of co-operation** over time.
- Undue political influence, favouritism; misuse of funds (Nepal’s savings-cooperative scandals).

Public corporation: features, advantages, disadvantages; vs joint stock company 4

PIN 1/29 81 Ba

- **Meaning:** autonomous body created by a **special act of parliament**, wholly owned by government, for public service. Nepal: Corporation Act 2021 BS and individual acts. Eg Nepal Electricity Authority, Nepal Oil Corporation, Nepal Airlines Corporation, Nepal Food Corporation.
- **Features:**
 - Statutory creation; separate legal entity.
 - Wholly government owned; board appointed by government.
 - Financial and managerial autonomy.
 - Service motive.
 - Public accountability to parliament.

Advantages

- Autonomy + public accountability: flexibility of business, control of government.
- Service motive, protects public interest; strategic sectors kept national.
- Large finance from government; continuity; limited liability.
- Adhikari's "Public Corporations" slides (pp111–114: domestic / foreign / alien, double taxation) describe the **American business corporation**, i.e. a JSC. The syllabus means the government statutory corporation above.

Disadvantages

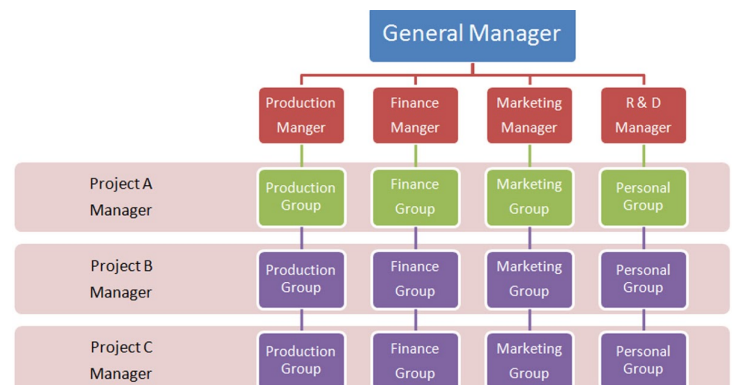
- Political interference; appointments by politics, not merit.
- Inefficiency, bureaucracy, losses (no profit pressure).
- Monopoly → poor service to customers.
- Autonomy often only on paper; delay in decisions.

Basis	JSC	Public corp.
Created by	Registration, Companies Act	Special act
Owners	Private shareholders	Government
Motive	Profit	Public service
Management	Elected board	Government-appointed board
Accountable to	Shareholders	Parliament, public
Capital	Shares	Government funds, loans

As CEO of a software company, which ownership and structure would you choose? 4+4

PIN 1/29 74 Ch · answer with a figure

- **Ownership: private limited company:**
 - Limited liability protects founders.
 - Can bring in investors and give employees shares (ESOP).
 - Continuity; credibility with foreign clients.
 - Converts to public later if it grows.
- **Structure: matrix:** functional departments (development, QA, design, marketing, finance) supply staff to project teams under project managers.
 - Many parallel client projects.
 - Shares scarce specialists.
 - Fast response, skill growth.



Matrix organization: functional managers across the top, project managers down the side. AJ Sir, Ch. 1 deck, slide 67.

1.5 Organizational Structure

Organization structure: meaning, factors, how it is designed for an enterprise 2 4

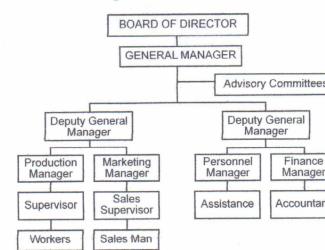
HOT 6/29 79 Bh, 75 Ch, 71 Ch, 71 Shr, 70 Asa, 68 Ch · short note in 79 Bh, 75 Ch, 71 Shr

- **Meaning:** the **framework** that divides total activities into groups, fixes superior-subordinate relations and authority, and shows who reports to whom. Shown on an **organization chart**.
- Covers division of labour, coordination, communication, workflow, formal power.
- **Depends on:**
 - Size of the firm; nature of product.
 - Technology; complexity of problems.
 - Environment; competence of staff.
- **Types:** line, functional, line & staff, committee, (matrix, project).
- The structure is the **skeleton**: it divides total activity into groups, prescribes each person's authority, and so shows the **hierarchy**: persons arranged by rank.
- An **organization chart** draws it: boxes = positions, lines = authority, dashed or side boxes = advisory (staff) bodies.
- Read the chart right: the board sets policy, the GM executes it, an advisory committee advises the GM without commanding anyone.

How it is designed (steps)

1. Set objectives.
 2. Identify and list activities.
 3. Group activities into departments (by function, product, area, customer).
 4. Decide levels and span of control.
 5. Assign authority and responsibility; delegate.
 6. Set coordination and communication links.
 7. Draw the chart; review as the firm grows.
- No single best structure: each firm evolves its own.

2.3.6 Organization Chart



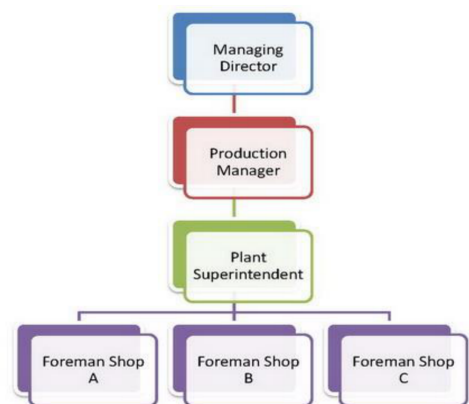
An organization chart with an advisory committee beside the general manager.
ioenotes Ch. 1, p20.

Line organization: features, advantages and disadvantages 4

HOT 4/29 76 Ch, 76 Ash, 71 Ch, 70 Asa

- Oldest, simplest; also **military** or **scalar** organization.
- Authority flows **vertically** top to bottom; responsibility bottom to top.
- Each head has full control of his department (finance, production, sales); each department is subdivided into sections and is a **self-supporting unit**.
- Lines of authority clearly defined; managers must be all-rounders.
- Suits small firms, routine work, few employees.

Advantages	Disadvantages
Simple, easily understood	Overload on top executives
Clear authority and responsibility	No specialization
Strong discipline	Dictatorial, one-man show
Unity of command	Depends on a few key persons
Quick decisions	Duplication of work across departments
Rapid communication, easy coordination	Unsuitable for large firms
Economical	Scope for favouritism; poor communication upward
Each department self-supporting	Undue reliance on key men; wastage of materials

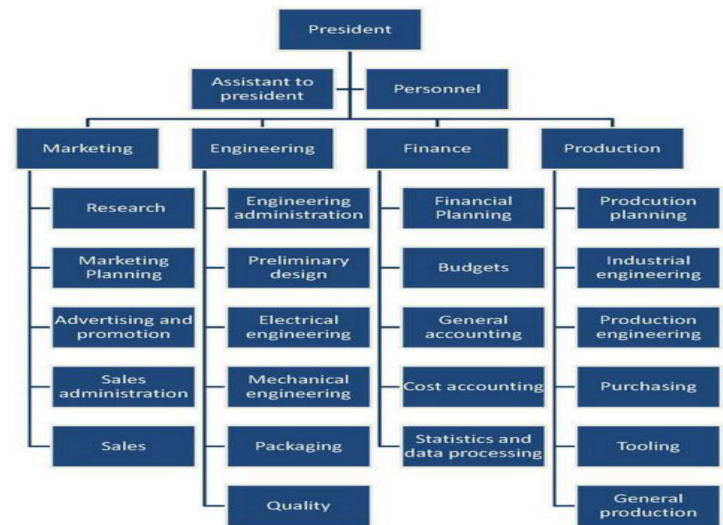


Line organization: a single vertical chain from managing director to foremen.

Functional organization: features, advantages and disadvantages (with sketch) 3 4

HOT 4/29 82 Bh, 68 Ch, 68 Ba, 67 Asa · short note in 67 Asa

- Proposed by **F.W. Taylor**: all-round foremen were hard to find.
- Work divided by **function**; each function under a **specialist** who commands workers **only in his field**.
- Taylor's **functional foremanship**: 8 foremen.
 - **Planning room (4)**: route clerk, instruction card clerk, time and cost clerk, disciplinarian.
 - **Shop floor (4)**: gang boss, speed boss, repair boss, inspector.
- Worker receives orders from several bosses → breaks unity of command.



Functional structure: marketing, engineering, finance, production, each under its expert.

Advantages

Specialization, expert guidance
 Efficiency, standardized operations
 Easy selection and training of specialists
 Reduced burden on top executives
 Lower cost through expertise; scope for growth
 Separation of planning from doing; lower prime cost

Disadvantages

Violates unity of command → indiscipline
 Overlapping authority, conflicts
 Shifting of responsibility
 Poor coordination between functions
 Delayed decisions needing several experts; costly
 Kills workers' initiative

Line and staff organization; its advantages over line and functional organization 4 5

HOT 4/29 82 Ba, 79 Bh, 74 Ch, 66 Bh

- Line structure + **staff specialists** (legal adviser, personnel, research, quality control) who **advise** line managers but have **no command authority**.
- “Staff are thinkers, line are doers.”
- Suits large plants: steel mills, electricity boards, big factories.
- Why staff exist: line heads are busy with routine production, so staff do the investigation, research, recording and advising, and supply expert information.

General advantages

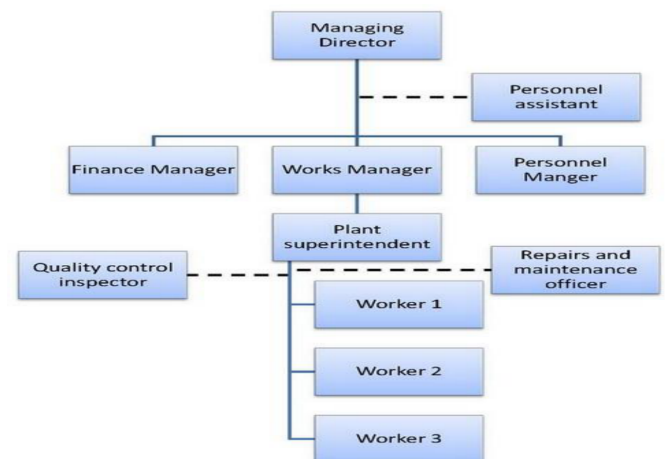
- Specialized knowledge available; suits progressive, growing business.
- Less wastage, improved quality, better decisions.

Advantages over line organization

- Expert advice → better decisions.
- Relieves overloaded line executives.
- Suits growth and complexity.

Advantages over functional organization

- Keeps unity of command and discipline.
- No overlapping authority.
- Clear accountability for results.



Line and staff: dashed lines are staff (advice only).

Disadvantages

- Line-staff friction; line resents advice.
- Staff without authority may be ineffective.
- Staff not accountable for results → lethargy.
- Expensive; confusion over duties.
- Line may lose initiative, over-rely on staff.
- Chances of misinterpreting staff advice.

Committee organization: types, merits and demerits 4

PIN 3/29 79 Bh, 75 Ash, 74 Ch

- Formal group of 2+ persons appointed to deliberate on problems one person cannot solve; members meet, discuss, recommend or decide.
- Not a separate structure: added to line or line & staff at any level.

Types

- **Ad hoc**: temporary, dissolves after its task.
- **Standing / permanent**: repetitive work (purchase, research committee).
- **Advisory (staff)**: recommend only.
- **Executive (line)**: has power to decide and enforce.
- **Formal / informal**: by charter or by convenience.

Merits

- Pooled knowledge; problem seen from several views.
- Coordination between departments.
- Democratic, participative; better acceptance.
- New ideas; checks concentration of power.

Demerits

- Slow decisions; costly (many people's time).
- Compromise, not the best decision; non-uniform final decisions.
- Divided responsibility (no one accountable).
- Dominated by a few members.

Matrix / project organization; which structure suits a (temporary) engineering project? 5 8

PIN 3/29 73 Ch, 72 Ch, 68 Ba · 68 Ba asks the matrix chart; 74 Ch's software-company version is in §1.4

- **Project organization**: a temporary team under a **project manager**, formed for one project, dissolved at completion.
- **Matrix**: project structure laid over a functional one; staff report to a functional manager (technical) **and** a project manager (schedule, cost). Two bosses. (Figure in §1.4.)
- **Merits**:
 - Efficient use of specialists across projects.
 - Clear project accountability.
 - Skill development, job rotation.
 - Fast response to clients.
- **Demerits**:
 - Violates unity of command (two bosses).
 - Power struggle between functional and project managers.
 - Complex, confusing; costly coordination.

Best for a temporary engineering project: project / matrix

- Project has fixed **start, end, budget, scope**: a dedicated project manager owns all three.
- Needs **many disciplines** (civil, electrical, mechanical, finance) together.
- Team **disbands** at completion; specialists return to departments.
- Single point of contact for client; faster decisions than line.
- **Line** would lack specialists, **functional** would scatter responsibility, **committees** are too slow.
- Large one-off project (hydropower, highway) → pure project organization; firm running many projects → matrix.

Authority and responsibility; span of control; policy group vs executive group 3 4 5

HOT 4/29 68 Ch, 68 Ba, 67 Asa, 65 Shr · span of control also a short note in 68 Ch

Authority vs responsibility

Basis	Authority	Responsibility
Meaning	Right to decide, command, use resources	Obligation to perform assigned work
Flow	Downward	Upward
Delegation	Can be delegated	Cannot be delegated fully
Source	Position (formal)	Superior-subordinate relation

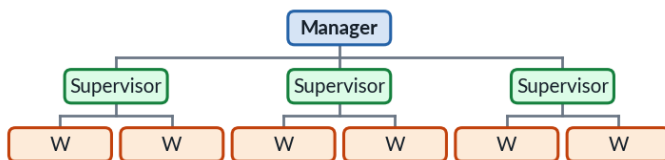
Principle: authority must be **equal** to responsibility.

Policy group vs executive group

- **Policy group**: board of directors / top management; frames objectives, policies, budgets (administration, thinking).
- **Executive group**: managers who carry policies out, direct day-to-day work (management, doing).

Span of control

- **No. of subordinates directly reporting** to one superior.
- **Narrow span:** close supervision, many levels, tall structure, slow communication.
- **Wide span:** flat structure, fast communication, more delegation, less control.
- Typical: 5–6 at top, 15–20 at shop floor.
- In a **line organization** span decides the no. of levels.



Narrow span (2 to 3 each): tall, 3 levels, close supervision.

Factors deciding the span:

- Ability of the manager and of subordinates.
- Nature of work: routine → wide; complex → narrow.
- Degree of delegation.
- Communication tools.
- Geographic spread of subordinates.
- **Graicunas:** with n subordinates the relationships to manage are $n(2^{n-1} + n - 1)$: 6 people → 222. Why span must stay limited.
- For $n = 1$ to 8: 1, 6, 18, 44, 100, 222, 490, 1080. Adding the 7th subordinate more than doubles the load.



Wide span (8): flat, 2 levels, more delegation.

1.6 Purchasing and Marketing Management

Purchasing: meaning, objectives (5 R's), principles, functions of the purchasing department 3 4 5 8

TOP 10/29 82 Bh, 81 Bh, 78 Bh, 76 Ch, 72 Ka, 71 Ch, 70 Ch, 70 Asa, 67 Asa, 65 Shr · short note "role of purchasing and marketing" in 82 Bh

- **Purchasing:** acquiring materials, supplies, machines, tools and services by payment, for the firm's operation. First phase of materials management.
- **Alford and Beatty:** "procuring of materials, supplies, machines, tools and services required for equipment, maintenance and operation of a manufacturing plant."
- **Objectives: the 5 R's:** right **quality**, right **quantity**, right **price**, right **time**, right **source** (+ right place).
- Continuous supply, minimum inventory, good vendor relations, low total cost.

Principles of purchasing

- Buy to specification and standards.
- Competitive quotations; buy on total value, not lowest price.
- Centralized purchasing for economy.
- Ethical dealing, no favouritism; prompt payment.
- Develop reliable, alternative sources.
- Keep records; review make-or-buy.

Purchasing vs procurement (71 Ch)

- **Purchasing:** the buying transaction (order, pay).
- **Procurement:** wider: need identification, sourcing, contracts, buying, logistics, receiving, vendor management; includes lease, hire, transfer.

Functions of purchasing department

1. Receive and check purchase requisitions.
2. Review specifications; standardize, simplify.
3. Keep records of materials, substitutes, suppliers, prices.
4. Call and analyse quotations; select vendor.
5. Negotiate; place orders; follow up.
6. Inspect: right quality and quantity.
7. Prepare purchase budget.
8. Buy at right time and economical rate.
9. Keep uninterrupted supply to production.
10. Liaison between vendors and production, QC, finance, stores.
11. Ensure prompt payment → good vendor relations.
12. Develop new sources; handle sub-contracts.
13. Keep inventory optimum; dispose of scrap and salvage.

Role of purchasing and marketing departments

- **Purchasing:** input side, supplies right materials at lowest total cost → cost and quality of product.
- **Marketing:** output side, finds needs, sells product, brings revenue.
- Together they link the firm to suppliers and customers.

Methods of purchasing; purchasing procedure (process outline) 2 4 5

HOT 6/29 81 Bh, 79 Ba, 76 Ash, 74 Ash, 72 Ch, 71 Shr

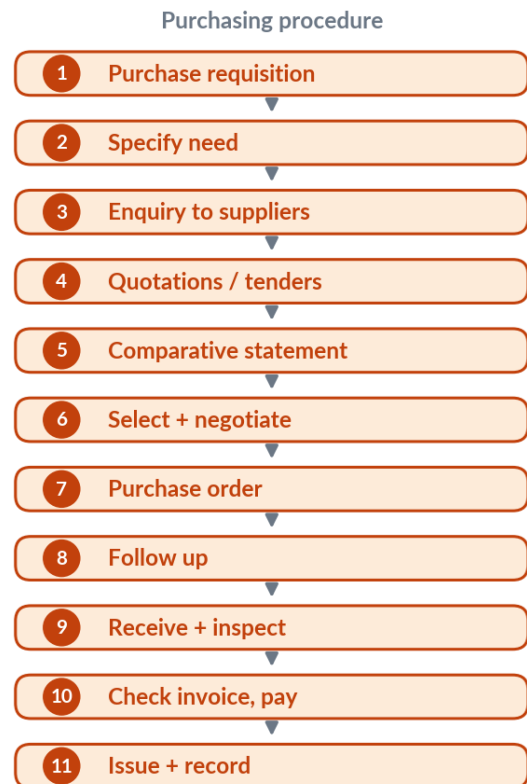
Methods (choice depends on firm size, nature and quantity of material)

1. **By requirement** (hand-to-mouth): buy whenever the need arises.
2. **For a specified period**: regular standard items (oils, coolants, stationery) bought for a future period set by production schedules.
3. **Market purchasing**: requirements from production planning + market trends → buy ahead of need when prices are low and likely to rise.
4. **Speculative**: buy excess at a price low for profit; price trend, not need, decides (unlike market purchasing).
5. **Rate contract**: suppliers supply at agreed prices for 2–3 years, reviewable on notice; basic materials (pig iron, steel, coal, coke).
6. **Group purchasing**: items bought in groups or lots in one order, not one order per item.
7. **Through a government supplies agency** (India's DGS&D): agency fixes annual rate contracts; departments order directly at those rates; firms certify they sell no cheaper elsewhere.

By procedure (AJ Sir)

- **Direct**: urgent, from a known supplier.
- **Quotation**: at least 3 quotations, pick best.
- **Tender**: public notice in newspapers; sealed bids; evaluate (large or public purchases; Public Procurement Act 2063 in Nepal).

Purchasing procedure (draw this for “outline of the purchasing process”, 76 Ash)



- **Purchase order** copies go to stores, accounts and the requesting department.
- **Inspection** ends in a goods-received report; only then is the invoice passed for payment.

Marketing: definition, concept, and its importance (in modern business and the digital era) 3 4 5 8

TOP 14/29 82 Bh, 82 Ba, 81 Bh, 81 Ba, 79 Bh, 79 Ba, 78 Bh, 76 Ch, 75 Ch, 74 Ash, 73 Ch, 73 Shr, 70 Ch, 68 Ba · asked almost every year

- **Stanton:** “a total system of interacting business activities designed to plan, price, promote and distribute want-satisfying products and services to present and potential customers.”
- **Kotler:** satisfying needs and wants through exchange processes.
- **Flow view:** business activities that direct the flow of goods and services from producer to consumer: the right product, right place, right quantity, right price, right time.
- Starts with identifying a consumer need, ends with satisfying it; the modern concept stresses consumer needs and freedom to choose.
- **4 P's (marketing mix):** product, price, place, promotion.

Marketing concept

- Evolution: production → product → selling → **marketing** → societal concept.
- Three pillars: **customer orientation**, **profit through satisfaction** (not volume), **integrated effort** of all departments.
- Selling concept: “sell what we make”. Marketing concept: “make what customers want”.

Marketing research (4 elements)

- **Current sales analysis:** what sells, where, to whom.
- **Market intelligence:** competitors' moves.
- **Market analysis:** reaction to price changes and new products.
- **Product evaluation:** how the product performs for users.

Functions of marketing 4 5

PIN 3/29 81 Ba, 71 Shr, 71 Ch

Classical grouping (Adhikari): marketing adds value through its functions

- 1. Exchange** (ownership transfer):
 - **Buying:** assembling goods for resale or use.
 - **Selling:** finding buyers, persuading, transferring ownership.
- 2. Physical distribution:**
 - **Transport:** moves goods to where they are needed (place utility).
 - **Storage / warehousing:** holds goods until needed (time utility).
- 3. Facilitating:**
 - **Standardization and grading:** uniform quality, easy buying.
 - **Financing:** credit to dealers and customers.
 - **Risk bearing:** price fall, spoilage, theft; insurance.
 - **Market information:** demand, prices, competitors.

Importance

- **To the firm:**
 - Brings revenue, the only source of profit.
 - Tells production what to make.
 - Builds brand and loyalty; survives competition.
- **To consumers:**
 - Right product, right place, right price.
 - Wider choice and product information.
- **To society / economy:**
 - Creates demand → production → jobs.
 - Raises living standard.
 - Distribution to remote areas; exports earn foreign exchange.
- Fundamental: without a customer nothing else in the firm is needed (Drucker: “marketing and innovation are the two basic functions”).

Importance in the digital era (81 Ba)

- Social media, search, e-commerce: reach millions at low cost.
- Data analytics: precise targeting, personalization.
- Measurable ROI; real-time campaigns.
- Two-way engagement, reviews, influencers build trust.
- Small firms and startups compete globally.
- Digital payments (eSewa, Khalti) enable online selling in Nepal.

Modern list (AJ Sir)

- **Product / service management:** what to make, features, quality.
- **Pricing:** cost, competition, demand.
- **Promotion:** advertising, sales promotion, publicity.
- **Distribution (channel):** wholesalers, retailers, online.
- **Selling and financing.**
- **Market information management,** buying.
- **Packaging and branding:** protects and identifies the product.
- **After-sales service:** installation, repair, warranty.

Advertising: definition, and why it is one of the best forms of marketing 1 3

PIN 3/29 78 Bh, 72 Ch, 70 Ch

- **Definition:** paid, non-personal communication of information about a product, service or idea through mass media by an identified sponsor, to create or maintain demand.
 - Media: newspaper, TV, radio, online, billboards.
 - From Latin *ad verito*, “to turn towards”: turning attention toward the product.
- Integral part of marketing; the **pivot** of modern trade: informs customers what and where, creates demand.
- Real goal: effective communication that **modifies attitudes or behaviour** of the receiver.
- **Why the best form:**
 - Mass reach at low cost per contact; repetition builds memory.
 - Creates awareness and new demand; launches new products.
 - Builds brand image and goodwill; supports salesmen and dealers.
 - Informs and educates consumers; counters competition.
 - Digital ads are targeted and measurable.

Salesmanship; marketing vs purchasing; marketing software products in Nepal 4

PIN 3/29 76 Ash, 73 Shr, 68 Ch

Salesmanship is an important ingredient of marketing? Agree (73 Shr)

- Personal selling persuades face to face, explains technical products, handles objections, **closes the sale**.
- Brings customer feedback to the firm; builds relationships.
- But it is **one ingredient**: product, price, place and research come first.

Marketing vs purchasing (68 Ch)

- Purchasing: **inputs**, firm is buyer, aims at low cost, deals with suppliers.
- Marketing: **outputs**, firm is seller, aims at revenue and satisfaction, deals with customers.

Challenges for marketing software in Nepal (76 Ash)

- Small domestic market; low willingness to pay for software.
- Piracy; free alternatives.
- Preference for foreign brands; trust deficit in local products.
- Low digital literacy and internet access outside cities.
- Limits on international payments; weak IP law enforcement.
- Skilled marketers scarce; brain drain.
- Response: freemium, local language, SaaS subscriptions, export markets, partnerships with banks and telecoms.

1.7 Topics Set Only by the 2065–2068 Papers

These came from the older ME708 syllabus and have not been set since 2068. Know one paragraph each. verified/added

Activities of production development 5

PIN 2/29 66 Bh, 65 Shr

- Idea generation (market, R&D, competitors).
- Screening; feasibility (technical, economic).
- Product design; prototype; testing.
- Pilot production; test marketing.
- Commercial launch; review.
- Supporting: standardization, simplification, specialization, value analysis.

Manufacturing methods; CIM 4 6

PIN 1/29 67 Asa · CIM short note

- **Job / project:** one-off to order (ships, dams).
- **Batch:** lots of similar items (medicine, garments).
- **Mass / flow:** standard products on lines (cars).
- **Continuous process:** round-the-clock flow (cement, refinery).
- **CIM** (computer-integrated manufacturing): the whole plant computer-controlled.
 - Parts: CAD, CAPP, CAM, CNC machines, robots, FMS, AGVs, one shared database.
 - Benefits: flexibility, consistent quality, short lead time, less inventory.
 - Limits: high cost, skilled staff needed, vulnerable to system failure.

1.8 PYQ Mapping

Theory questions, covered in this document

- Define organization; need, importance, role in society; can we sustain without it → §1.1. 16 papers; the rider on most 8-mark questions.
- Organization as open / social system; system approach → §1.1. 81 Bh 2, 79 Ba 2+2, 70 Ch 4
- Role in employee's professional growth → §1.1. 81 Bh 4, 75 Ch 4
- Principles of organization → §1.1. 9 papers, 4 to 8.
- Formal vs informal; need for informal; informal organization in a family → §1.1. 11 papers.
- Historical development 79 Bh 3; characteristics / elements 73 Ch 4, 68 Ch 4; organizational behaviour 68 Ch 4, 68 Ba 4, 67 Asa 4, 66 Bh 4 → §1.1.
- Functions of management (incl. "two principal functions", 79 Ba) → §1.2. 15 papers: the most asked list in the subject.
- Levels, need for levels, functions at each level → §1.2. 9 papers.
- Managerial skills, degree per level, qualities of a manager → §1.2. 10 papers.
- Models of management → §1.2. 81 Bh 4, 80 Bh 4, 76 Ash 4, 70 Ch 3+5
- Roles of a manager 76 Ash 4; importance of management 71 Shr 4, 70 Asa 3; science or art 78 Bh 3; organization vs management 79 Ba 2, 76 Ash 2, 71 Ch 2; best theory today / for Nepal 80 Ba 4, 74 Ash 8 → §1.2.
- Scientific management (principles, experiments, still applicable; vs management science; HR manager implementing it) → §1.3. 11 papers.
- Fayol's administrative theory, 14 principles, applicability → §1.3. 9 papers.
- Behavioural approach; its advantages over scientific / modern; comparisons of the three approaches → §1.3. 11 papers.
- System and contingency approach 82 Ba 8, 68 Ch 5, 67 Asa 7, 66 Bh 4; MBO 68 Ch 4 → §1.3.
- Forms of ownership; changing form; CEO of a software company → §1.4. 82 Ba 4, 74 Ch 4+4, 73 Shr 4+4, 71 Shr 4
- Single ownership 80 Bh 8, 71 Shr 4, 65 Shr 4; partnership 81 Bh 3, 74 Ash 3+5, 67 Asa 4 → §1.4.
- Joint stock company (features, formation in Nepal, merits, vs partnership, vs public corporation, private vs public) → §1.4. 10 papers.
- Cooperatives 82 Bh 5, 81 Bh 5, 79 Bh 8, 72 Ka 8; public corporation 81 Ba 4 → §1.4.
- Organization structure → §1.5, including short notes in 79 Bh, 75 Ch, 71 Shr.
- Line 76 Ch 4, 76 Ash 4, 71 Ch 4, 70 Asa 4; functional 82 Bh 3, 68 Ch 3, 68 Ba 4, 67 Asa 4; line & staff 82 Ba 4, 79 Bh 4, 74 Ch 4, 66 Bh 5; committee 79 Bh 4, 75 Ash 4, 74 Ch 4 → §1.5.
- Matrix; structure for an engineering project 73 Ch 8, 72 Ch 8, 68 Ba 5; authority, responsibility, span, policy vs executive group 68 Ch, 68 Ba, 67 Asa, 65 Shr → §1.5.
- Purchasing, 5 R's, principles, functions → §1.6. 10 papers.
- Methods of purchasing; purchasing procedure → §1.6. 6 papers.
- Marketing: definition, concept, importance → §1.6. 14 papers.
- Functions of marketing; advertising; salesmanship; marketing vs purchasing; software marketing in Nepal → §1.6.
- Production development, manufacturing methods, CIM → §1.7. 67 Asa, 66 Bh, 65 Shr

Short-note prompts placed here (2065–2068 ones from the Detailed PYQ's "Extra" section)

- Scientific management, span of control, MBO 68 Ch; contingency management theory 66 Bh; marketing concept 68 Ba; partnership, functional organization, CIM 67 Asa; single ownership, informal organization 65 Shr; team effort model, managerial skills 81 Bh; organizational structure 79 Bh, 75 Ch, 71 Shr.
- Industrial relations, collective bargaining, job design, incentive programs and manpower planning go to Ch. 2; leadership style, contingency approach to leadership, satisfaction-progression vs frustration-regression and entrepreneurial characteristics go to Ch. 3.

Numerical content

- This chapter has no numerical component: no paper in 29 sets a calculation. The only formula here (Graicunas' span relationships) is there to strengthen a theory answer.

Corrections to the source notes

- Adhikari p17: WTO was founded in 1995 (not 1955); GATT was signed in 1947.
- Adhikari p105: a private company in Nepal may have up to 101 shareholders (Companies Act 2063), not 50; p106 says "maximum number of persons to form a company is seven": seven is the **minimum** for a public company.
- Adhikari pp111–114: "public corporation" there is the US business corporation; the syllabus means a government statutory corporation.

- AJ Sir slide 50: cooperatives “at least 10 and no more than 100 members” is not Nepal law; the Cooperative Act 2074 sets a minimum of 30 for a primary cooperative.
- AJ Sir slide 31: Fayol was born in 1841.
- Syllabus file prints course code CT 710; every paper prints ME708.

Figures

- Only diagrams an answer would draw are kept: org charts, process cycles, step flows, pyramids, matrices. No photos, portraits or decorative charts.
- Cropped, never redrawn: Adhikari's notes (pp36, 37, 53, 57, 59, 62, 65, 66, 120, 124, 128) and the ioenotes Ch.1 PDF (pp1, 20).
- Generated (`ch1fig.py`) only where no source has a picture.

2. Personnel Management

8 hrs · asked in **all 29 papers** · 7 syllabus topics · **404 marks, 17.1% of the archive** → about 9 of every 80 marks, nearly always one full 8-mark question.

READ THIS FIRST

Chapter 2 follows the life of an employee: **plan** how many people are needed → **analyse** the job → **recruit and select** → **train** → **appraise** → **pay**. Every question picks two stops on that line.

1. **Four lists carry the chapter:** definition + functions of personnel management (17 papers), recruitment and selection steps (13), factors affecting wage and salary structure (12), job analysis with job description and specification (10).
2. **Personnel policy** (7 papers) is always asked from the new employee's side: "why discuss the handbook at hiring?"
3. **No numerical has ever been set**, but "how are wages calculated?" (75 Ch) is answered by the incentive-plan formulas in §2.7, each with one worked example.

Tier chips count **papers out of 29**: **TOP n/29** 8+ **HOT n/29** 4-7 **PIN n/29** 1-3 inside an 8-mark question **LOW n/29** 1-3, short note only. Bold year = Regular exam.

2.1 Personnel Management: Meaning, Functions, Role

Define personnel management. Explain its functions and its role in the organization **1 2 3 4 5 8**

TOP 17/29 82 Bh, 81 Ba, 80 Bh, 80 Ba, **76 Ch**, 76 Ash, **75 Ch**, 75 Ash, 74 Ash, 72 Ka, **71 Ch**, 71 Shr, 70 Ch, 70 Asa, 68 Ch, 68 Ba, 65 Shr · **the most asked question of the chapter**

- **Flippo**: "the planning, organizing, directing and controlling of the **procurement, development, compensation, integration and maintenance** of personnel, to accomplish organizational, individual and social goals."
- Simple: the part of management that **obtains, develops, uses and keeps a satisfied workforce**. Also called human resource management.
- **Three-part view**: the part of management dealing with human resources:
 - their **procurement**;
 - their **development** in skills, knowledge, attitude;
 - their **motivation** toward objectives, by keeping a climate that favours such development.

Aims

- Get the best out of people.
- Maximum individual development.
- Better service to society.
- Good labour relations.

Three aspects (AJ Sir)

- **Welfare**: working conditions, canteen, housing, health, safety.
- **Labour / personnel**: recruitment, pay, promotion, training.
- **Industrial relations**: unions, disputes, collective bargaining.

Characteristics

1. Part of general management.
2. Has its own principles and policies.
3. Humanistic: deals with people, not things.
4. A philosophy of management: how the firm treats its people.
5. Carries **advisory, executive and administrative** responsibilities.

Managerial functions

- **Planning:** manpower needs, policies, programmes.
- **Organizing:** structure of the HR department, tasks.
- **Directing:** motivating, guiding employees.
- **Coordinating and controlling:** audit of HR results against plan.

Role in the organization

- Supplies the **right people** at the right time.
- Builds **skills** (training) → productivity.
- **Fair pay**, incentives → motivation, low turnover.
- **Industrial peace:** handles grievances, unions.
- **Welfare and safety** → morale, fewer accidents.
- Advises line managers; keeps records; ensures labour-law compliance (Labour Act 2074 in Nepal).
- Aligns people with strategy → growth and development of the firm.

Adhikari's list of functions (write it for “functions” at 5 marks)

1. Develop personnel policy.
2. Manpower planning.
3. Recruitment and selection.
4. Analysis, description and valuation of work.
5. Compensation and appraisal schemes.
6. Personnel records.
7. Welfare and safety programmes.
8. Training and development.
9. Wage and salary administration.
10. Collective bargaining with unions.
11. Promotion, transfer, retirement.

Operative functions

1. **Procurement:** manpower planning, job analysis, recruitment, selection, placement, induction.
2. **Development:** training, executive development, career planning, promotion.
3. **Compensation:** job evaluation, wage and salary administration, incentives.
4. **Integration:** reconcile employee and organizational interests: communication, grievance handling, collective bargaining.
5. **Maintenance:** health, safety, welfare, social security, ergonomics.
6. **Motivation:** monetary and non-monetary incentives.
7. **Records and separation:** personnel records, transfers, retirement, exit.

Principles of personnel management

1. **Scientific selection:** right person for the right job.
2. **Fair reward:** pay matched to work and market.
3. **High morale:** satisfied, willing workers.
4. **Dignity of labour:** respect every job.
5. **Team spirit** among employees.
6. **Effective communication** both ways.
7. **Joint management:** workers consulted on decisions affecting them.
8. **Effective utilization** of each person's ability.
9. **Maximum individual development.**
10. **National prosperity** through productive industry.

How does human resource management differ from personnel management? 2

PIN 1/29 71 Ch verified/added

Basis	Personnel management	Human resource management
View of people	Cost, a tool	Asset, a resource to develop
Nature	Routine, administrative, reactive	Strategic, proactive
Focus	Rules, records, compliance, pay	Development, motivation, culture, performance
Job design	Division of labour	Teamwork, empowerment
Responsibility	Personnel department only	All line managers share it

2.2 Personnel Policy

Personnel policy: what it must include, its importance, and why it is discussed with new employees

3 4 5 8

HOT 7/29 82 Bh, 80 Ba, 79 Bh, 75 Ash, 74 Ch, 74 Ash, 73 Shr · 79 Bh asks it twice (policies; discussing at hiring)

- **Personnel policy (employee handbook):** written statement of what the employer expects from employees and what they may expect in return. A **guide for quick, consistent decisions**; a means to an end, not an objective.
- Eg a “promotion from within” policy serves the objective of employee development.
- Objectives: maximum development and use of workers' abilities; good labour relations; protect interests of labour and customers.

Terms a good personnel policy includes

1. Welcome message; company name, history, values, objectives, structure.
2. **Recruitment and selection:** sources, procedure, probation.
3. **Working hours,** attendance, leave (sick, annual, holidays).
4. **Pay:** salary structure, allowances, overtime, incentives, bonus.
5. **Benefits:** insurance, provident fund, gratuity, pension, education aid.
6. **Training** and development; **promotion and transfer** policy.
7. **Code of conduct:** discipline, dress, harassment, conflict of interest.
8. **Health and safety;** accident procedure.
9. **Grievance procedure;** union and collective bargaining.
10. **Separation:** resignation notice, retirement, dismissal, exit process.

Importance

- **Consistency and fairness:** same rule for everyone → trust.
- Quick decisions by managers without referring upward.
- Reduces conflict and grievances; legal protection for both sides.
- Attracts and retains good staff; builds culture.

Why discuss it at hiring / with new employees

- Employee knows **rights and duties** from day one: pay, leave, benefits, discipline.
- **No surprises** later → fewer disputes, lower early turnover.
- Part of **induction:** faster adjustment, sense of belonging.
- Written acknowledgement protects the employer legally.
- Clarifies expectations on conduct, safety, harassment.

2.3 Manpower Planning

Manpower (human resource) planning: meaning, steps, and why it is needed 3 4 8

TOP 8/29 81 Bh, 81 Ba, 80 Ba, 79 Bh, 76 Ch, 76 Ash, 75 Ch, 73 Shr · short note in 75 Ch

- **Definition:** the process of having the **right number** of the **right kind** of people at the **right place** and **right time**, doing the jobs they suit. Formally: “the scientific process of allocating the right quantity of right men required in future at right time on the right job”.
- **Manpower** = total skills, knowledge, talent, creativity and values of the workforce.
- Two objectives: **use present employees fully; fill future needs**.
- Types: macro (national) vs micro (firm); short, medium, long term.
- **Factors** affecting the estimate:
 - Working hours and shifts.
 - Nature of production and product mix.
 - Performance rate of workers.
 - Hours lost to absenteeism; labour turnover.

Steps of the HR planning process

Manpower planning process



Importance / need

- **No shortage, no surplus:** avoids idle labour and overload.
- **Cuts labour cost;** right people raise productivity.
- Anticipates **retirement, turnover, expansion:** vacancies filled in time.
- Basis for **recruitment, training, career planning**.
- Copes with **technological change:** new skills planned ahead.
- Attracts talent; supports succession planning.
- Nepal: high outmigration of skilled youth → planning replacements and retention is essential.

Notes on the steps (left)

- **Demand** depends on workload, expansion, technology, budget.
- **Supply:** internal (promotion, transfer, retirement, turnover) and external (labour market).
- **Net requirement** = demand – supply: shortage → recruit or train; surplus → redeploy or downsize.

Adhikari's version of the steps

1. Understand the business activities.
2. Identify the tasks.
3. Forecast manpower.
4. Job analysis.
5. Review existing manpower.
6. Anticipate supply from outside.
7. Manpower management (act on the gap).

2.4 Recruitment and Selection

Explain the process of recruitment and selection of manpower (steps of hiring); scientific selection 3

5 6 8

TOP 13/29 82 Ba, 81 Ba, 80 Bh, 79 Bh, 78 Bh, 72 Ch, 71 Ch, 71 Shr, 69 Ch, 68 Ch, 67 Asa, 66 Bh, 65 Shr

- **Recruitment:** searching for prospective employees and **stimulating them to apply**; creates a pool (positive process).
Dale S. Beach: “the development and maintenance of adequate manpower resources. It involves the creation of a pool of available labor upon whom the organization can depend when it needs additional employees.”
- **Selection:** **choosing** the most suitable from the pool by eliminating the rest (negative process).
- **Hiring:** the final act of appointing, with offer, contract, placement.

Steps of recruitment and selection

Selection procedure



Notes on the steps

- **Before step 1:** job analysis, job description and a staff requisition fix what is needed; the vacancy is **advertised** (title, duties, qualification, deadline).
- **Screening and preliminary interview** drop the plainly unfit early and cheaply.
- **Employment tests:** intelligence, trade, psychological, aptitude, interest, personality.
- **Employment interview:** HR first, then the department head.
- **Reference check:** former employers, teachers; background verification.
- **Final selection:** offer letter, appointment, contract.
- **Placement and induction:** orientation, handbook (add these when the question says “steps of hiring”).
- Each step **eliminates**: selection is a series of hurdles, which is why it is called a negative process.

Sources of recruitment

- **Internal:** promotion, transfer, internal posting, former employees, employee referrals.
- **External:** advertisement (newspaper, websites like merojob, LinkedIn), campus recruitment, employment agencies, walk-ins, unions, consultants.

Scientific selection

- Taylor’s principle: **right person for the right job**, one who maintains the output with least effort.
- Selection by **objective tests and standards**, not favour or guess.
- **Methods:** psychological and aptitude tests, trade tests, structured interviews, work samples, assessment centres, medical tests.

Fitness factors checked

- **Physical:** sound body, limbs, height, weight, eyesight.
- **Personal:** age, sex, marital status, experience, family.
- **Proficiency:** skill and ability now: the basic fit.
- **Competency:** potential to learn and become proficient.
- **Temperament, character; vocational interest.**

Outsourcing 2 3

PIN 2/29 80 Bh, 69 Ch verified/added

- Contracting a function to an **outside firm** instead of hiring staff for it.
- Eg security, cleaning, payroll, IT support, call centres; Nepal’s IT firms do outsourced work for foreign clients.
- In recruitment: an agency does the hiring, or staff are supplied by a manpower company.

Recruitment vs selection vs hiring

Basis	Recruitment	Selection / hiring
Aim	Attract many applicants	Pick the best, appoint
Nature	Positive	Negative (eliminating)
Order	First	After recruitment
Output	Pool of candidates	Employee with contract

Advantages

- Lower cost: no salaries, benefits, training for that function.
- Specialist skill and technology of the vendor.
- Management focuses on core work.
- Flexible headcount: scale up or down by contract.

Referral recruitment: pros and cons 4

PIN 1/29 74 Ch verified/added

- Existing employees recommend candidates, often for a referral bonus.

Pros

- Fast and cheap: no advertising cost.
- Pre-screened, trusted candidates.
- Better cultural fit; referred staff stay longer.
- Referral bonus motivates existing staff.
- Use it with the same tests and interviews as other candidates.

Disadvantages

- Less control over quality and timing.
- Confidentiality and data risk.
- Dependence on one vendor.
- Staff morale falls if jobs are cut.

Cons

- Nepotism and favouritism.
- Less diversity: people refer people like themselves.
- Conflict if the referral fails or is rejected.
- Closes the door to better outside talent.

Role of interview in hiring; the interviewing process 3

PIN 2/29 75 Ch, 73 Ch

- **Role:** get the maximum information about the candidate's suitability that tests cannot show: communication, attitude, confidence, motivation, fit with team.
- Two-way: candidate learns about the job and firm.
- Types: preliminary, structured, unstructured, panel, stress, technical, online.

Interviewing process

1. **Preparation:** study job description and application; set questions.
2. **Reception:** put candidate at ease.
3. **Information exchange:** questions on knowledge, experience, attitude.
4. **Close:** candidate's questions, next steps.
5. **Evaluation:** rate on a standard form.

2.5 Training and Development

Training and development: meaning, need, methods, and how training needs are identified 2 3 4 5

HOT 5/29 82 Ba, 81 Bh, 79 Bh, 72 Ch, 71 Shr

- **Training:** planned learning of **specific knowledge and skills** to do the **present job** better; short term; mainly for workers.
- **Development:** long-term education for **future roles**, conceptual and managerial growth; mainly for executives.
- **Training, formally:** "a well thought and well planned process of conscious learning of new knowledge and skills for improving the learner's ability to perform certain tasks more efficiently and effectively, in the immediate or near future, with active support of some other individuals."
- **Development, formally:** long-term, systematic process by which managers gain conceptual and theoretical knowledge; an executive development programme raises their capability to achieve objectives.
- Objectives: disseminate knowledge, develop skills, change attitudes.

Need / role (esp. for new employees)

- New hires become productive in the **minimum time**; no learning by mistakes.
- **Higher productivity**, quality; less wastage and spoilage.
- **Fewer accidents**, less machine breakdown.
- **Less supervision** needed.
- **Morale, job satisfaction**; lower turnover and absenteeism.
- Keeps pace with **new technology**; flexibility.
- Career growth, higher earnings; builds future managers.

Identifying training needs

- **Organization analysis**: goals, problems (low output, high scrap, accidents).
- **Task analysis**: job description vs skills required.
- **Person analysis**: performance appraisal gaps, supervisor reports, self-assessment.
- Other signals: complaints, new machines, promotions, surveys, exit interviews.

Methods: workers

- **Demonstration**: instructor actually shows how to do the task; small shops, auto repair workshops (Eg how to do a job on a centre lathe).
- **On-the-job**: supervisor instructs a new worker at the machine: nature of work, tools, safety; shows how an experienced worker does it. Most realistic.
- **Vestibule**: a separate training school duplicating, as nearly as possible, the real equipment, materials and conditions; many trainees, no production pressure; costly.
- **Apprenticeship**: long, for all-round craftsmen (machinists, tool makers, millwrights, fitters, welders); best for large firms with steady demand for skill.
- **Special courses**: knowledge-based, in-house or outside institutes.

Methods: supervisors

- Induction and orientation.
- Lecture (classroom); conference.
- Written instruction.
- Training within industry (TWI).

Methods: executives

- **On the job**:
 - Understudy (appointed assistant to a manager).
 - Committee membership; job rotation.
 - Job enlargement / enrichment; MBO.
- **Off the job**: lectures, case study, business games, role playing, seminars, e-learning.

HR problems: talent poaching; short-term employee development programme; appropriate vs inappropriate human resources 4 8

PIN 2/29 79 Ba, 73 Shr verified/added

Tackling talent poaching (79 Ba)

- Competitive pay, reviewed against the market.
- Retention bonus, ESOP, bond or notice clauses.
- Clear career path, promotion from within.
- Training, recognition, good culture, flexible work.
- Stay interviews; fix reasons found in exit interviews.
- Succession planning so no one person is critical.

Short-term development programme

1. Identify skill gaps (appraisal).
2. Set 3–6 month objectives per employee.
3. Workshops, mentoring, job rotation, online courses.
4. Review, reward, plan the next cycle.

Appropriate vs inappropriate human resources (73 Shr)

- **Appropriate**: people whose skills, qualification, attitude and number **match** job requirements → productive, satisfied.
- **Inappropriate**: a mismatch.
 - Over-qualified: bored, leave soon.
 - Under-qualified: errors, need close supervision.
 - Surplus: idle cost.
 - Wrong attitude: conflict, low morale.
- **Ideas to get appropriate HR**:
 - Manpower planning and job analysis.
 - Scientific selection.
 - Training and appraisal.
 - Redeployment of surplus staff.
 - Fair pay to keep the right people.

2.6 Job Analysis, Job Evaluation and Merit Rating

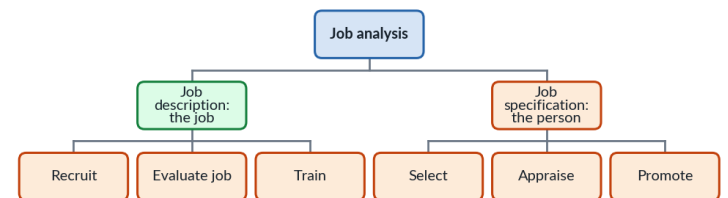
Job analysis: meaning, methods, job description and job specification 2 3 4 5 6

TOP 10/29 82 Bh, 81 Bh, 81 Ba, 79 Bh, 78 Bh, 71 Ch, 68 Ch, 67 Asa, 66 Bh, 65 Shr · job design as a short note in 67 Asa

- **Job:** group of tasks, duties and responsibilities done by one person for pay.
- **Job analysis:** systematic study of a job to find its **tasks** and the **skills, knowledge, abilities and responsibilities** needed to do it well. Answers what, how, why, with what tools, under what conditions.
- Outputs: **job description** (about the job) and **job specification** (about the person). Used for recruitment, training, job evaluation, pay, appraisal.

Data collected in a job analysis

1. Job identification.
 2. Nature of the job: major and other duties.
 3. Operations performed.
 4. Materials and equipment used.
 5. Skill: education, training, responsibility, job knowledge, mental capability, accuracy → degree of difficulty.
 6. Physical demands: activities, working conditions, hazards → effort.
 7. Relation with other jobs.
- **Job:** a regular assignment to one employee, with duties, responsibilities and conditions different from other assignments.
 - The analyst (or a trained supervisor) collects information, prepares the description, then works up the specification.



Job analysis yields the job description and job specification, and each feeds HR decisions.

Steps

1. Collect background info (charts, old descriptions).
2. Select jobs to analyse.
3. **Collect job data** (methods below).
4. Process and verify with job holder and supervisor.
5. Prepare job description.
6. Prepare job specification.

Methods of collecting data

- **Questionnaire:** holders fill structured forms; cheap for many jobs; misunderstanding possible.
- **Interview:** holder describes duties, even unobservable ones; may exaggerate.
- **Observation:** analyst watches the work; firsthand; best for manual jobs; workers may change behaviour.
- **Diary / log, critical incident, technical conference** with experts.

Most suitable in Nepal? (82 Bh)

- **Interview + observation:**
 - Many small firms have no written records.
 - Jobs are manual or mixed, easy to watch.
 - Literacy varies → questionnaires return poor data.
- Large firms and banks can add questionnaires.

Job description (what the job is): identification, summary, work performed

- Job title, code, department, location.
- Job summary (purpose).
- Duties and responsibilities: what, how, why.
- Reporting relationships.
- Machines, tools, materials.
- Working conditions: shifts, hazards.
- Pay grade, benefits.

Job specification (who can do it)

- Education, training, experience.
- Skills, knowledge, abilities.
- Physical and mental requirements.
- Personal qualities (communication, attitude).

Example: Lecturer, engineering college (81 Ba)

- **JD:** title Lecturer, Dept of Electronics; reports to HoD.
 - Teach 16–18 periods/week; prepare lessons.
 - Supervise labs; set and evaluate exams.
 - Guide projects; research and publish; departmental duties.
- **JS:**
 - Master's in relevant engineering, first division; NEC registration.
 - Teaching or industry experience preferred.
 - Communication and presentation skill, lab skills, ethics, research aptitude.

Job design (short note)

- Structuring tasks of a job for productivity and satisfaction:
 - **Simplification:** narrow, easy tasks.
 - **Rotation:** move between jobs.
 - **Enlargement:** more tasks at the same level.
 - **Enrichment:** more responsibility and control.
 - **Ergonomics:** fit the job to the body.

Job evaluation: meaning, methods, and comparison with merit rating 2 3

PIN 2/29 81 Ba, 79 Bh

- Systematic determination of the **relative worth of a job** compared with others → fair, uniform **wage structure**.
- Principles:
 - **Rate the job, not the man.**
 - Uniformity of understanding of the factors.
 - Co-operation from employees.
- **War Manpower Commission (USA):** an accurate job study answers four questions: what does the worker do? how? why? what skill does it take?
- **Non-quantitative methods:**
 - **Ranking:** jobs ranked whole, most to least important; each head ranks his department's jobs, a central committee merges them into grades.
 - **Classification / grading:** jobs slotted into predefined grades.
- **Quantitative methods:**
 - **Factor comparison:** skill, mental effort, physical effort, responsibility, working conditions compared factor by factor with key jobs.
 - **Point rating:** points per factor degree, totals set the grade.

Basis	Job evaluation	Merit rating
Rates	The job	The person doing it
Purpose	Wage structure for the job	Reward, promotion over the base wage
Timing	Before hiring, on job change	Periodically, after hiring
Basis	Job analysis	Actual performance, traits
Output	Job grades	Ranking of employees

Uses of job evaluation

- Fair, consistent wage structure; removes pay anomalies.
- Fewer wage disputes and grievances.
- Basis for wage negotiation with unions.
- Helps place new jobs into existing grades.

Merit rating / performance appraisal: meaning, need, traditional and modern methods 2 4 5 6

HOT 7/29 82 Ba, 81 Bh, 79 Bh, 78 Bh, 70 Ch, 68 Ba, 66 Bh

- **Merit rating:** systematic, periodic assessment of an **employee's worth** in terms of job performance, integrity, leadership, behaviour, aptitude.
- Factors: quality and quantity of work, job knowledge, dependability, attitude, initiative, versatility, leadership.

Why necessary

- Basis for **promotion, transfer, pay rise, bonus**.
- Feedback to employee → improvement.
- Identifies **training needs**.
- Validates selection; objective, fair decisions; motivation.

Traditional methods (trait-based)

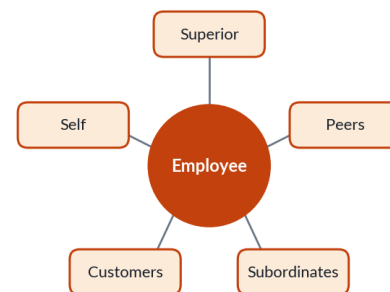
- **Ranking:** best to worst, overall.
- **Paired / person-to-person comparison:** each with every other.
- **Man-to-man comparison:** against key men on 5 traits (physical qualities, personal qualities, intelligence, general value to service, leadership), 5 degrees each.
- **Grading:** predefined grades (outstanding, satisfactory, unsatisfactory).
- **Graphic rating scale:** traits rated on a scale; most common.
- **Checklist:** yes/no statements.
- **Free essay:** supervisor's written description.
- **Critical incident:** record of good and bad events.

Modern methods (result-based)

- **MBO / appraisal by results:** manager and employee jointly set goals, appraise against them.
- **Assessment centre:** simulations, group exercises, in-basket tasks observed by trained assessors; for promotion.
- **360-degree feedback:** from superiors, peers, subordinates, customers, self. verified/added
- **BARS** (behaviourally anchored rating scale): scale points tied to real behaviours. verified/added
- **Human resource accounting:** employee's value in money. verified/added
- **KPI / balanced scorecard** dashboards. verified/added

Merits of modern methods

- Objective, measurable; less bias.
- Employee participates → commitment.
- Links appraisal to goals and development.



360-degree feedback: ratings from every side of the employee.

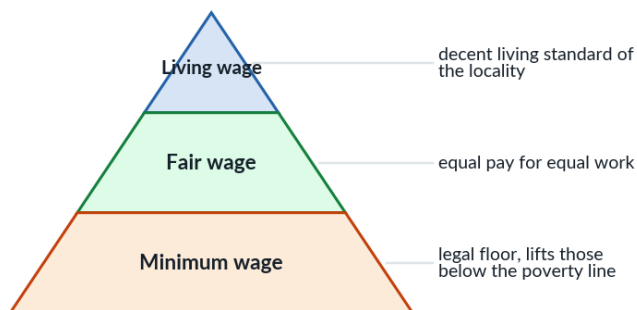
2.7 Wages and Incentives

Wages and salary: difference, factors affecting the wage/salary structure, how wages are calculated, salary and benefits you expect 2 3 4 5 8

TOP 12/29 82 Bh, 82 Ba, 80 Bh, 78 Bh, 76 Ch, 75 Ch, 74 Ash, 73 Ch, 72 Ch, 72 Ka, 70 Asa, 68 Ba

- **Wage:** remuneration for labour, paid per hour, day, week or per unit of output; production and manual workers.
- **Salary:** fixed monthly or annual pay; office, professional, managerial staff.
- **Nominal (money) wage:** cash only. **Real wage:** what the cash buys, plus benefits.
- A wage should meet ordinary needs, give some comforts, and keep the worker's standard of living.
- **Structure:**
 - Job classes / grades from job evaluation.
 - Flat rate or rate range per grade.
 - **Direct pay:** basic, overtime, bonus.
 - **Indirect pay:** insurance, provident fund, gratuity, allowances.

Three layers of wages



Minimum wage is the legal floor (Nepal revises it periodically).

Salary and benefits I would expect (82 Ba, 75 Ch)

- Basic salary at or above market rate for the post, with annual increment.
- Allowances: dearness, transport, communication, festival (Dashain) allowance.
- Provident fund, gratuity / SSF contribution; health and accident insurance.
- Paid leave; overtime; performance bonus.
- Training, career growth; flexible hours. Justify with qualification and market data.

Factors affecting wage / salary structure

1. **Ability of the firm to pay** (profit, size).
2. **Demand and supply** of labour.
3. **Prevailing market rate** for comparable jobs.
4. **Cost of living** (inflation index).
5. **Legal provisions:** minimum wage, Labour Act, SSF.
6. **Job requirements** from job evaluation: skill, effort, responsibility, hazards.
7. **Productivity** (output per man-hour).
8. **Bargaining power** of trade unions.
9. Employee's qualification, experience, seniority.
10. Regularity of employment, working hours, promotion prospects, training cost.

Examples (80 Bh)

- **Supply-demand, market rate:** an IT engineer in Kathmandu earns more than a same-degree engineer in a small town.
- **Legal:** garment workers' pay rises when the government revises the minimum wage.
- **Bargaining power:** a union-negotiated increase at NEA.
- **Cost of living:** dearness allowance raised after inflation.

Incentives: meaning, need, types 2 3 4

TOP 10/29 78 Bh, 76 Ch, 74 Ash, 73 Ch, 72 Ch, 72 Ka, 71 Shr, 70 Asa, 68 Ba, 66 Bh · “incentive programs” short note in 68 Ba, 66 Bh

- **Incentive:** payment or benefit **over and above** normal wage, linked to output or performance above a standard, to motivate higher productivity.
- **Incentive scheme:** a system of wage payment in which the earnings of an employee, a group, or all employees are directly related to output of acceptable quality above a standard, by a predetermined formula.

Why needed

- Raises productivity, lowers unit cost.
- Rewards efficient workers fairly.
- Motivation, morale, job satisfaction.
- Reduces supervision, absenteeism, turnover.
- Attracts and retains talent.

Types

- **Financial:** bonus, profit sharing, commission, piece-rate premium, ESOP.
- **Non-financial:** praise, recognition, promotion, job security, training, flexible hours, better conditions.
- **Semi-financial:** subsidized lunch, medical, recreation, pension.
- **Individual:** based on one person's output (Taylor, Halsey, Rowan).
- **Group:** based on team or plant output (Scanlon, Priestman, profit sharing): efficient workers help slow ones.

How wages are calculated: wage plans and incentive formulas 3

PIN 1/29 75 Ch · no numerical has been set in 29 papers; each formula shown once

Basic plans

- **Time rate:** $\text{wage} = T_a \times R$ (hours worked $\times$ hourly rate). Simple, secure; no push for output.
- **Piece rate:** $\text{wage} = \text{units} \times \text{rate per unit}$. Rewards output; no security.
- **Combined:** guaranteed time wage + piece earnings above it.

Taylor's differential piece rate

- By F.W. Taylor: rewards efficient workers, penalizes sluggish ones.
- Two rates: low (Eg 80%) below standard, high (Eg 120%) at or above.
- Example: standard 80 jobs/day (8 h), Rs 12/h $\rightarrow$ standard rate $12/10 = \text{Rs } 1.20$.
 - A makes 75: $75 \times 0.96 = \text{Rs } 72$.
 - B makes 85: $85 \times 1.44 = \text{Rs } 122.40$.

Halsey premium plan (1891)

- By Frederick A. Halsey, American engineer.
- Minimum wage guaranteed; standard time fixed per job.
- Time saved earns a fixed share (usually 50%) as bonus.
- $\text{Bonus} = \frac{1}{2}R(T_s - T_a)$; $\text{wage} = T_a R + \frac{1}{2}R(T_s - T_a)$.
- $T_s = 8$, $T_a = 6$, $R = \text{Rs } 10$: bonus Rs 10, wage **Rs 70**.

Rowan plan (1901)

- By James Rowan of Glasgow.
- Guaranteed minimum wage + bonus on part of the time saved.
- The share paid is the ratio time saved / standard time.
- $\text{Bonus} = \frac{T_s - T_a}{T_s} T_a R$. Peaks when $T_a = T_s/2$.

T_a	7	6	5	4	3	2	1
Rowan	8.75	15.00	18.75	20.00	18.75	15.00	8.75
Halsey	5.00	10.00	15.00	20.00	25.00	30.00	35.00

$T_s = 8$ h, $R = \text{Rs } 10$; bonus in Rs. Computed here (source table p184 is blank in the scan).

- Rowan pays more for small savings, Halsey more once over half the time is saved.
- Rowan protects against rate-cutting from careless standards.

Emerson efficiency plan

- By Harrington Emerson. Daily wage guaranteed; standard (100%) from past records.
- $\text{Efficiency} = \text{actual} / \text{standard output} \times 100$.
- Bonus scale:
 - Below 66.67%: time wage only.
 - 66.67–100%: bonus rising fast to 10% at 90% and 20% at 100%.
 - Above 100%: 20% + 1% per 1% extra.
- Example: Rs 40/day, standard 50 pins.
 - X 30 pins (60%): Rs 40. Y 50 (100%): Rs 48. Z 60 (120%): **Rs 56**.

Scanlon plan (group)

- By Joe Scanlon (USA). Guaranteed time wage.
- The standard labour-cost-to-sales ratio indexes total labour effectiveness; bonus = the % fall in that ratio.
- $\text{Bonus \%} = \frac{L_b/S_b - L_a/S_a}{L_b/S_b} \times 100$ (labour cost / sales).
- Example: base $1.35 / 12.40 = 10.887\%$; month $1.45 / 15.50 = 9.355\% \rightarrow$ bonus **14.07%**.
- With 2.15 lakh labour cost: $13.87\% >$ base, no bonus.

Priestman plan (group)

- $\text{Bonus \%} = \frac{P_a/W_a - P_b/W_b}{P_b/W_b} \times 100$ (output per worker).
- Example: standard 1200 cars / 500 workers = 2.4. Next month 1400 cars:
 - Same 500 workers $\rightarrow 2.8$, bonus **16.67%**.
 - 600 workers $\rightarrow 2.333$, -2.78% , no bonus.

Profit sharing

- Agreed % of annual profit shared with employees (Nepal's Labour Act and Bonus Act require a share of profit as bonus).

2.8 Industrial Relations and Collective Bargaining

Industrial relations; collective bargaining and its process 3 4 6

HOT 4/29 68 Ch, 67 Asa, 66 Bh, 65 Shr · 2065–2068 papers only verified/added

Industrial relations

- Relations between **management, workers (and their unions) and government** in an industry.
- Aim: **industrial peace**, higher productivity, fair treatment, no strikes or lockouts.
- **Tools:**
 - Collective bargaining; grievance procedure.
 - Joint consultation; workers' participation.
 - Conciliation, arbitration, labour courts.
- **Causes of poor relations:**
 - Low wages, bad working conditions.
 - Unfair dismissal.
 - Politically affiliated unions.
 - Poor communication.
- **Nepal:** Labour Act 2074, Trade Union Act 2049; frequent politically affiliated strikes.

Collective bargaining

- **Negotiation between employer and a union representing workers** on wages, hours, conditions, benefits.
- Result is a written **collective agreement**.

Process

Collective bargaining process



- Deadlock → conciliation, mediation, arbitration.

2.9 PYQ Mapping

Theory questions, covered in this document

- **Definition, functions, role of personnel management** → §2.1. 17 papers; HRM vs personnel management 71 Ch 2.
- **Personnel policy / employee handbook** → §2.2. 7 papers, 3 to 5; 75 Ash prints no marks against it.
- **Manpower / HR planning** → §2.3. 8 papers.
- **Recruitment and selection, steps of hiring, scientific selection, recruitment vs selection / hiring** → §2.4. 13 papers.
- **Outsourcing** 80 Bh 3, 69 Ch 2; **referral recruitment** 74 Ch 4; **interview** 75 Ch 3, 73 Ch 3 → §2.4.
- **Training: need, types, methods, training needs** → §2.5. 82 Ba 4, 81 Bh 3+2, 79 Bh 3, 72 Ch 4, 71 Shr 5
- **Talent poaching, development programme, appropriate HR** 79 Ba 4+4, 73 Shr 8 → §2.5.
- **Job analysis, job description, job specification, lecturer example, method for Nepal, job design** → §2.6. 10 papers.
- **Job evaluation vs merit rating** 81 Ba 3, 79 Bh 2; **merit rating / performance appraisal methods** (7 papers) → §2.6.
- **Wage vs salary, factors of wage structure, how wages are calculated, expected salary** → §2.7. 12 papers.
- **Incentives, why needed, incentive programs** → §2.7. 10 papers.
- **Industrial relations, collective bargaining** → §2.8. 68 Ch 4, 67 Asa 4+6, 66 Bh 6, 65 Shr 4+3

Short-note prompts placed here (2065–2068 ones from the Detailed PYQ's "Extra" section)

- Industrial relation 67 Asa, 65 Shr; collective bargaining 68 Ch; job design and work efficiency 67 Asa; incentive programs 68 Ba, 66 Bh; manpower planning 75 Ch.
- The Detailed PYQ files 66 Bh's "What do you mean by industrial relation? [6]" under Ch. 1 (organizational structure); it is answered here.

Numerical content

- **No paper in 29 sets a wage or incentive calculation.** The formulas in §2.7 answer the 3-mark “how are wages calculated?” (75 Ch) and are worth knowing in case a numerical appears.

Corrections to the source notes

- Adhikari p190, Priestman example (b): $(1400/600 - 1200/500)/(1200/500) = -2.78\%$, not -2.91% . The conclusion (no bonus) stands.
- Adhikari p181 spells the plan “Hasley”; it is the **Halsey** plan (F.A. Halsey).

Figures

- Adhikari’s Ch. 2 slides have no diagrams, so the few process figures kept are generated (ch2fig.py). Only diagrams an answer would draw are kept; no photos or decorative charts.

3. Motivation, Leadership and Entrepreneurship

10 hrs · asked in **all 29 papers** · 3 units, 21 syllabus topics · **591 marks, 25.0% of the archive** → about 13 of every 80 marks: usually two full 8-mark questions.

READ THIS FIRST

Asked in every paper, like Ch. 1 and Ch. 2. Three blocks, each with a fixed pattern:

1. **Motivation** = “define motivation” (2–3 marks) + one named theory (5–6 marks). Five theories cover it: **Maslow** (9 papers), **Herzberg** (10), **Vroom** (7), **McGregor X-Y** (6), **ERG** as a comparison (6). Learn one figure or table per theory. **Content** theories (Maslow, ERG, McClelland, Herzberg) say *what* motivates; **process** theory (Vroom) says *how*.
2. **Leadership** = qualities of a good leader (12) or styles (11) + an opinion: “which style suits an engineering project / Nepal / a startup?”. Answer: situational, leaning democratic, with a reason for each context (§3.3).
3. **Entrepreneurship** = definition + characteristics (8) or “why promote it in Nepal” (7). Always tie the second half to Nepal: jobs, remittance dependence, IT.

Tier chips count **papers out of 29**: **TOP n/29** 8+ **HOT n/29** 4–7 **PIN n/29** 1–3 inside an 8-mark question **LOW n/29** 1–3, short note only syllabus, no PYQ yet never set. Bold year = Regular exam.

3.1 Motivation: Needs, Types, Techniques

Define motivation. What are human needs, and how is a need used to motivate? Types of motivation

2 3 4

TOP 16/29 82 Bh, 79 Bh, 79 Ba, 76 Ch, 76 Ash, 75 Ch, 74 Ch, 74 Ash, 73 Ch, 72 Ka, 71 Ch, 71 Shr, 70 Ch, 70 Asa, 69 Ch, 68 Ch · the 2–3 mark opener to nearly every theory question

- From Latin *movere*, “to move”; **motive** = need, want, drive within a person.
- **Definition**: the process of **stimulating people to act** toward goals by creating conditions in which **their needs and the organization’s goals are satisfied together**.
- Forces within a person that decide the **direction, level and persistence** of effort at work.
- Two aspects: knowing human needs; inspiring people to do their best for the enterprise.

Human needs

- **Primary** (physiological, safety): food, shelter, security.
- **Secondary** (social, esteem, self-actualization): belonging, status, growth.
- Needs are unlimited, change over time, differ between people, and several operate at once.

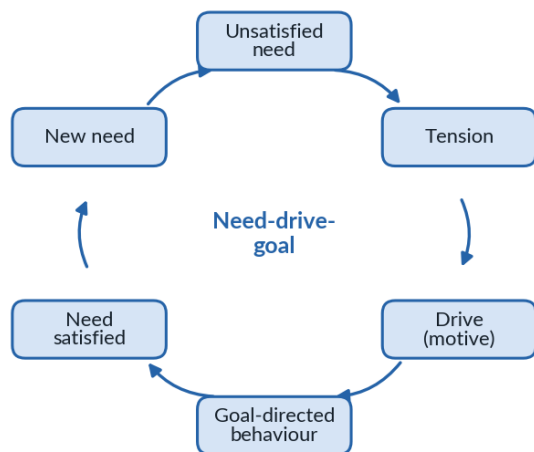
Types of motivation

- **Intrinsic**: from within; interest, enjoyment, sense of achievement in the task itself. “I want to complete this task.” Lasting.
- **Extrinsic**: from outside; pay, bonus, promotion, fear of penalty. “Complete it and be rewarded.”
- **Positive**: reward, praise, incentive (carrot).
- **Negative**: fear, punishment, threat of dismissal (stick).
- **Financial vs non-financial**.

Adhikari's six types, by drive

- **Incentive:** rewards, monetary or not, for reaching a target (bonus, promotion): the **carrot**.
- **Fear:** consequences; used when incentives fail: the **stick**.
- **Achievement:** drive for **competency**; tackle challenges, prove oneself; mostly intrinsic.
- **Growth:** burning desire for self-improvement and learning.
- **Power:** autonomy over one's own life, or control over others.
- **Social:** belong, be accepted, connect with others.
- No single type works for everyone: personalities differ.

How a need motivates: the need-drive-goal cycle



- Manager's job: **link the employee's unmet need to the organization's goal**, so working for the firm satisfies it.
- A satisfied need no longer motivates; a new one takes its place.

Human needs at work

- **The worker expects:**
 - Goals to fulfil.
 - The right to choose a job.
 - Responsibility to reach reasonable goals.
- **The manager must:**
 - Understand behaviour.
 - Merge employees' needs with the firm's objectives.
 - Know what satisfies each employee.
 - Build a co-operative climate: an unco-operative, indifferent one kills motivation.

Why motivation is complex

- Same need → different behaviour in different people.
- Needs change in kind and intensity.
- People cannot always express their needs.
- Many needs act at once.

Importance

- Higher productivity and quality; better use of resources.
- Low absenteeism and turnover; fewer grievances.
- Willing acceptance of change; good relations.
- Helps employees reach personal goals.

The motivation equation

- **Inputs** from the member: time, effort, skills, knowledge.
- Become **performance** toward the organization's goals.
- Which returns **outcomes** the member values: pay, security, autonomy.
- Motivation holds only while the outcomes feel worth the inputs.

Techniques of motivation; attitude, group and executive motivation; role of management 2 3 4 5 8

TOP 9/29 82 Ba, 81 Bh, 81 Ba, 76 Ch, 75 Ash, 73 Shr, 72 Ka, 70 Asa, 69 Ch

Techniques

1. **Financial:** fair wages, bonus, incentives, profit sharing, ESOP.
2. **Meaningful, challenging work:** job enrichment, rotation.
3. **Clear targets** and measured performance (MBO).
4. **Regular, specific feedback.**
5. **Recognition and praise:** public appreciation, awards.
6. **Participation** in decisions; **delegation.**
7. **Training, career growth,** promotion.
8. **Job security,** good working conditions, quality of work life.
9. **Status,** healthy competition, flexible hours.

Role of management in motivation

- Understand each employee's needs; people differ.
- Create an environment where personal and firm goals meet.
- Fair pay system; fulfil promised rewards.
- Give information, training, feedback; act on employee suggestions.
- Lead by example; make work interesting.

Necessities of group motivation

- Fairness, truthfulness, honesty, loyalty, stability, tolerance.
- Ability to take responsibility and co-operate.
- Constructive criticism; compromise when necessary.
- Orderliness in mind and action; a sense of humour.

Attitude (self) motivation

- How a person thinks and feels: self-confidence, belief, positive outlook.
- Falls with monotonous work, pushy boss, poor health, bad relations, money worries.
- Cure: interesting job, constructive thinking, build on strengths, adapt.

Group motivation

- Motivating a **team** as a unit; team productivity decides success.
- Needs fairness, honesty, loyalty, tolerance, shared responsibility, constructive criticism.
- Tools: team goals, group incentives, team building, social recognition.

Executive motivation

- Keeping **managerial talent** motivated and in the firm.
- Executives seek achievement, esteem, self-actualization more than money.
- Tools: challenging responsibility, autonomy, profit sharing, stock options, status perks, public recognition.
- Low when: poor boss, blocked advancement, poor pay, insecurity.

Difference (75 Ash)

- **Attitude:** the individual's inner state.
- **Group:** the team as a whole.
- **Executive:** top talent, driven by higher-order needs.

What motivates executives

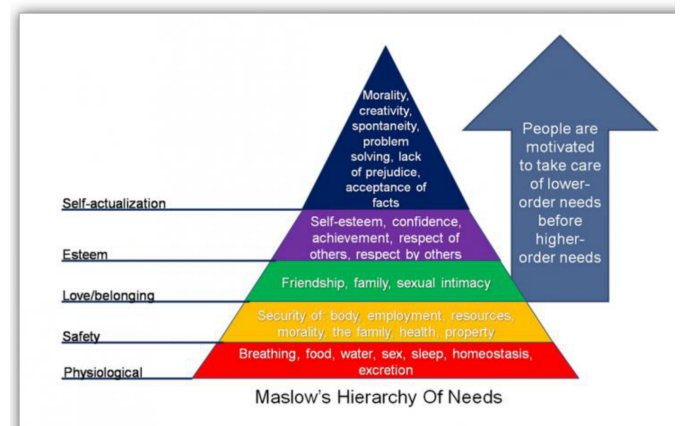
- Doing something they can call their **own**; tasks they set themselves.
- **Seeing** what they have done; feeling the job is **important.**
- Security; challenge; belief in the value of the work.
- Higher need for achievement; actualization, esteem, social needs matter more.

3.2 Motivation Theories

Maslow's hierarchy of needs: features, satisfaction-progression, examples 3 4 5

TOP 9/29 82 Bh, 82 Ba, 81 Bh, 78 Bh, 76 Ash, 75 Ch, 74 Ash, 70 Ch, 69 Ch

- **Abraham Maslow, 1943.** Three propositions:
 - Man is a **wanting being**: needs are endless.
 - A **satisfied need does not motivate**; only unmet needs do.
 - Needs form a **hierarchy**: lower must be largely met before higher ones motivate.
- **Five levels** (bottom up), with what the firm gives:
 1. **Physiological**: food, water, shelter, sleep → adequate wage, rest breaks.
 2. **Safety**: physical and economic security → safe conditions, job security, insurance, pension.
 3. **Social**: belonging, friendship → teams, informal groups, events.
 4. **Esteem**: self-respect, status, recognition → titles, awards, promotion.
 5. **Self-actualization**: realizing full potential → challenging, creative work, autonomy.



Maslow's pyramid. Lower needs must be met before higher ones motivate.

Satisfaction-progression process (75 Ch)

- When a need is **satisfied**, the person **progresses** to the next higher need, which now motivates.
- Eg a fresh graduate first wants a salary (physiological), then a permanent post (safety), then a good team (social), then a senior title (esteem), then to start something of their own (self-actualization).

Features and limitations

- + Simple, intuitive; widely used by managers.
- - Order is not fixed for everyone (an artist may skip safety).
- - Several needs act together.
- - Hard to measure satisfaction; culture-bound (US study).
- - Only one direction (no regression) → ERG fixes this.

Can Maslow explain Laxmi Prasad Devkota's tireless quest? (69 Ch)

- **Only partly.** Devkota wrote *Muna Madan* and his epics while in poverty, debt and grief, and kept writing on his deathbed with cancer.
- Pure Maslow says unmet physiological and safety needs should have blocked self-actualization.
- So he **contradicts the strict order**: self-actualization (creative expression, intrinsic motivation) drove him even with lower needs unmet.
- Better explained by **ERG** (needs act together) and intrinsic motivation.

Why some fall from grace while climbing the pyramid (78 Bh)

- Fraud, crime, suicide among the successful are the **irony** of the pyramid.
- **Esteem sought by any means**: status, wealth, power without ethics → corruption, fraud.
- **Frustration-regression (ERG)**: blocked at the top, people fall back to greed for material things or to despair.
- **Personal power need** (McClelland) outgrows social good.
- Pressure, isolation, loss of social support → depression, suicide.
- Maslow's own top level includes **morality**: those who fall never truly reached self-actualization, they stalled at esteem.
- Remedy: values and ethics education, social support, fair systems, counselling.

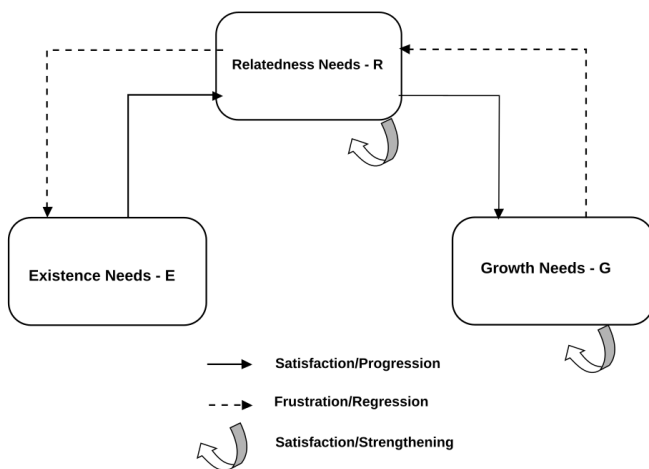
Maslow vs Alderfer's ERG; ERG and Herzberg compared with Maslow; satisfaction-progression vs frustration-regression 3 4 5 6

HOT 6/29 82 Ba, 78 Bh, 75 Ch, 71 Shr, 69 Ch, 68 Ch · satisfaction-progression vs frustration-regression as a short note in 75 Ch, 69 Ch

Alderfer's ERG theory (1969)

- Collapses Maslow's five into three:
 - **Existence:** physiological + safety.
 - **Relatedness:** social + external esteem.
 - **Growth:** internal esteem + self-actualization.
- **Satisfaction-progression:** satisfied lower need → move up (as Maslow).
- **Frustration-regression:** if a higher need is **blocked**, the person **falls back** and wants more of the lower need.
 - Eg an engineer denied promotion (growth) demands higher pay (existence).
- Several needs can motivate **at the same time**.

Alderfer's ERG Theory



Solid arrows: satisfaction-progression; dashed: frustration-regression; curled: satisfaction-strengthening. The 75 Ch / 69 Ch short note in one picture. Wikimedia Commons (Upul, vectorized by Sushant Savla), CC BY-SA 4.0.

Basis	Maslow	Alderfer ERG
Levels	5	3
Order	Strict, step by step	Flexible
Needs at once	One dominant	Several together
Direction	Up only (progression)	Up and down (regression)
Basis	Clinical observation	Empirical research

Herzberg against Maslow and ERG

- Herzberg's **hygiene factors** = Maslow's lower needs = ERG's existence and relatedness.
- Herzberg's **motivators** = Maslow's esteem + self-actualization = ERG's growth.
- Difference: Maslow and ERG say every need motivates when unmet; Herzberg says lower needs only **prevent dissatisfaction**, never motivate.
- Maslow and ERG: general human needs. Herzberg: **job factors**.

ERG at work

- **Existence:** basic pay level to buy daily items.
- **Relatedness:** good relations, feedback from boss and peers.
- **Growth:** a worker who keeps improving skills, creative work.

McGregor's Theory X and Theory Y 5 6 8

HOT 6/29 81 Bh, 81 Ba, 79 Ba, 73 Shr, 72 Ch, 65 Shr

- **Douglas McGregor**, *The Human Side of Enterprise* (1960): two sets of **assumptions managers hold about workers**, which shape how they manage.

Theory X (traditional)

- Average person dislikes work, avoids it.
- No ambition, avoids responsibility, prefers to be led.
- Self-centred, indifferent to organizational goals.
- Resists change; motivated only by money and fear.
- → Manager must **coerce, control, direct, threaten**: autocratic style, close supervision.

- McGregor showed **Theory Y managers** create more **employee engagement**.

Link to leadership:

- X ↔ autocratic style, close supervision, fear and punishment.
- Y ↔ democratic style, delegation, job enrichment.
- In the exam, write X and Y as two columns, 6–7 points each.

Theory Y (modern)

- Work is as natural as play or rest.
- People exercise **self-direction and self-control** when committed.
- Commitment comes from rewards of achievement.
- They **seek responsibility**; creativity is widespread.
- → Manager **delegates, involves, enriches jobs**: democratic style.

X vs Y against the other theories (72 Ch)

- X assumes people live at Maslow's **lower needs** / Herzberg's hygiene level; fits fear-and-punishment theory.
- Y assumes **higher needs** and Herzberg's motivators drive people.
- McGregor favoured Y; modern view: situation decides (contingency).

Herzberg's motivation-hygiene (two-factor) theory: where motivation at work comes from 4 5 6

TOP 10/29 82 Ba, 80 Bh, 79 Bh, 78 Bh, 74 Ch, 73 Ch, 71 Ch, 68 Ba, 67 Asa, 66 Bh

- **Frederick Herzberg**, 1959: interviewed about 200 engineers and accountants in Pittsburgh on times they felt very good or very bad at work.
- Finding: **satisfaction and dissatisfaction come from different factors**. The opposite of satisfaction is **no satisfaction**, not dissatisfaction.
- Man has two sets of needs: as an animal, to **avoid pain**; as a human, to **grow psychologically**.

Hygiene (maintenance) factors: job context

- Company policy and administration; supervision.
- Salary; job security; working conditions.
- Relations with superiors, peers, subordinates; status.
- Absent → **dissatisfaction**. Present → only **no dissatisfaction**, not motivation.

Motivators: job content

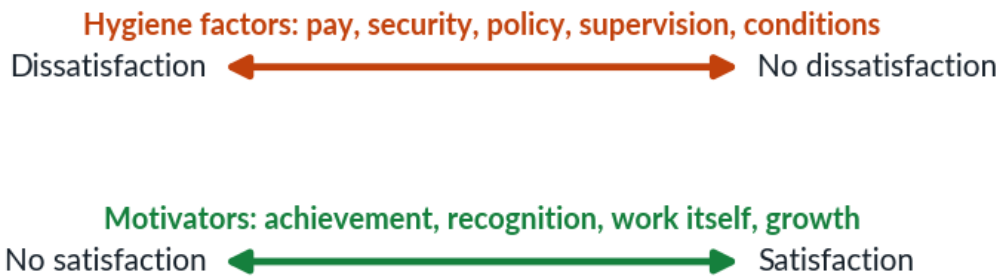
- Achievement; recognition.
- The work itself (interesting, challenging).
- Responsibility; advancement; growth.
- Present → **satisfaction and motivation**. Absent → no satisfaction (but not dissatisfaction).

So where does motivation at work come from? (80 Bh)

- From the **job itself**: achievement, recognition, responsibility, growth. Pay and conditions only remove complaints.
- Managers: first fix hygiene (else unhappiness), then **enrich jobs** to motivate.

Criticism

- Small, professional sample; people credit themselves for good events and blame others for bad (method bias).
- Pay can be a motivator for low-income workers.

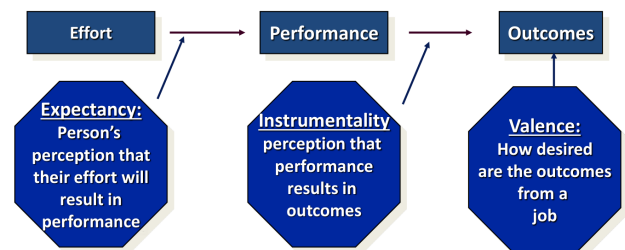


Two separate scales, not one: fixing hygiene moves a worker only to “no dissatisfaction”; only motivators move them toward satisfaction.

Vroom’s expectancy (VIE / valency) theory; why it is a process theory; importance of reward 4 5 8

HOT 7/29 82 Bh, 80 Bh, 80 Ba, 78 Bh, 76 Ch, 74 Ch, 70 Asa · short note in 82 Bh

- **Victor Vroom, 1964.** A person chooses the behaviour they **expect** to bring the outcome they **value**.
-
- $Motivation\ M = E \times I \times V$
- **Expectancy (E):** belief that **effort** → **performance** (0 to 1). “If I try, can I do it?”
- **Instrumentality (I):** belief that **performance** → **reward**. “If I do it, will I be rewarded?”
- **Valence (V):** **value** the person places on that reward. “Do I want it?”
- Product: if **any one is zero**, motivation is zero; high only when **all three** are high.
- Eg an engineer offered a Dubai posting (V high) for finishing a project early:
 - No effort if the deadline is impossible (E low).
 - No effort if the boss rarely keeps promises (I low).
- Managers must link rewards to performance and find out which outcomes each worker wants.



Effort → performance → outcomes, and the belief at each link.

Why it is a process theory (80 Bh)

- It does not list *which* needs motivate (content theories do).
- It explains **how** a person **decides**: a mental calculation of E, I, V among alternatives.

Importance of reward (74 Ch)

- Reward sits in two of three terms: it must be **valued** (V) and **reliably linked** to performance (I).
- Managers: offer rewards people want; keep promises; make targets reachable by training and resources (E).

McClelland's theory of learned needs

syllabus, no PYQ yet

- David McClelland (1961): needs are **learned** from life experience, not inborn; everyone ranks them differently.
- **Achievement (nAch):** drive to excel; moderate risks, want feedback, personal responsibility → entrepreneurs.
- **Power (nPow):** make others behave as they otherwise would not; personal or social (institutional) power → managers.
- **Affiliation (nAff):** friendly, close relations; avoid conflict → team roles.
- Each need has its own work preferences → managers **tailor the environment**: high achievers get responsibility, feedback, intermediate risk.

Fear and punishment theory 4

LOW 1/29 82 Bh · short note

- People work to **avoid penalties**: warnings, fines, demotion, dismissal.
- Assumes workers are lazy and need pushing (Theory X); the **stick** of carrot-and-stick.
- Linked to Skinner's reinforcement: punishment and negative reinforcement.
- **Deterrence**:
 - **Specific**: corrects the offender, stops a repeat.
 - **General**: punishment made public warns everyone else.

Merits

- Quick results for discipline, attendance, safety rules.
- Clear limits: everyone knows the consequence.

Demerits

- Resentment, fear, low morale.
- Only minimum effort: people avoid penalty, not seek excellence.
- Higher turnover; hidden mistakes.
- Use only as a last resort, fairly and consistently.

3.3 Leadership

Define leadership. Why is it needed? Qualities of a good leader 3 4 5 8

TOP 12/29 81 Bh, 81 Ba, 80 Bh, 78 Bh, 76 Ch, 75 Ash, 74 Ash, 73 Ch, 72 Ka, 70 Ch, 70 Asa, 68 Ch

- **Leadership**: the process of **influencing people** to work willingly toward group or organizational goals, and providing an environment for it.
- Koontz: "the art or process of influencing people so that they strive willingly and enthusiastically toward the achievement of group goals."

Why needed

- Gives **vision and direction**.
- **Motivates**, builds morale and confidence.
- Coordinates effort; resolves conflict.
- Leads **change**; builds culture and teamwork.
- Represents the group; develops future leaders.

What sets a leader apart from regular employees (80 Bh)

- Vision beyond own job; takes initiative unasked.
- Owns results, including failures.
- Influences without formal authority; others follow willingly.
- Develops others; thinks of the team first.

Adhikari's qualities, with his meaning

- **Guiding vision**: knows what to do; pursues it against opposition and failure; sets achievable goals.
- **Passion**: believes in the goals; positive outlook others follow.
- **Integrity**: knows own weaknesses; promises only what can be kept.
- **Honesty** at work and in private life; **trust**: acts for followers.
- **Curiosity**: a learner; **risk**: calculated, learns from mistakes.
- **Dedication**; **charisma**: maturity, respect, compassion, humour.
- **Listening**: "the most important attribute of all".

Qualities of a good leader

1. **Vision**: knows where to go, sets achievable goals.
2. **Honesty, integrity**: same inside and out; trust.
3. **Self-confidence**, courage, decisiveness.
4. **Communication**: clear speaking, writing, **listening**.
5. **Human skill**: respect, fairness, empathy; credit to others.
6. **Initiative and innovation**.
7. **Knowledge**, intelligence, judgement.
8. **Discipline**, commitment to excellence; leads by example.
9. **Emotional stability**, patience, sense of humour.
10. **Calculated risk-taking**; learns from mistakes.
11. **Charisma**, passion.
12. **Adaptability**: changes style with the situation.

Types of leaders, by the source of their lead

- **By position**: lead because of the post they hold.
- **By personality**: charisma draws followers.
- **By moral example**: goodness and integrity inspire.
- **By power held**: control over rewards or punishment.
- **Intellectual**: lead through knowledge and ideas.
- **By ability to get things done**: results earn the lead.

Leadership styles (autocratic, democratic, free-rein); which style suits an engineering project, industry, Nepal, a startup? 2 3 4 5 6 7 8

TOP 11/29 82 Ba, 80 Ba, 78 Bh, 75 Ch, 74 Ash, 73 Ch, 72 Ch, 68 Ch, 68 Ba, 67 Asa, 65 Shr · “leadership style” short note in 68 Ba · **opinion half every time**

- **Style** = the behaviour pattern a leader adopts toward followers.

Factors affecting style

- The **task**: urgent, routine or creative.
- The **tradition** of the organization.
- The **type of labour force**: skilled or unskilled, followers’ maturity.
- The leader’s **personality**.
- The **time** available.
- Also “gender?”: research finds style differences small and situational.

	Autocratic Leadership	Democratic Leadership	Free-Rein Leadership
Concentration of Power	With Manager	Shared between Manager and Employees	With Employees — Minimal Supervision
Decision Making	By Manager	By Manager and Employees	By Employees
Communication Flow	From Top Down	Two Way	Widespread and as Needed

Power, decision making and communication under the three styles.

- **Autocratic (authoritarian):**
 - Leader decides alone, trusts only own competence, gives orders, expects obedience; one-way communication.
 - + quick decisions, clear control; suits emergencies, unskilled or new staff, military.
 - – low morale, no creativity, dependence on one person, resentment.
- **Democratic (participative):**
 - Leader consults, delegates as far as experience allows, keeps final responsibility; praise and blame shared; two-way.
 - + better decisions, commitment, morale, creativity; usually most productive.
 - – slow; needs capable, willing followers.
- **Laissez-faire (free-rein):**
 - Leader gives goals and freedom; group decides and self-controls.
 - + suits experts, researchers; high autonomy.
 - – no direction, chaos with inexperienced staff.

Other styles

- **Paternalistic**: fatherly care for workers’ welfare, leader still decides.
- **Bureaucratic**: leads strictly by rules and procedures.
- **Transformational**: inspires change through vision.
- **Transactional**: reward for results, punishment for failure.

Participative management (65 Shr)

- Employees share in decisions:
 - Suggestion schemes; quality circles.
 - Work committees; worker directors on the board.
- Importance: commitment, better ideas, fewer disputes, job satisfaction.

Which style for a (modern) engineering project? (78 Bh, 74 Ash)

- **Democratic / participative** as the base: engineers are skilled professionals; design choices gain from many views; commitment matters over months.
- **Autocratic** moments: safety, deadlines, site emergencies.
- **Free-rein** for expert sub-teams (design, R&D).
- → **Situational leadership**, democratic by default.

Most effective style in an industrial organization (80 Ba, 68 Ba, 67 Asa)

- Determine it by: task structure, followers' skill and maturity, leader's power, time pressure (Fiedler, Hersey-Blanchard).
- Shop floor, routine work, new workers → directive; skilled staff → participative; experts → delegating.
- Most effective overall: **team (9,9) style** on Blake-Mouton's grid.

Blake and Mouton's managerial grid: styles, examples, significance for growth 4 5 8

HOT 6/29 82 Bh, 80 Bh, 79 Ba, 76 Ash, 75 Ash, 70 Ch

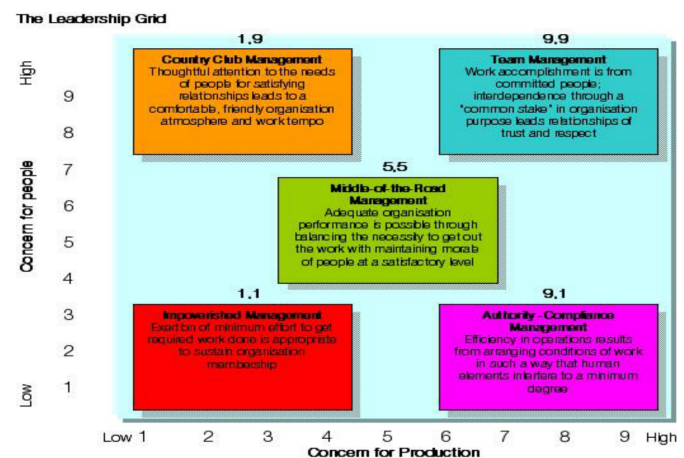
- Robert Blake and Jane Mouton (1964). Two axes, each 1 (low) to 9 (high): **concern for production** (x) and **concern for people** (y).
- Five key styles:
 - **(1,1) Impoverished**: minimum effort for either; avoids blame. Eg a manager waiting for retirement.
 - **(1,9) Country club**: comfort and friendliness, weak output. Eg a boss who never enforces deadlines.
 - **(9,1) Authority-compliance (task)**: output through rules and pressure, people ignored. Eg a sweatshop supervisor.
 - **(5,5) Middle of the road**: balance, adequate results, neither fully met.
 - **(9,9) Team**: high on both; commitment, trust, shared stake. Eg an agile software team lead.

Most efficient in Nepal? (73 Ch)

- Tradition is autocratic and paternalistic (hierarchy, respect for seniority).
- Educated young workforce, migration of talent → **democratic with a paternalistic touch** works best: consult, care for welfare, decide firmly.
- Government offices need less bureaucracy and more participation.

Nepalese start-ups (82 Ba)

- Small teams of young, skilled people, fast change, scarce funds.
- Best: **democratic + transformational**: shared vision, open ideas, quick consensus, ownership (ESOP), freedom for experts.
- Founder must switch to decisive (autocratic) mode in a cash or deadline crisis.



The leadership grid: concern for production vs concern for people.

Significance for organizational growth (76 Ash)

- Diagnoses a manager's present style objectively.
- Shows the target: move toward **(9,9)**.
- Basis for leadership training (grid seminars).
- Shows people and production are not a trade-off: both together raise productivity and morale → growth.

Leadership approaches / theories: trait, behavioural, contingency (situational), integrated 4 5 8

HOT 7/29 79 Ba, 76 Ch, 75 Ch, 75 Ash, 69 Ch, 66 Bh, 65 Shr · contingency approach of leadership as a short note in 66 Bh

Trait approach

- Leaders are **born, not made** (Great Man theory).
- Identifies traits of effective leaders: drive, leadership motivation, honesty, self-confidence, intelligence, business knowledge, self-monitoring.
- – No trait list fits all leaders; ignores followers and situation.

Behavioural approach

- Leaders are **made**: what they **do** matters, and can be learned.
- Two independent behaviours:
 - **Task-oriented**: assign work, set standards, push deadlines.
 - **People-oriented**: trust, respect, welfare.
- Ohio State and Michigan studies; Blake-Mouton grid.
- Effective leaders score high on both.

Adhikari's three approaches (write if the question says "approach")

1. **Trait-oriented**: leaders born, not made; more intelligent, better educated, stronger power needs, independent, take some risks, self-actualized, hard working, self-confident.
2. **Situational**: leadership is a function of the situation, culture, context and customs of the group.
3. **Functional**: considers both the individual and the situation; the leader must solve the task problems.

Contingency / situational approach

- **No one best style**: it depends on leader, followers and situation.
- **Fiedler**: task- or relationship-motivated leader fits situations of different favourableness (leader-member relations, task structure, position power).
- **House's path-goal**: leader chooses **directive, supportive, participative or achievement-oriented** behaviour to clear the path to goals.
- **Hersey-Blanchard**: telling, selling, participating, delegating as follower maturity rises.

Integrated approach

- Combines three sets of variables to choose the effective style:
 - **Leader**: traits, behaviour, experience.
 - **Followers**: ability, maturity.
 - **Situation**: structure, technology, objectives, environment.
- Most realistic of the four.

Adhikari's eight leadership theories

1. **Great Man**: born to lead; heroic, destined; "man" because leadership was then seen as male, military.
2. **Trait**: inherited traits (extraversion, self-confidence, courage).
3. **Contingency**: environment variables decide the best style; none best everywhere.
4. **Situational**: choose the action by situation (authoritarian when the leader knows most; democratic with skilled experts).
5. **Behavioural**: leaders are **made**; learnable by teaching and observation.
6. **Participative**: takes others' input; leader keeps the right to decide.
7. **Management (transactional)**: supervision, organization, group performance; rewards and punishments.
8. **Relationship (transformational)**: inspires by showing the higher good of the task; wants each member to fulfil potential; high ethics.

Leader vs manager; boss vs leader; “a reader is a leader”; challenges for a modern leader; you as a future leader 2 3 4 5 8

HOT 4/29 75 Ch, 75 Ash, 73 Ch, 71 Ch

Basis	Manager	Leader
Power from	Position (authority)	Personal influence
Focus	Things, systems, control	People, vision, inspiration
Motto	Do things right	Do the right things
Work	Plan, organize, budget	Set direction, align, motivate
Change	Maintains order	Drives change
Followers	Subordinates	Followers

- A good manager should be a leader; a leader need not be a manager.

Boss vs leader (73 Ch)

- **Boss** = autocratic style:
 - Commands, uses fear; says “I”; says “Go!”.
 - Takes credit, blames others for failure.
- **Leader** = democratic / transformational style:
 - Guides, inspires; says “we”; says “Let’s go!”.
 - Shares credit, owns failure.

Managers vs leaders, function by function

- Both do the **same four activities**: planning, organizing, directing, controlling. They do them differently (right).
- Manager: focus on things, do things right, follows the rules.
- Leader: focus on people, do the right things, inspires, shapes entities.

“A reader is a leader” (71 Ch)

- Reading builds:
 - **Knowledge** (expert power).
 - **Vision** and **judgement** from others’ experience.
 - **Communication**: vocabulary, clear expression.
- Informed leaders can switch styles wisely: consult as a democrat, decide as an autocrat when needed.
- Eg Gandhi, Mandela, Gates: lifelong readers.

Challenges for a modern leader (75 Ch)

- Rapid technological change; remote and diverse teams.
- Talent retention (migration in Nepal); ethics, transparency.
- Uncertainty (pandemic, politics); information overload.

Me as a future leader (75 Ash, 75 Ch)

- Start as a technically strong engineer; build human skills.
- Lead by example, listen, share credit, keep learning.
- Aim for a (9,9) team style, adapting to the situation.

	Manager	Leader
Planning	budgets, targets, detailed steps, allocates	strategy, direction, vision
Organizing	structure, job descriptions, staffing, delegates	gets people on board; communicates, networks
Directing	solves problems, negotiates, builds consensus	empowers people; cheerleader
Controlling	control systems, measures, fixes variances	motivates, inspires, sense of accomplishment

Authority and power 5

PIN 1/29 66 Bh

- **Authority**: **legitimate right** to command, from the position.
 - Flows downward; accepted as rightful.
 - Lost when the post is left.
- **Power**: **ability to influence** behaviour, from any source.
 - Can exist without authority (an informal leader).
 - Can flow in any direction.

Sources of power (French and Raven)

- **Legitimate**: from the position.
- **Reward**: can give pay, promotion, praise.
- **Coercive**: can punish.
- **Expert**: knowledge and skill.
- **Referent**: personal charisma; people want to identify with the leader.
- Leaders rely most on expert and referent power.

3.4 Entrepreneurship

Entrepreneurship: meaning, development, characteristics of an entrepreneur 3 4 5 8

TOP 8/29 79 Bh, 76 Ch, 75 Ash, 74 Ash, 72 Ch, 71 Shr, 70 Asa, 69 Ch · entrepreneurial characteristics as a short note in 79 Bh

- From French *entreprendre*, “to undertake”.
- **Entrepreneurship**: the capacity and willingness to **spot an opportunity, organize resources and run a new venture, bearing its risk**, for profit and value.
- **Schumpeter**: entrepreneurs are **innovators**: new products, new methods, new markets, new sources of supply, new organization.
- **Schumpeter**: “Entrepreneurs are business leaders and not simple owners of capital. They are men of vision, drive and talent, who spot out opportunities and promptly grasp them for exploitation.”
- **McClelland**: a nation’s growth depends on its people’s need for achievement more than on natural resources.
- The word first meant an organizer of musical or other entertainments; from the early 18th century it meant business.

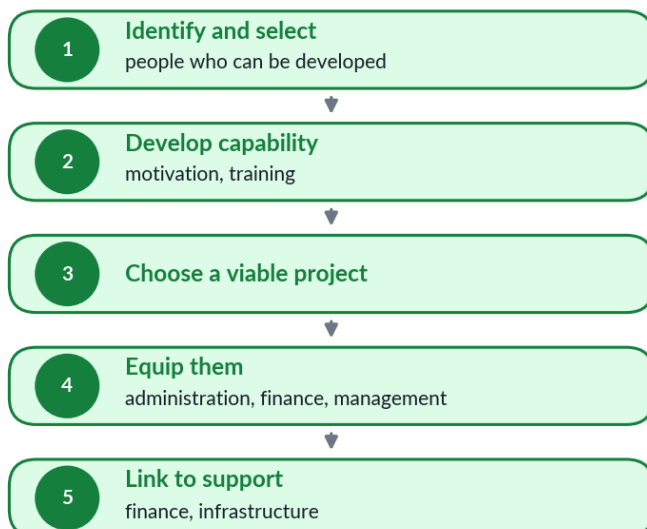
Functions of an entrepreneur

1. See an opportunity.
2. Organize the enterprise.
3. Raise capital.
4. Hire labour.
5. Arrange materials.
6. Select managers.
7. Bear the risk and uncertainty.

Entrepreneurship development

- Entrepreneurs are **made**: Entrepreneurship Development Programmes (EDPs), in steps:

Entrepreneurship development programme



Factors influencing it

- Components of creativity: need for achievement, independence, power; family background.
- Originality; flexibility; fluency of ideas.
- Sensitivity to problems and the ability to redefine them.

Characteristics of an entrepreneur

1. **Initiative** and self-starting.
2. **Opportunity seeking**; foresight.
3. **Calculated (moderate) risk-taking**: not a gambler.
4. **Creativity and innovation**.
5. **High need for achievement**; competitive.
6. **Self-confidence**, optimism.
7. **Commitment, persistence**: learns from failure.
8. **Decision making**; administrative ability.
9. **People skills**, communication, leadership.
10. **Technical and business knowledge**.
11. **Hard work**, discipline, strong work ethic.
12. **Flexibility**; handles uncertainty.

Adhikari's list of 12 (short-note version, 79 Bh)

- Administrative capability; creativity.
- Self-confidence; optimism.
- Human relations; communication.
- Foresight; clarity of objectives.
- Technical knowledge; secrecy.
- Decision making; willingness to take risk.

Steps for establishing a small scale unit 4 5

PIN 3/29 81 Bh, 79 Bh, 71 Shr

1. **Self-assessment:** will, skills, risk appetite.
2. **Product / service identification:** market research, competition, demand.
3. **Feasibility study and project report:** technical, market, financial.
4. **Form of ownership:** sole, partnership, private limited.
5. **Location:** raw materials, market, transport, labour, power, water, waste disposal.
6. **Registration and approvals:** Department / Office of Cottage and Small Industries (Nepal), PAN / VAT, local ward, environmental clearance.
- **Why small units:** easy to set up, create jobs, can produce export-quality goods (India's example).
7. **Finance:** fixed capital (land, building, machines) + working capital; own funds, bank loan, government subsidized loans.
8. **Land, building, machinery;** utility connections.
9. **Recruit and train** manpower.
10. **Raw material** arrangement; **production** layout, trial run, quality control.
11. **Marketing:** pricing, distribution, promotion.
12. **Monitor and improve** the plan at every stage.

Need for promotion of entrepreneurship; its role in economic development; importance for Nepal; conducive environment and law enforcement 4 8

HOT 7/29 82 Bh, 82 Ba, 81 Ba, 79 Ba, 75 Ash, 73 Ch, 73 Shr

Role in economic development / why promote

- **Employment** and self-employment.
- **Capital formation:** mobilizes idle savings.
- **Innovation**, new products, technology.
- **Balanced regional development;** decentralization.
- **Equitable income distribution.**
- **Exports**, import substitution → foreign exchange.
- Better use of local resources; higher GDP and tax revenue.
- Raises living standards; inspires others.

Why Nepal and its government must promote it

- Over a million youths leave for foreign jobs; remittance-dependent economy.
- Huge trade deficit; low industrial base.
- Unused resources: hydropower, tourism, agriculture, herbs, IT.
- Tools: startup loans, incubation, tax holidays, one-window registration, skills training, access to credit.

Creativity alone is not enough: a conducive environment (79 Ba, 73 Shr)

- **Political stability** and consistent policy.
- **Access to finance:** banks, venture capital, collateral-free loans.
- **Infrastructure:** power, roads, internet.
- **Easy registration**, low red tape, fair tax.
- **Market access**, skilled labour, education, incubators.
- **Culture** that accepts failure and risk.

Significance of law enforcement (73 Ch)

- Protects property and **intellectual property**.
- Enforces **contracts** → trust in business deals.
- Fair competition; controls cartels and corruption.
- Investor confidence → capital flows in.

Entrepreneurship in Nepal's IT sector; would you start a startup; risks and challenges; Silicon Valley

3 5 8

PIN 3/29 76 Ash, 74 Ch, 69 Ch

Role in Nepal's IT sector (76 Ash)

- IT services exports (outsourcing, software) earn foreign currency with little capital.
- Jobs for graduates at home, slowing brain drain.
- Digital services: payments (eSewa, Khalti), ride sharing (Pathao), e-commerce (Daraz), ed-tech.
- Solutions for local problems in agriculture, health, government.

Would I start one after graduation? (74 Ch)

- **Yes, with preparation:** gain 2–3 years' industry experience and savings, validate the idea, find a co-founder, start lean.
- **Or no:** if family needs steady income or the idea is untested. Either answer works if justified.

Risks and challenges for an aspiring entrepreneur in Nepal

- Small domestic market; low purchasing power.
- Scarce seed and venture capital; banks want collateral.
- Restrictions on international payments and foreign investment in small amounts.
- Policy instability, bureaucracy, strikes.
- Talent loss to migration; social stigma of failure.
- Weak IP protection; infrastructure gaps outside cities.

Silicon Valley: the best example (69 Ch)

- Stanford University research + venture capital + risk-taking culture + skilled migrants + network effects.
- Produced Apple, Google, Intel, HP; failure accepted as learning.
- Lesson for Nepal: link universities, investors and startups in clusters.

3.5 PYQ Mapping

Theory questions, covered in this document

- **Define motivation; human needs; how needs motivate; intrinsic motivation** → §3.1. 16 papers, the opener to most theory questions.
- **Techniques of motivation; attitude, group, executive motivation; role of management** → §3.1. 9 papers.
- **Maslow** (features, satisfaction progression, Devkota 69 Ch, falling from grace 78 Bh) → §3.2. 9 papers.
- **Maslow vs ERG; ERG and Herzberg vs Maslow; satisfaction-progression vs frustration-regression** → §3.2. 82 Ba 6, 78 Bh 3, 75 Ch 4, 71 Shr 5, 69 Ch 4, 68 Ch 5
- **McGregor X-Y** 81 Bh 5, 81 Ba 5, 79 Ba 6, 73 Shr 5, 72 Ch 8, 65 Shr 5 → §3.2.
- **Herzberg** (10 papers) and **Vroom** (7 papers) → §3.2.
- **Fear and punishment** 82 Bh 4; **McClelland**: syllabus, never set → §3.2.
- **Leadership: definition, need, qualities** → §3.3. 12 papers.
- **Styles; best style for engineering project / industry / Nepal / start-ups; participative management** → §3.3. 11 papers.
- **Managerial grid** 82 Bh 4, 80 Bh 4, 79 Ba 4, 76 Ash 8, 75 Ash 8, 70 Ch 5 → §3.3.
- **Leadership approaches: trait, behavioural, contingency, integrated** → §3.3. 7 papers.
- **Leader vs manager / boss; reader is leader; modern challenges; career vision** → §3.3. 75 Ch, 75 Ash, 73 Ch, 71 Ch
- **Authority and power** 66 Bh 5 → §3.3.
- **Entrepreneurship: meaning, characteristics** (8 papers); **small scale unit steps** 81 Bh 5, 79 Bh 4, 71 Shr 5 → §3.4.
- **Promotion of entrepreneurship, economic role, Nepal, environment, law** (7 papers); **IT sector, startups, Silicon Valley** 76 Ash, 74 Ch, 69 Ch → §3.4.

Short-note prompts placed here (2065–2068 ones from the Detailed PYQ's "Extra" section)

- **Leadership style** 68 Ba; **contingency approach of leadership** 66 Bh; **satisfaction-progression vs frustration-regression** 75 Ch, 69 Ch; **entrepreneurial characteristics** 79 Bh.
- 81 Bh's "State Maslow's need theory with examples [3]" sits in Ch. 1 of the Detailed PYQ; answered here.

Numerical content

- **This chapter has no numerical component.** Vroom's $M = E \times I \times V$ is the only formula, used qualitatively.

Corrections to the source notes

- AJ Sir slide 10: Maslow is 1943 ("A Theory of Human Motivation"), not "the 1940s"; slide 25: the grid's co-author is **Jane Mouton**, not "Jake".
- Adhikari p213: fear and punishment slide is criminal deterrence (workplace version above); p249: McClelland was at Harvard, not "Horward".

Figures

- Only diagrams an answer would draw are kept; no photos, clip art or decorative charts.
- Cropped: Adhikari's notes (pp197, 219, 238, 241).
- Generated (`ch3fig.py`) for processes with no source picture.
- ERG diagram: Wikimedia Commons (Upul, vectorized by Sushant Savla, CC BY-SA 4.0).

4. Case Studies

2 hrs · asked in **14 of 29 papers** · 5 syllabus topics · **92 marks, 3.9% of the archive** → small, but a near-certain 4-mark short note since 2079.

READ THIS FIRST

Only two things are ever asked, in rotation: **objectives of case study** (8 papers) and **steps / phases of case study** (5 papers, all since 2079). Learn the **6 steps** and the **5 phases** as two separate numbered lists; examiners use both words. Twice a paper has set an applied 8-marker (prepare a case study of NEA or Nepal Telecom; recommend improvements for your college): answer it by filling in the **report structure** of §4.3. §4.4 and §4.5 are those two cases **written out in full**, as models. §4.6 is a **complete sample report** of an industrial visit: the 10-mark internal assessment report.

Tier chips count **papers out of 29**: **TOP n/29** 8+ **HOT n/29** 4–7 **PIN n/29** 1–3 inside an 8-mark question. Bold year = Regular exam.

4.1 Introduction and Objectives

What is a case study? Why are case studies conducted? Objectives and value of case study **2 3 4**

TOP 8/29 81 Ba, 80 Ba, 73 Ch, 71 Ch, 71 Shr, 70 Asa, 69 Ch, 66 Bh · short note in 71 Shr, 69 Ch, 66 Bh ("value of case study")

Meaning

- A **case** = written description of a **real management situation**: facts, reports, even opinions, but **no evaluation**.
- **Case study method**: learners analyse the case, identify problems, weigh alternatives and **decide**, as a manager would.
- Belief behind it: managerial competence comes from studying and discussing **actual situations**, not only theory.
- Word means different things to different users:
 - lawyer: a legal case;
 - researcher: a vehicle for **testing a hypothesis**;
 - management education: an **authentic problem** to train analysis and decision making.
- Aim secured by facing the learner with a **whole series** of cases, not one.
- Harvard Business School made it the standard method of management teaching (1920s) **verified/added**.

Why conducted (AJ Sir)

- Teach managerial skills: **analyse** a situation, **frame** the right questions, **find answers**, **make decisions**.
- Link the three foundations of management practice: **research, theory, practice**.

Value (short note, 66 Bh)

- + realistic, active learning.
- + develops analytical thinking; learner participates.
- + many solutions welcome → no single "right" answer.
- + bridges classroom and industry.
- – time consuming: gathering, arranging and writing up the facts is long and tedious.
- – needs a skilled discussion leader.
- – one case may not generalize.

Objectives

1. **Impart knowledge and facts** about real organizations.
2. Develop **problem analysis** skill: separate symptoms from causes.
3. Develop **decision-making** skill under incomplete information.
4. Improve **communication**: discuss, defend, write reports.
5. Shape **attitude**: nothing in human behaviour is absolutely right or wrong; respect other views.
6. Apply **theory to practice**; test principles against reality.
7. Build **teamwork** and judgement through group discussion.

Case study vs lecture **verified/added**

	Lecture	Case study
Learner	listens	analyses, decides
Content	principles	real situation
Answer	one correct	several defensible
Skill built	recall	judgement, communication

Types of case studies 2

PIN 1/29 80 Ba

By research purpose

- **Illustrative:** descriptive; one or two instances show what a situation is like; makes the unfamiliar familiar; gives readers a common language.
- **Exploratory (pilot):** short study before a large investigation, to find the right questions and measures. Risk: early findings released as conclusions.
- **Cumulative:** combines past studies from several sites and times → wider generalization at no new cost.
- **Critical instance:** one or a few sites of unique interest, or to challenge a universal claim; answers cause-effect questions.

Teaching cases

- **Incident:** a short event to react to.
- **Problem / decision:** choose an action.
- **Appraisal:** evaluate a situation.
- **Descriptive:** shows how a firm handled something.

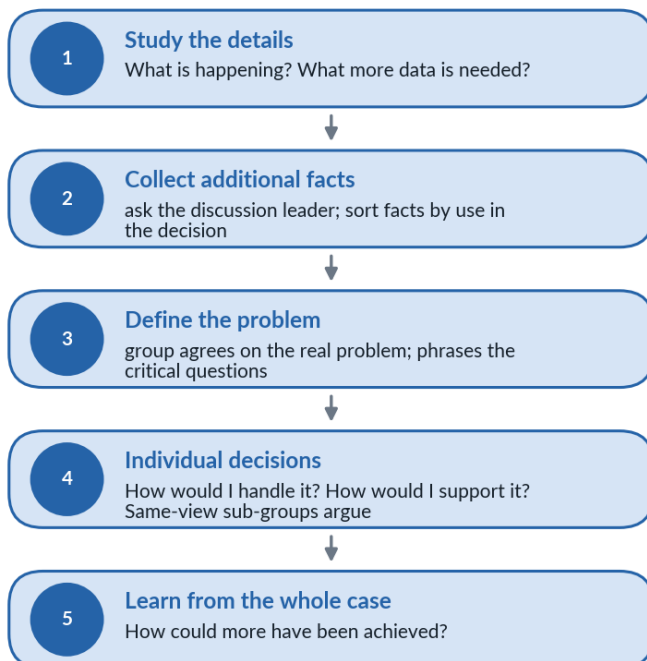
4.2 Phases and Steps of Case Study

Explain the phases and steps of case study, with an example from your field 4 5 8

HOT 5/29 82 Bh, 81 Bh, 81 Ba, 80 Bh, 79 Bh · short note in 82 Bh, 81 Bh · every paper since 2079 except 80 Ba

Five phases (group discussion)

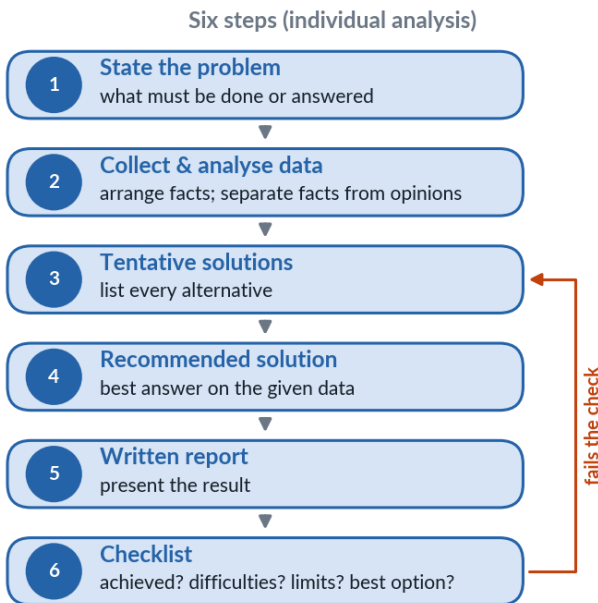
Five phases (group discussion)



Notes on the phases

- **Phase 1:** every member reads the case alone first; a group that has not read cannot discuss.
- **Phase 2:** facts are organized by relevance to the decision; irrelevant detail is dropped.
- **Phase 3:** some cases supply the critical questions; otherwise the group phrases them. A wrongly defined problem wastes the rest.
- **Phase 4:** each sub-group presents its strongest argument; decisions are then debated openly.
- **Phase 5:** the lesson is transferred beyond this one case.
- The **discussion leader** (teacher) guides but does not give the answer.

Six steps (individual analysis)



Notes on the steps

- **Step 1:** read the case and its questions, then write in one line what the case is about and what action or answer is required.
- **Step 2:** arrange facts systematically; mark which are facts and which are opinions of people in the case.
- **Step 3:** treat first ideas as tentative; give every alternative due consideration before rejecting it.
- **Step 4:** choose **one** solution, the best answer on the given data.
- **Step 6 checklist:** what does it achieve? what difficulties? what limits (time, money, equipment, manpower)? is it the best alternative?
- A solution that fails the checklist sends you back to step 3.

Phases vs steps: which to write

- **Phases** = how a **group** works through a case in a class or workshop (discussion method).
- **Steps** = how **one analyst** solves it and writes it up (report method).
- Question says “phases” → 5 phases; “steps” → 6 steps; “phases and steps” (81 Bh) → both lists, 2 marks each.
- Always add one line of example from engineering (below).

Analysing a case problem (AJ Sir): the reading order before step 1

1. Read the **title** of the case.
2. Note the **company** and the case **problems** as given.
3. **Careful, detailed reading** of the whole case.
4. **Meet the central characters:** who decides, who is affected.
5. **Define the problems**, then develop the solution.

Example from engineering: why did a hydropower project finish two years late? verified/added

Step	Applied
1 State the problem	60 MW project, planned 4 years, took 6; cost overrun 40%. Why, and how to prevent it next time?
2 Collect and analyse data	Schedule vs actual (Gantt), change orders, land-acquisition dates, contractor payments, site logs, interviews with engineers and locals. Finding: land dispute (8 months), tunnel geology surprise (10 months), late payments → contractor slowdown.
3 Tentative solutions	(a) better geological survey; (b) settle land before tender; (c) escrow payment to contractor; (d) matrix project organization with one accountable manager.
4 Recommended solution	Combine (a)+(b) as pre-construction gates, (c) for cash flow; (d) as the structure.
5 Written report	Background, problem, data, analysis, recommendations, implementation plan.
6 Checklist	Achieves on-time completion; needs upfront survey budget; limit: political land issues; still the best alternative.

4.3 Case Report Writing, and Preparing a Case Study

Prepare a case study of an organization (NEA / Nepal Telecom; your college) following the case structure 8

PIN 2/29 74 Ch, 72 Ch · applied 8-markers: fill in the structure · full models in §4.4 (72 Ch) and §4.5 (74 Ch)

Case report writing methodology

- 1. Organizational background:**
 - goals and objectives; formation;
 - structure; social needs and value;
 - historical relation; financial position.
- 2. Identification of the problem.**
- 3. Hypothesis:** the suspected cause.
- 4. Case research design:** what data, from whom, how.
- 5. Collection and analysis of data.**
- 6. Generalization and interpretation.**
- 7. Conclusions and recommendations.**

Typical report titles (a full O&M case study)

- 1.** Acknowledgement.
- 2.** Organization: history, objectives, input, process, output.
- 3.** Organization structure (chart).
- 4.** Form of ownership.
- 5.** Personnel policies; manpower planning.
- 6.** Recruitment and selection; training and development.
- 7.** Job evaluation; merit rating; wages and incentives.
- 8.** Motivation; marketing; leadership; MIS.
- 9.** Drawbacks; suggestions and recommendations.
- 10.** Conclusion; references.
 - The titles are the syllabus itself: a case study is where Ch. 1–3 and 5 are applied to one real organization.

Writing the 8-mark version in the exam

- Use the **7 headings** of the methodology as your headings; 1–3 lines each.
- Spend the marks on **what the question names**: 72 Ch names planning horizon, leadership, motivation, HRD → one analysis paragraph each.
- Give **one or two real numbers** (loss %, load-shedding hours, pass rate); they show the data step was done.
- Draw **one** figure: an org chart, a cause-and-effect (fishbone) diagram, or a time-bound plan.
- End with **recommendations + time frame**, then the **checklist** line (achieves, difficulties, limits, best alternative).
- Roughly: background 1, problem + hypothesis 1, design + data 2, analysis 2, recommendations 2 marks.

4.4 Mock Case Study 1: Nepal Electricity Authority (72 Ch)

PIN 1/29 72 Ch · written out in full as a model · figures are NEA's published figures; staff-survey results are illustrative verified/added

Question (72 Ch, 8 marks)

If you are asked to prepare the case study considering the planning horizon, leadership, motivation and human resource development for either Nepal Electricity Authority or Nepal Telecom to improve the existing performance of these institutions, how do you prepare the case study following its structure?

Title

Improving the performance of Nepal Electricity Authority: a case study of planning horizon, leadership, motivation and human resource development. Period studied: FY 2072/73 (the crisis year) to FY 2080/81.

Acknowledgement

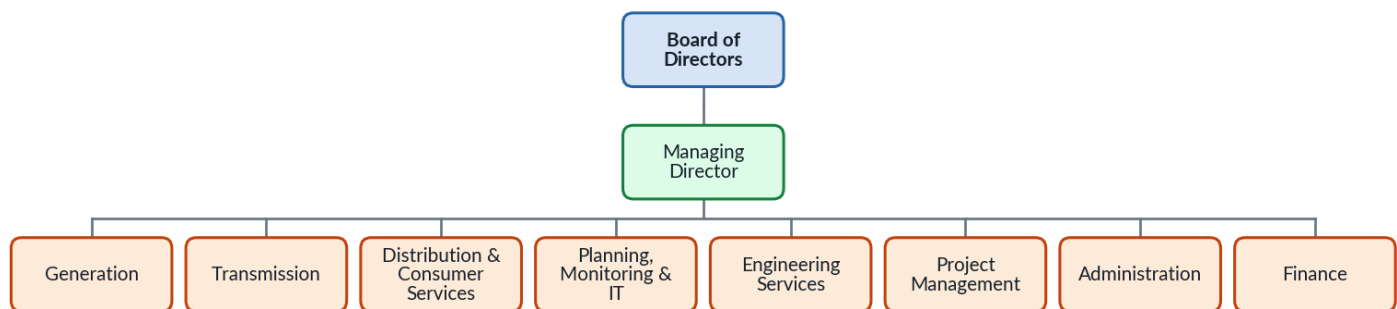
Thanks to NEA's Planning, Monitoring and IT Directorate for the annual reports, to the distribution-centre chiefs who gave interviews, and to the course teacher for guidance.

1. Organizational background

- **Formation:** established 16 August 1985 (Bhadra 1, 2042) under the **NEA Act 2041**, merging the Electricity Department, Nepal Electricity Corporation and related development boards.
- **Form of ownership:** **public corporation**, fully government owned (§1.4): own legal entity, govt control, service + commercial motive.
- **Goals:** generate, transmit and distribute adequate, reliable and affordable power; plan and develop the national grid; trade power with India.
- **Structure:** Board of Directors chaired by the Energy Minister → Managing Director → Deputy Managing Directors heading functional directorates (figure) → regional offices and distribution centres. A **line and staff** structure (§1.5).
- **Social value:** power drives industry, health, education, irrigation, telecom; every household is a customer.
- **Financial position:** accumulated loss of **Rs 34.61 bn** up to FY 2072/73; first net profit **Rs 1.46 bn** in FY 2073/74, profitable every year since.

Input, process, output

Stage	NEA
Input	water (river flow), own plants, power bought from IPPs, imports from India, staff, capital, tariff revenue
Process	generation → transmission (132/220/400 kV) → distribution → metering and billing
Output	electricity to domestic, industrial, commercial consumers; export in the wet season; revenue



NEA organization structure (simplified to directorate level).

2. Identification of the problem

- In FY 2072/73 NEA showed every sign of a failing public enterprise:
 - **load-shedding** up to **18 hours a day** in the dry winter;
 - **system loss 25.78%**: a quarter of energy lost, much of it theft;
 - accumulated **loss Rs 34.61 bn**; industries on diesel generators;
 - low staff morale; public anger.
- **Problem statement:** why did NEA perform so poorly, and what should it do in planning, leadership, motivation and HRD to improve?

3. Hypothesis

- **Main:** poor performance is as much a **management** failure as a shortage of generation.
- H1: planning horizon was short → fire-fighting, no storage.
- H2: leadership was weak and unstable → no accountability.
- H3: staff lacked motivators → leakage tolerated.
- H4: HRD was neglected → skill gaps in new technology.

4. Case research design

Data	Source and method
Secondary	NEA annual reports FY 2072/73–2080/81: loss, demand, supply, finance; news reports
Interviews	20 officers: MD office, 2 directorates, 6 distribution centres
Staff survey	200 staff, 5-point scale: pay, recognition, training, leadership
Consumer survey	300 households and 30 industries: supply hours, complaints
Observation	2 substations and 1 control centre (dispatch)
Analysis	trend charts; before/after 2073; fishbone of causes

5. Collection and analysis of data

- **System loss**, FY 72/73 to 80/81 (%): **25.78** → 22.90 → 20.45 → 15.32 → 15.27 → 17.18 (COVID year) → 15.38 → 13.46 → **12.73**.
- The sharpest fall came in the first three years after the new MD (2073 Bh).
- Each point of loss is roughly a billion rupees a year: NEA says leakage control earns it about Rs 12 bn a year.



Cause-and-effect analysis of NEA's performance in FY 2072/73.

(a) Planning horizon

- Found: planning mostly **short term**, one dry season at a time.
- Storage and peaking plants delayed for years → winter deficit grew with demand.
- Fix by three horizons:
 - short** (1 yr): manage peak, cut leakage, import in winter;
 - medium** (2–5 yr): transmission lines and substations, smart meters;
 - long** (10–20 yr): storage projects, cross-border lines, export.
- H1 **accepted**.

(b) Leadership

- A new MD took charge on 14 Sept 2016 (Bhadra 2073).
- Within about two months Kathmandu Valley was declared load-shedding free (Tihar, Oct 2016); nationwide by **May 2018**.
- Mostly the **same** generation capacity: the change came from **dispatch management**, ending **dedicated-feeder** privileges, leakage raids, peak-hour imports.
- Style: high concern for task **and** people → 9,9 team management on Blake–Mouton's grid (§3.3); visible, decisive, accountable.
- H2 **accepted**: leadership was the turning factor.

6. Generalization and interpretation

- The turnaround came **before** big new plants: management, not only megawatts, explains most of it → main hypothesis supported.
- Supply growth (new plants, imports) then made the gains last.
- Generalization**: other Nepali public enterprises (Nepal Telecom, Nepal Airlines, Nepal Oil Corporation) can improve the same way: stable accountable leadership, performance targets, a long planning horizon, merit-based HRD.
- Limit**: the gains depend on one leader unless built into systems; political interference is the main risk.

(c) Motivation

- Staff survey (illustrative): pay and job security rated fair; recognition and growth rated poor.
- Herzberg (§3.2): **hygiene** factors present, **motivators** absent → no dissatisfaction, but no drive either.
- What worked: loss **targets per distribution centre**, monthly ranking, public recognition, action against poor performers.
- Maslow: esteem needs met by recognition; Vroom: effort → visible result → reward raised expectancy.
- H3 **accepted**.

(d) Human resource development

- Found: little structured training; promotion by seniority; no succession plan.
- New needs: smart metering, SCADA, grid operation, cross-border trade, project management.
- Needed: training-needs analysis (§2.4), skill-based training, merit rating linked to promotion, succession planning for key posts.
- H4 **partly accepted**: training exists but is not tied to career growth.

7. Conclusions and recommendations

Area	Recommendation	Horizon
Planning	rolling 1/5/20-year plan; load forecast; storage first	all three
Leadership	fixed MD tenure; performance contract with the Board	short
Motivation	loss and service targets per centre; ranking, rewards	short
HRD	needs analysis; training academy; merit promotion; succession	medium
Technology	smart meters, SCADA, MIS dashboards (§5)	medium
Governance	shield operations from political transfers	long

Checklist (step 6)

- **Achieves:** reliable supply, loss near 10%, sustained profit, motivated staff.
- **Difficulties:** union resistance to performance ranking; political transfers.
- **Limits:** capital for storage plants; dry-season hydrology; import capacity.
- **Best alternative?** Yes: cheaper and faster than building capacity alone.

Drawbacks of this study; references

- Survey figures are illustrative; a real study would sample properly.
- References: NEA annual reports; Kathmandu Post (2016, 2017, 2019); Khabarhub (2024).

4.5 Mock Case Study 2: Recommendations to Your College (74 Ch)

PIN 1/29 74 Ch · written out in full as a model · “XYZ Engineering Campus” and all its numbers are illustrative; put your own campus’s facts in **verified/added**

Question (74 Ch, 8 marks)

Having spent more than 3 years in a particular college/campus of yours, what are the recommendations you wish to propose to the college management for the future improvement that would in turn boost up overall academic performance and image of the college?

1. Organizational background

- **XYZ Engineering Campus**, affiliated to TU, IOE; founded 2060 BS; private (company-owned) college.
- **Programmes:** BCT, BEI, BEL, BCE; about 960 students, 72 faculty (40 full time).
- **Goal:** produce competent engineers; top-10 IOE results; placement.
- **Structure:** Board → Principal → Heads of Department, Exam Section, Admin and Accounts (line and staff).
- **Financial position:** fee-funded; stable, little surplus for labs.

2. Identification of the problem

- Board pass rate fell from 74% to 62% over three years.
- Admissions fell 15%; few campus placements.
- **Problem:** what should management change to raise academic performance and the college’s image?

3. Hypothesis

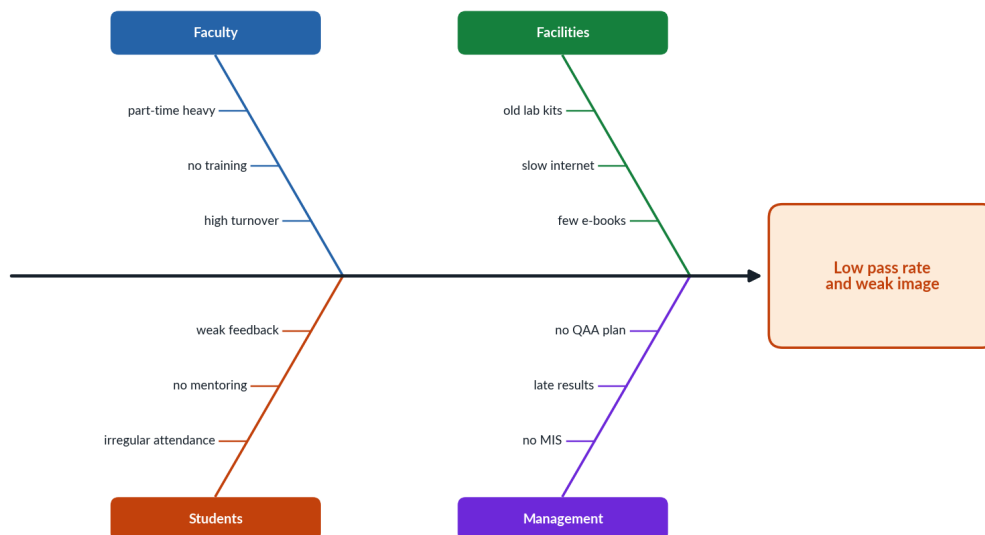
- Weak results come mainly from **faculty instability** and **poor facilities**, not from weaker students.

4. Case research design

Data	Source and method
Records	3 years of results, attendance, faculty turnover, admissions
Survey	240 students, 5-point scale on six areas
Interviews	12 faculty, 3 HoDs, the principal
Observation	labs, library, internet, hostel
Benchmark	two better-ranked IOE colleges

5. Collection and analysis of data

- Student survey, mean score out of 5:
 - Lowest: **career cell 1.8, internet 2.1, labs 2.3.**
 - Middling: library 2.9, hostel 2.6.
 - Strongest: teaching 3.4.
- Faculty turnover 25% a year; 45% of classes taken by part-timers.
- Students with over 80% attendance passed at 81%; the rest at 44%.



Cause-and-effect analysis: why results and image are falling.

6. Generalization and interpretation

- Attendance and faculty stability explain most of the pass-rate gap → hypothesis supported.
- Facilities (labs, internet) hurt practical courses and the college's image.
- Image follows results and placement: fix those, then publicize.
- Holds for most private affiliated colleges: **people and systems** before buildings.

Checklist

- Achieves:** higher pass rate, stronger admissions, better placement.
- Difficulties:** cost of full-time faculty; staff resistance to evaluation.
- Limits:** fee ceiling; TU calendar; space.
- Best alternative:** yes, most items cost little and repay through admissions.

7. Recommendations (by priority)

Area	Recommendation	When
Faculty	more full-time posts; training; merit-based pay and promotion	1 yr
Academic	attendance rule; mentoring; internal tests; remedial classes	now
Labs	replace kits; lab manuals; industry-sponsored labs	1–2 yr
Library, IT	e-journals; campus Wi-Fi; LMS for notes and assignments	1 yr
Students	career cell, internships, alumni talks, clubs	now
Research	small grants; MoUs with industry; project exhibitions	2 yr
Management	MIS for attendance, results, feedback; UGC QAA accreditation	1–3 yr
Image	website, social media, results and placement publicity	ongoing
Close: "Priorities 1–2 first; review every semester with student feedback."		

4.6 Sample Case Study Report: Industrial Visit

internal assessment · **case study report = 10 of the 20 internal marks** · the course requires one after visiting an industrial organization · written out in full as a model · "ABC Cement Industries Ltd." and every number in it are illustrative: put your own visit's facts in **verified/added**

How to use this model

- Structure = the **7-step methodology** and the **report titles** of §4.3, merged into one report.
- Chapters 2–7 = the syllabus applied to one plant:
 - Ch. 1: ownership, structure, purchasing, marketing;
 - Ch. 2: manpower, recruitment, training, job evaluation, wages;
 - Ch. 3: motivation, leadership;
 - Ch. 5: MIS.
- Chapter 8 = the **case** proper: one problem, hypotheses, data, test.
- Real report: typed, 20–30 pages, figures and tables numbered, 2–3 visit photos if the plant allows.
- Keep every number traceable: say where it came from (log, report, survey).

Front matter order

- Title page.
- Acknowledgement.
- Abstract.
- Table of contents, list of figures and tables.
- List of abbreviations.

Table of contents, and the §4.3 step each chapter fills

Chapter	Contents	Step
1 Introduction	background of visit, objectives, methodology, limitations	4
2 Organization profile	history, objectives, products, capacity, process	1
3 Ownership, structure	public limited company, org chart	1
4 Personnel management	manpower, recruitment, training, job evaluation, merit rating, wages	5
5 Motivation, leadership	employee survey, theories applied	5
6 Purchasing, marketing	materials, spares, channels, promotion	5
7 MIS	reports by level, software	5
8 The case problem	problem, hypotheses, stoppage analysis, test	2, 3, 5
9 Findings	strengths, drawbacks, generalization	6
10 Recommendations	action table, plan, checklist	7
11 Conclusion	summary; references; appendices	7

Steps: 1 background, 2 problem, 3 hypothesis, 4 research design, 5 data and analysis, 6 generalization, 7 conclusions and recommendations.

Title page

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
[Campus name]

A Case Study Report on ABC Cement Industries Ltd.

Makwanpur, Nepal

Submitted as a requirement of
Organization and Management (ME 708)

Submitted by	Submitted to
[Name], [Roll no.]	Department of [...]
[Name], [Roll no.]	[Campus]
[Date of submission]	

Acknowledgement

We thank the management of ABC Cement Industries Ltd. for permission to visit the plant on [date], and the Plant Manager, the HR Officer, the Stores Officer and the shift engineers who guided us and answered our questions. We are grateful to [teacher], Department of [...], for guidance throughout, and to the campus for arranging the visit.

Abstract

This report studies ABC Cement Industries Ltd., an integrated cement plant with a 1,200 TPD kiln. It describes the company's ownership, structure, personnel practices, motivation, leadership, marketing and MIS. The case problem is low capacity use: 64–71% over four years. Kiln logs for FY 2082/83 show 1,490 hours of unplanned stops, 69.1% of them from power trips and breakdowns. The report recommends a standby power supply, preventive maintenance, spares planning, training and regularizing contract operators, and an ERP, aiming at 80% capacity use in two years.

Chapter 1. Introduction

1.1 Background of the visit

- One-day visit on [date]: 32 students of IV-I and 2 teachers.
- Why this plant: a **continuous-process** industry → every O&M topic is visible: structure, shifts, manpower, maintenance, marketing.

1.2 Objectives of the study

1. Study the organization: history, ownership, structure.
2. Study personnel practices: manpower planning to wages.
3. Study motivation, leadership and MIS in practice.
4. Identify the plant's main management problem, analyse its causes, recommend solutions.

1.4 Limitations

- One-day visit; sample of 60 out of 412 employees.
- Some cost and wage data confidential → published figures only.
- Respondents chosen by the HR section → possible bias.

1.3 Methodology (case research design)

Method	What and from whom
Observation	plant tour: quarry, crusher, kiln, central control room, packing, stores
Interviews	Plant Manager, HR Officer, Stores Officer, Marketing Officer, 2 shift engineers
Questionnaire	60 employees (40 permanent, 20 contract), 5-point scale; Appendix A
Records	annual reports FY 2079/80–2082/83, kiln stop-page log, maintenance log, HR records
Analysis	tables, bar chart, cause-and-effect (fishbone) diagram, hypothesis test

Chapter 2. Organization Profile

2.1 History

- **2065 BS:** registered as a private limited company (Companies Act 2063).
- **2068:** commercial production, grinding purchased clinker.
- **2072:** own 1,200 TPD kiln commissioned → integrated plant.
- **2076:** converted to public limited; IPO of 30% of shares; listed on NEPSE.
- **2080:** cement mill expanded to 1,500 TPD.

2.2 Vision and objectives

- **Vision:** most trusted cement brand of central Nepal.
- **Objectives:**
 - produce cement to Nepal Standard (NS) quality;
 - run the plant at 85% of capacity or more;
 - earn profit, pay regular dividend;
 - employ local people, substitute imports.

2.4 Input, process, output

Stage	ABC Cement
Input	limestone (own quarry, 12 km), clay, iron ore, gypsum and coal (imported via Birgunj), fly ash, power (NEA grid, 33 kV), water, 412 staff, capital
Process	quarry → crusher → raw mill → preheater and kiln (1,450 °C) → clinker → cement mill → packing (figure)
Output	OPC and PPC in 50 kg bags; dust (held by bag filters), waste heat; revenue, taxes, jobs

2.3 Products, capacity, performance

Item	Value
Products	OPC (53 grade), PPC; brand “ABC Strong”
Clinker capacity	1,200 TPD = 396,000 t a year (330 days)
Cement capacity	1,500 TPD = 495,000 t a year
Employees	412 (248 permanent, 164 contract)
Working	3 shifts of 8 h; kiln runs round the clock
Finance, FY 82/83	sales Rs 4.2 bn; net profit Rs 180 million; 10% cash dividend

FY (BS)	Cement, '000 t	Capacity used
2079/80	351	70.9%
2080/81	332	67.1%
2081/82	318	64.2%
2082/83	342	69.1%

Capacity used = production ÷ 495,000 t.

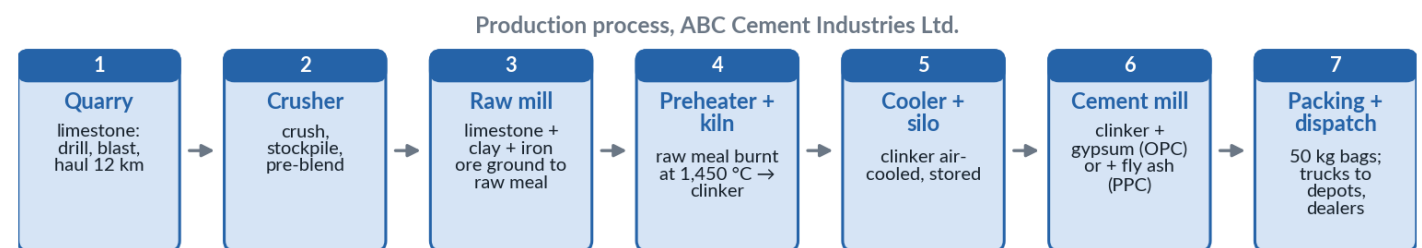


Figure 2.1: Dry-process production flow of the plant.

Chapter 3. Form of Ownership and Organization Structure

3.1 Form of ownership: public limited company (§1.4)

- Registered with the OCR under the Companies Act 2063; listed on NEPSE.
- Shares: promoters 70%, general public 30%.
- Separate legal entity, limited liability, perpetual succession.
- AGM elects a Board of 7 directors (5 promoter, 2 public).
- Board appoints the MD as chief executive.
- Disclosure to SEBON and NEPSE; yearly external audit.
- Suits the plant: large capital raised from the public, professional management.

3.2 Organization structure (§1.5)

- **Type:** line and staff.
 - line: MD → Plant Manager → department heads → shift engineers → supervisors → operators;
 - staff: Safety and Environment advises the MD (dashed in the chart).
- **Departmentation:** by function.
- **Span of control:** MD 7; shift engineer about 12 operators.
- **Centralization:** purchases above Rs 5 lakh and all hiring decided by the MD; shift decisions delegated.
- **Levels:** 6, from Board to operator.

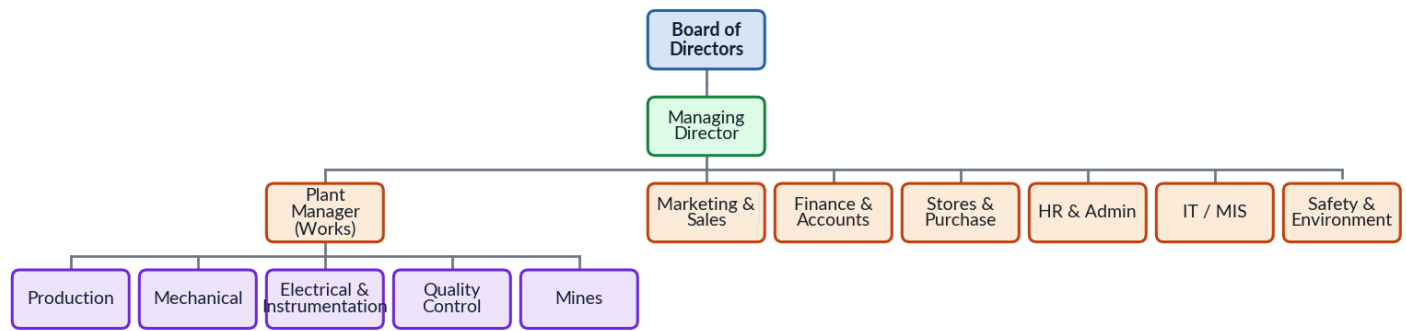


Figure 3.1: Organization chart of ABC Cement Industries Ltd.

Chapter 4. Personnel Management

4.1 Personnel policy and manpower planning (§2.2, §2.3)

- Written HR bylaws under the Labour Act 2074:
 - 8 h a day, 48 h a week;
 - overtime at 1.5 times the wage rate;
 - probation up to 6 months.
- Manpower plan made yearly with the budget, from the production target.
- No long-term forecast: posts filled only after a vacancy arises.

4.2 Recruitment and selection (§2.4)

- **Internal:** promotion to supervisor and officer posts.
- **External:**
 - newspaper and website advertisements;
 - campus (IOE) for trainee engineers;
 - employee referrals;
 - labour contractor for packing and loading labour (contract).
- **Selection:** screening → written test (engineers, officers) → panel interview (Plant Manager, HR, department head) → medical test → appointment → 6-month probation.
- 60% of staff from Makwanpur, per agreement with local community.

Manpower, FY 2082/83

Department	Staff
Production	142
Mechanical, Electrical and Instrumentation	88
Mines (quarry)	46
HR and Admin (incl. security, drivers)	32
Marketing and Sales	26
Quality Control (laboratory)	24
Finance and Accounts	20
Stores and Purchase	18
Safety and Environment	8
IT / MIS	8
Total	412
Level	Staff
Top (MD, Plant Manager, GMs)	6
Middle (department heads, engineers, officers)	38
Supervisory	72
Operators and workers	296

4.3 Training and development (§2.5)

- **Induction:** 3 days, including safety.
- **On the job:** new operator works under a senior operator for 3 months.
- **Off the job:** equipment supplier trains engineers on the kiln and DCS; plant visits abroad for 2 engineers a year.
- FY 2082/83: 14 programmes, 186 participants.
- Contract workers: safety briefing only.
- No training-needs analysis, no skill matrix.

4.4 Job evaluation and merit rating (§2.6)

- **Job evaluation:** grading method → 10 grades, G1 (helper) to G10 (department head).
- Written job descriptions for officer posts only.
- **Merit rating:** yearly, graphic rating scale, rated by immediate supervisor on 6 factors:
 - quality and quantity of work;
 - attendance, safety;
 - teamwork, initiative.
- Used for the annual increment only: promotion mostly by seniority.

4.5 Wages, incentives and welfare (§2.7)

- **Time wage:** monthly salary by grade + yearly grade increment.
- **Allowances:** night-shift, dearness, festival allowance of one month's basic salary (Dashain).
- **Production incentive:** paid when monthly output exceeds 90% of target → earned in only 3 of 12 months of FY 2082/83.
- **Bonus:** 10% of net profit under the Bonus Act 2030 → Rs 18 million shared in FY 2082/83.
- **Social security:** SSF contributions under the Contribution-based Social Security Act 2074; medical insurance.
- **Welfare:** canteen, staff bus, quarters for 90 families.
- Contract workers paid by the contractor at about the minimum wage, few benefits.

4.6 Industrial relations (§2.8)

- 2 unions of permanent staff; collective agreement every 2 years.
- 3-day strike in 2081 over regularizing contract workers → settled with a promise to review.

Chapter 5. Motivation and Leadership

5.1 Employee survey (Appendix A)

Factor (mean, 1-5)	Permanent (40)	Contract (20)
Job security	4.1	1.9
Pay and allowances	3.4	2.6
Supervision	3.3	2.9
Working conditions, safety	3.2	2.8
Training and growth	2.7	1.6
Recognition	2.4	2.0
HR records	Permanent	Contract
Turnover a year	4%	28%
Absenteeism	3%	9%

5.2 Analysis (§3.2)

- **Herzberg:**
 - permanent staff: hygiene factors met (security 4.1, pay 3.4), motivators weak (recognition 2.4, growth 2.7) → not dissatisfied, but not driven;
 - contract staff: even hygiene factors unmet (security 1.9) → dissatisfaction → 28% turnover.
- **Maslow:** contract workers stuck at safety needs; permanent staff seek esteem.
- **Vroom:** incentive earned in 3 months of 12 → effort not seen to bring reward → low expectancy.

5.3 Leadership (§3.3)

- **MD:** sets targets, decides fast; autocratic on purchase and hiring, participative in the monthly production review.
- **Shift engineers:** task-oriented, 9,1 on Blake-Mouton's grid: output first, little coaching.
- **Plant Manager:** visits every shift, listens; rated highest by staff → closest to 9,9 team management.

Chapter 6. Purchasing and Marketing (§1.6)

6.1 Purchasing and stores

- **Limestone:** own quarry, reserve for about 30 years.
- **Clay, iron ore:** local suppliers on yearly contract.
- **Coal:** imported by tender from 2 suppliers, lead time 6–8 weeks.
 - stock target 45 days;
 - fell to 12 days twice in FY 2082/83 → kiln slowed.
- **Spares:** mostly imported, lead time 3–4 months.
 - bought only after a breakdown: no minimum stock → long stops.
- Purchase committee for orders above Rs 5 lakh; quotations or tender.

6.2 Marketing

- **Channel:** plant → 4 depots → 180 dealers → retailers → users.
- **Direct sales** to projects (hydropower, roads, bridges): 20% of volume.
- **Market:** Nepal self-sufficient in cement, many plants → oversupply, price competition.
- **Price:** follows market leader; volume discounts to dealers.
- **Promotion:** hoardings, FM radio, dealer meets, mason training programmes.
- **Seasonality:** demand falls in monsoon (Ashad to Bhadra) and at Dashain–Tihar → clinker yard fills → kiln stopped 240 h.

Chapter 7. Management Information System (Ch. 5)

7.1 Reports by level

Level	Report	How often
Operational	shift log, DCS trends, stores issue slips	every shift
Middle	daily production report (DPR), weekly maintenance report	daily, weekly
Top	monthly MIS report to MD; board report; annual report	monthly, quarterly, yearly

7.2 Findings

- DCS in the central control room logs every kiln parameter.
- Accounts on accounting software; stores, maintenance and sales on separate spreadsheets.
- No link → maintenance cannot see spares stock → learns of a shortage only after a breakdown.
- DPR reaches the MD at 10 am next day: no real-time KPI.

Chapter 8. The Case Problem: Low Capacity Use

8.1 Identification of the problem

- Capacity used 64–71% for four years (section 2.3), against the 85% the budget assumes.
- Every idle kiln hour loses 50 t of clinker (1,200 TPD ÷ 24): the loss cannot be made up later.
- **Problem statement:** why does the plant use only about two-thirds of its capacity, and what should management do?

8.2 Hypotheses

- **H1:** unplanned kiln stops from power trips and breakdowns are the main cause.
- **H2:** weak off-season demand adds to it.
- **H3:** manpower practices (contract turnover, little training) cause many of the breakdowns.
- **H4:** raw-material shortage is the main cause.

8.3 Data: kiln stops, FY 2082/83

Cause	Hours	% of unplanned
Power trips, voltage dips	610	40.9
Mechanical breakdowns	420	28.2
Clinker yard full (market)	240	16.1
Coal, material shortage	150	10.1
Others	70	4.7
Unplanned total	1,490	100.0
Planned shutdown	360	
All stops	1,850	

8.4 Analysis

- Kiln ran $8,760 - 1,850 = 6,910$ h $\rightarrow$ run factor 78.9%.
- Clinker lost to unplanned stops: $1,490$ h $\times 50$ t = $74,500$ t = 18.8% of the 396,000 t capacity.
- Power + breakdowns = $1,030$ h = **69.1%** of unplanned stops.
- **Power:** 146 trips in the year.
 - each trip $\rightarrow$ about 4 h to restart and stabilize the kiln;
 - a 5-minute trip costs hours of output.
- **Breakdowns:** 29 in the maintenance log.
 - 11 traced to operating error (missed lubrication, overloading);
 - 9 of those 11 on shifts run mainly by contract operators;
 - 6 waited for imported spares $\rightarrow$ 180 of the 420 h.
- **Market:** yard-full stops all fell in monsoon and festival months.

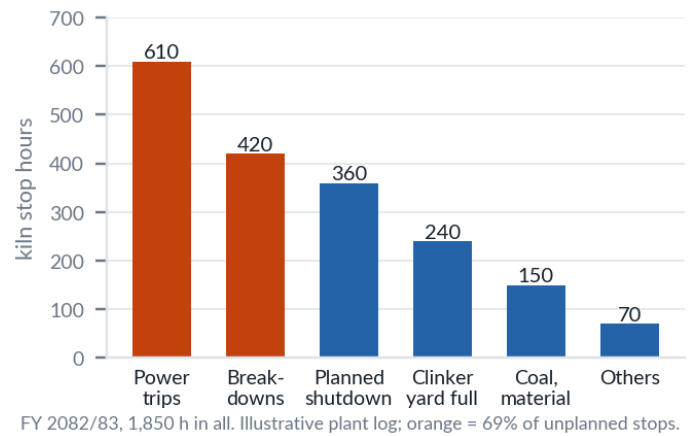


Figure 8.1: Kiln stop hours by cause.

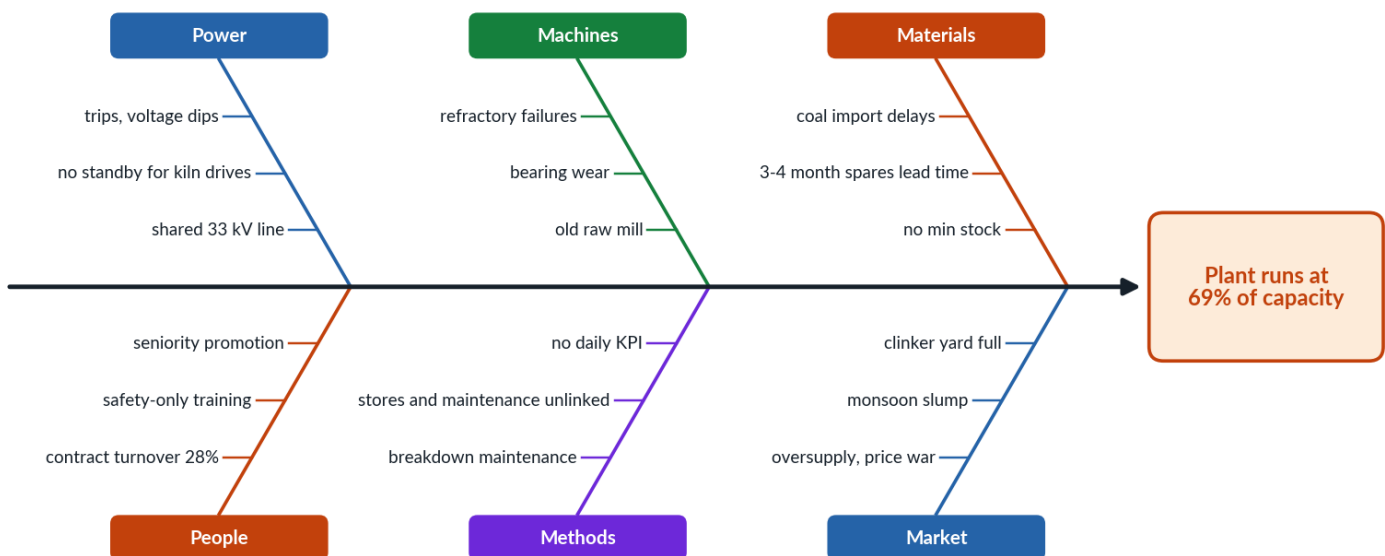


Figure 8.2: Cause-and-effect analysis of low capacity use.

8.5 Testing the hypotheses

H	Evidence	Result
H1	power trips + breakdowns = 1,030 h = 69.1% of unplanned stops	accepted
H2	clinker yard full 240 h = 16.1%, all in monsoon and festival months	partly accepted
H3	11 of 29 breakdowns from operating error, 9 of them on contract-run shifts; contract turnover 28%	accepted
H4	material shortage 150 h = 10.1%	rejected as main cause

Chapter 9. Findings, Generalization and Interpretation

9.1 Strengths

- Integrated plant with own quarry → low raw-material cost.
- NS-marked product; brand known in 30 districts.
- Profitable; regular 10% dividend.
- Good pay and welfare for permanent staff; lawful HR bylaws.
- Modern kiln with DCS control.

9.3 Generalization and interpretation

- Capacity is lost mainly **inside the plant gate** (69.1% of unplanned stops), not in the market → a **management** problem.
- Holds for continuous-process plants (cement, steel, paper): a lost kiln hour is never recovered → reliability of power, maintenance and people decides output.
- Limit: one plant, one year of logs.

9.2 Drawbacks (problems found)

1. Breakdown maintenance instead of preventive.
2. No standby power for kiln drives.
3. Spares bought after failure, not stocked.
4. Two-tier workforce: contract operators insecure, untrained, 28% turnover.
5. Promotion by seniority; merit rating used for increments only.
6. Production incentive too hard to earn (3 of 12 months).
7. MIS not integrated; no real-time KPI.
8. No off-season marketing plan.

Chapter 10. Recommendations

Area	Recommendation	Horizon	Responsible
Power	dedicated industrial line from NEA; standby DG set for kiln drive and fans; voltage stabilization	short to medium	Electrical
Maintenance	preventive schedule; vibration monitoring and kiln shell scanner; annual shutdown in monsoon, when demand is low anyway	short	Mechanical
Stores	VED and ABC analysis of spares; min-max stock of critical spares; rate contracts with suppliers	short	Stores and Purchase
People	training-needs analysis, skill matrix, operator certification; regularize long-serving contract operators; merit-weighted promotion	medium	HR
Motivation	shift incentive on kiln run hours; best-shift recognition; suggestion scheme	short	HR, Plant Manager
MIS	ERP linking stores, maintenance, production and sales; daily KPI dashboard	medium	IT / MIS
Marketing	new districts, project sales, pre-monsoon dealer stocking schemes, mason meets	medium	Marketing

Expected result

- Unplanned stops 1,490 h → under 800 h a year.
- Capacity used 69.1% → **80%** within two years.

Checklist (step 6)

- **Achieves:** about 80% capacity use, fewer breakdowns, stable workforce.
- **Difficulties:** capital for DG set and ERP; contractor and union resistance.
- **Limits:** NEA grid capacity, market demand, import lead times.
- **Best alternative?** Yes: more output from the existing plant costs far less than new capacity.

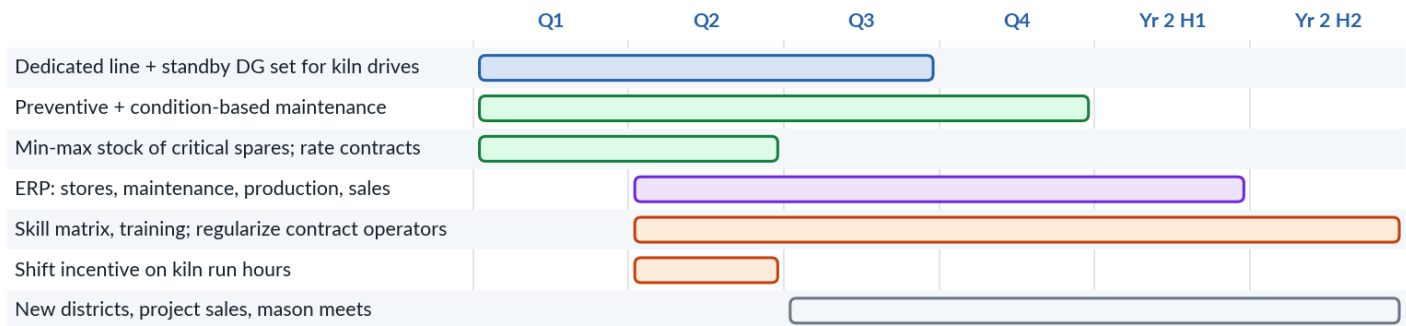


Figure 10.1: Implementation plan over two years.

Chapter 11. Conclusion

- ABC Cement = profitable public limited company, line-and-staff structure, functional departments.
- Personnel practices meet the law for permanent staff; the contract workforce is left out.
- Main problem: only about 69% of capacity used.
 - caused mainly by unplanned kiln stops (power, breakdowns);
 - not by demand or materials.
- Remedy lies in management: preventive maintenance, spares planning, trained and stable operators, integrated MIS.
- The visit showed O&M theory at work in a real plant.

References

1. ABC Cement Industries Ltd., Annual Reports FY 2079/80–2082/83.
2. Government of Nepal: Labour Act 2074; Bonus Act 2030; Companies Act 2063; Contribution-based Social Security Act 2074.
3. Adhikari, B., *Organization and Management* slide notes.
4. Lecture notes, Organization and Management (ME 708), IOE.

Appendix A: Questionnaire (sample)

- **Part A:** department; permanent or contract; years of service.
- **Part B:** rate 1 (very poor) to 5 (very good):
 - job security; pay and allowances;
 - supervision; working conditions and safety;
 - training and growth; recognition.
- **Part C** (open):
 - What stops the kiln most often, in your experience?
 - What one change would most improve your work?

Appendix B: Interview checklist

- **Plant Manager:** capacity, stops, maintenance practice.
- **HR Officer:** manpower, recruitment, training, appraisal, wages.
- **Stores Officer:** lead times, stock policy.
- **Marketing Officer:** channels, prices, seasons.
- **IT Officer:** reports, software.

4.7 PYQ Mapping

Theory questions, covered in this document

- **What is a case study, why conducted, objectives, value** → §4.1. 8 papers; three of them as a 4-mark short note.
- **Types of case studies** 80 Ba 2 → §4.1.
- **Phases and steps, with an example from your field** → §4.2. 82 Bh 4, 81 Bh 4, 81 Ba 5, 80 Bh 8, 79 Bh 8
- **Prepare a case study of NEA / Nepal Telecom** 72 Ch 8 → §4.3 structure, §4.4 full model.
- **Recommendations for your college** 74 Ch 8 → §4.3 structure, §4.5 full model.
- **Case study report** (internal assessment, 10 marks, not examined) → §4.6 complete sample.
- 73 Ch and 71 Ch pair case-study objectives with MIS needs; the MIS half is in Ch. 5.

Numerical content

- This chapter has no numerical component.

Sources

- Adhikari pp259–272 and AJ Sir's 7-slide deck cover all five syllabus topics between them. S.K. Joshi's textbook follows an older syllabus and has no case-study chapter, so nothing was taken from it here.
- No local source has a case-study diagram: every figure in this chapter is generated (ch4fig.py). Only the process flows, the NEA org chart and the two fishbones are kept: the diagrams an answer would draw.

- NEA facts: NEA annual reports as reported by Kathmandu Post (16 Aug 2016, 18 Aug 2017, 15 Feb 2019) and Khabarhub (Nov 2024).

5. Management Information System

5 hrs · asked in **25 of 29 papers** · 8 syllabus topics · **248 marks, 10.5% of the archive** → about 6 of every 80 marks: one 8-mark question, split 3+5 or 4+4.

READ THIS FIRST

MIS questions almost always open with “**define MIS / why is MIS needed?**” (17 papers) and add one of three second halves:

1. **Types of information systems** (TPS, MIS, DSS, ESS) and DSS vs MIS (7 papers) → one table in §5.3.
2. **Information support for functional areas** (5 papers) or **at different levels** → one pyramid figure in §5.3 answers both.
3. **Planning and decision making** (7 papers), **computers and MIS**, **database** and **networking** systems → §5.2 and §5.4.

The 2065–2068 papers asked “hierarchy of information needs” and “information architecture”; only S.K. Joshi’s textbook covers them, and they are summarized from it.

Tier chips count **papers out of 29**: **TOP n/29** 8+ **HOT n/29** 4–7 **PIN n/29** 1–3 inside an 8-mark question **LOW n/29** 1–3, short note only syllabus, no PYQ yet never set. Bold year = Regular exam.

5.1 Data, Information and MIS

Define MIS. Why is MIS needed? Its functions and importance **2 3 4 5**

TOP 17/29 82 Ba, 81 Ba, 80 Bh, 79 Ba, **76 Ch**, 76 Ash, 74 Ash, 73 Shr, 71 Ch, 70 Ch, 70 Asa, 69 Ch, 68 Ch, 68 Ba, 67 Asa, 66 Bh, 65 Shr · “needs for MIS” short note in 68 Ch, 66 Bh, 65 Shr; also the second half of 73 Ch and 71 Ch (filed with case studies)

- The three words:
 - **Management**: directing an enterprise toward objectives (planning, organizing, coordinating, controlling).
 - **Information**: processed, orderly selected data that gives knowledge for decisions.
 - **System**: interacting procedures forming an organized whole.
- **MIS**: “a system of obtaining, abstracting, storing and analysing data to produce information for planning, controlling and decision making, given to managers at the time they can best use it.”
- Henry C. Lucas: “a set of organized procedures which, when executed, provide information to support decision making.”
- Uses computing and communication technology; draws on internal and external sources; covers past, present and projected activity.

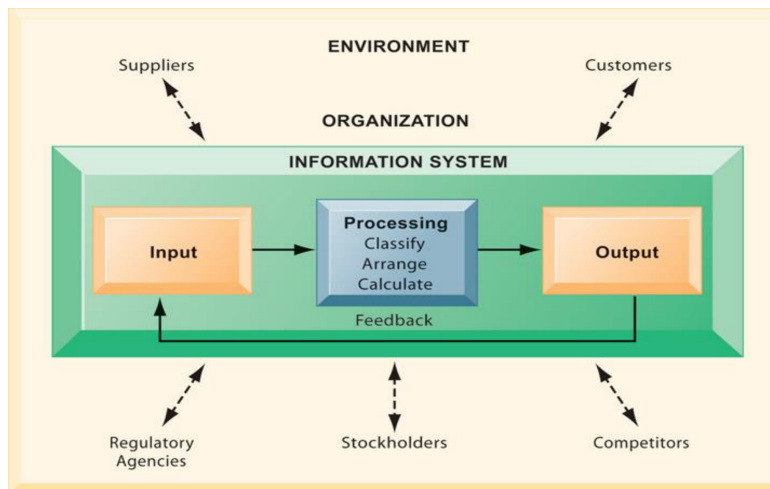
Functions

1. **Data capture**: from internal and external sources.
2. **Processing**: classify, calculate, compare, summarize.
3. **Storage and retrieval**.
4. **Determining information needs**: what, how much, when, for whom.
5. **Evaluation and abstraction**: judge reliability, edit, condense.
6. **Dissemination**: right information, right manager, right time.
7. **Prediction, planning, control** support.

Adhikari’s four, with his questions

- **Determination** of needs: what? how much? how, when, by whom?
- **Evaluation**: how much confidence can be placed in it?
- **Abstraction**: editing and reducing incoming information.
- **Dissemination**: right information, right manager, right time.

The information system inside the organization, inside its environment



The organization's information system within its environment.

Need for MIS

- Every manager decides on information; more complex firms → more data than a person can handle.
- **Production managers:** production, labour, machine, overhead costs; capacity.
- **Marketing managers:** sales trends, market research, selling costs, new products.
- **Personnel managers:** turnover, absenteeism, skills, wage levels, labour market.
- **Why MIS in addition to task software (79 Ba):** payroll, billing, inventory packages each serve one task; MIS **integrates** their data into summary and exception reports across the firm, for decisions no single package supports.

Importance

- The "heart" of the firm: timely, relevant information for **decisions**.
- **Measures performance against goals** → control.
- Supports **planning**, forecasting, coordination across departments.
- Reduces uncertainty and cost of information; keeps it current.
- Stores confidential information securely for future use.
- Competitive advantage: faster response to market.
- Eg a bank's MIS flags branches whose loan defaults exceed target; an airline's reservation data sets fares by demand.
- Also:
 - Brings new facts to management's notice.
 - Supplies information at a **reasonable cost**.
 - Focuses on decisions with an attractive **benefit/cost ratio**.
 - Supplies reliable, logical data for every management function.

Data vs information vs knowledge vs wisdom; how data and information are used in an office 4

PIN 3/29 82 Ba, 78 Bh, 75 Ash

- **Data:** raw, unprocessed facts and figures; no meaning alone. Eg 45, 62, 38.
- **Information:** data processed into a meaningful, useful form. Eg “average daily sales this week: Rs 48,000, down 12%”.
- **Knowledge:** information combined with experience and understanding: knows *how*. Eg “sales drop every exam month”.
- **Wisdom:** applying knowledge with judgement to decide well: knows *why* and *what is best*. Eg “run a student discount in exam months”.
- DIKW pyramid (right): each level adds context and value.

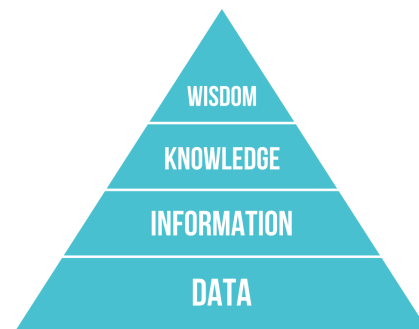
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The contrast in detail

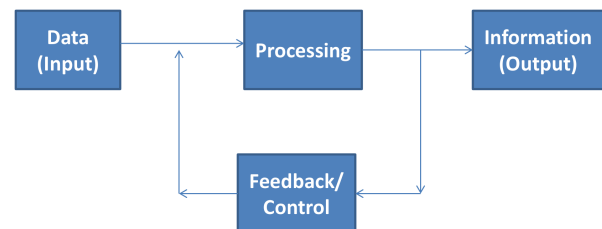
- **Data:**
 - Uninterpreted raw statement of facts.
 - Non-random symbols (words, values, figures) representing what has happened.
 - From research or observation. Eg payroll, bank statement.
- **Information:**
 - Facts organized to give the desired knowledge.
 - Data interpreted and understood by the recipient.
 - Can mean different things to different people.

Qualities of good information

- Accurate; timely; complete; concise.
- Relevant; at the right frequency; understandable.
- Worth its cost: costs less than its value.



DIKW pyramid. Wikimedia Commons (Longlivetheux), CC BY-SA 4.0.



Data becomes information through processing, with feedback.

Use in an office (78 Bh, 75 Ash)

- **Data** recorded daily: attendance, invoices, receipts, stock movements, letters.
- Processed into **information**: payroll, monthly accounts, stock reports, pending-file lists.
- Used for: paying staff, ordering supplies, answering customers, monitoring work, reporting to management, planning budgets.

5.2 MIS for Planning, Decision Making and Management Levels

Role of information in the planning process and in decision making; information system is vital for planning and control 4 5 6 8

HOT 7/29 76 Ch, 76 Ash, 68 Ch, 68 Ba, 67 Asa, 66 Bh, 65 Shr · short note “information system for planning process” in 68 Ba

Information system for planning

- Planning answers: **where are we? where do we want to go? how do we get there? when? who? how much will it cost?**
- MIS supplies:
 - Past performance: where we are.
 - Forecasts of demand, costs, market, technology: where to go.
 - Resource data: how, who, at what cost.
- Long-range (3–5 year) plan → yearly operating plan → individual targets: MIS tracks each.
- Planning is continuous; MIS keeps the plan a “living document” by feeding actual results back.

Vital for control (68 Ch)

- Control = compare actual with plan → correct. Without timely information there is nothing to compare.
- Exception reports highlight only deviations (management by exception).

Information system for decision making

- Decision = choosing the best alternative; more and better information → better choice.
- Steps supported:
 1. Identify the problem: exception reports.
 2. Generate alternatives: data, models.
 3. Evaluate: what-if analysis, DSS.
 4. Choose the best.
 5. Monitor: feedback reports.
- **Strategic:** future-oriented, long range, unstructured: new plant, new market.
- **Tactical:** short term: budgets, fund flow, personnel problems, product improvement.
- **Operational:** structured, routine: inventory, scheduling, allocating workers.

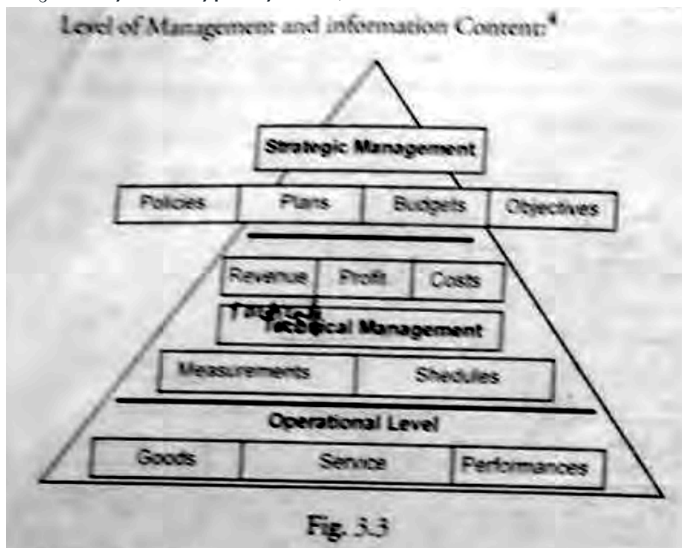
Value of MIS in the planning process (76 Ch, 76 Ash)

- **Accurate forecasts** of demand, cost, prices.
- **Realistic targets** based on past performance.
- **Scenario analysis:** what if sales fall 10%?
- **Early warning** when plans slip.
- **Coordination** of departmental plans from one shared database.

Hierarchy of information needs; organizational structure and MIS; MIS at different levels of management 5 8

HOT 4/29 80 Ba, 68 Ch, 66 Bh, 65 Shr · from S.K. Joshi, Ch. 3.1.1

- Managers at different levels need **different kinds** of information at **different intervals**: the **hierarchy of information needs**. MIS is a pyramid matching the organization's levels.
- Operational (lower)**: day-to-day production of goods and services.
 - Detailed, internal, precise, historical data.
 - Hourly or daily: raw material needs, work schedules, output vs schedule.
- Tactical (middle)**: puts top's plans into operation.
 - Summaries: schedules, revenue, profit, cost, other economic measures.
 - Weekly or monthly; used to set controls and allocate resources.
- Strategic (top)**: corporate policy, growth, survival.
 - Aggregated, future-oriented, much of it **external** (competitors, suppliers, consultants, media).
 - Quarterly or yearly: are goals met, is the firm profitable, what should change.
- Bottom of the pyramid: **transaction processing** data feeding all levels.
- For "MIS at different levels" (80 Ba), draw the pyramid in §5.3: system type by level, across the functional areas.



Level of management and information content: policies, plans, budgets at the top; revenue, profit, costs, measurements, schedules in the middle; goods, services, performance at the bottom. S.K. Joshi, Fig. 3.3.

Why the hierarchy is necessary (68 Ch)

- Each level gets **only what it needs**: no overload at top, no gaps below.
- Detail is condensed upward (reports), direction flows downward (plans).
- Matches decision type to information type.

Organizational structure and MIS

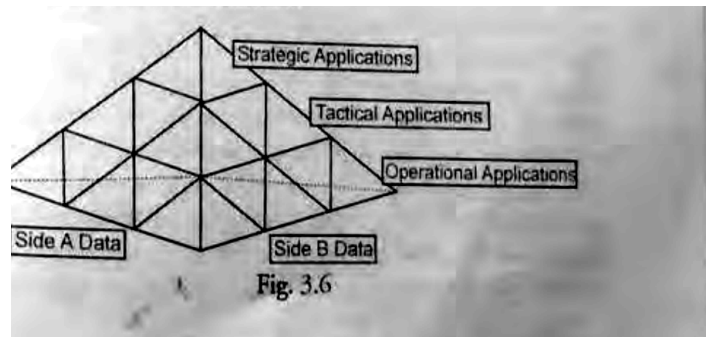
- Nature of the firm (profit or not, goods or services, single site or global) decides its information needs and systems.
- MIS draws on organization concepts: structure, communication, power, decision making, motivation, leadership.
- Adhikari: MIS is a pyramid of **four levels** of information resources: strategic, tactical, operational control, transaction processing. Since decisions are specific to levels, so are information needs.

Level	Scope	Source	Frequency
Strategic	Wide, summary, future	Mostly external	Quarterly, yearly
Tactical	Summaries, trends	Internal	Weekly, monthly
Operational	Narrow, detailed, current	Internal	Hourly, daily

Information architecture 3 4

PIN 2/29 68 Ba, 65 Shr · from S.K. Joshi, Ch. 3.2

- From Greek *archos* (chief) + *tekton* (carpenter): design and structure.
- Information architecture = **how the parts of an information system interconnect and communicate**: the general **flow** of information, how **databases interrelate**, and how applications make access easier for managers.
- Also (AJ Sir): organizing, structuring and labelling content so users find information and complete tasks.
- **Anthony's triangle**: strategic, tactical, operational applications stacked by level.
- Add **divisions** (production, marketing, accounting) → a two-dimensional structure (figure below).
 - Each division keeps its unique data.
 - Common events (a sale) create **shared data** → a **shared corporate database**.
- Traditional bureaucracy: information flows up, over, down; divisions rarely talk directly. Good architecture removes these barriers.



Anthony's triangle extended across divisions: side A and side B data meet in shared data. S.K. Joshi, Fig. 3.6.

5.3 Types of Information Systems and Functional Support

Classification of information systems (TPS, MIS, DSS, ESS) and their support to managers; DSS vs MIS; EIS for top managers 2 3 4 5 6

HOT 7/29 82 Bh, 80 Bh, 79 Ba, 78 Bh, 75 Ash, 71 Ch, 69 Ch

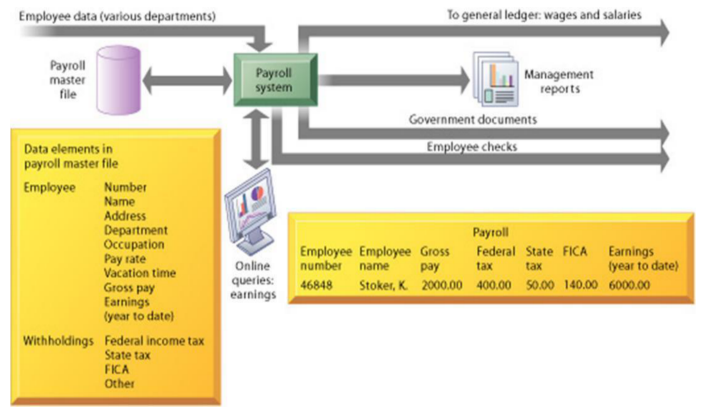
- **Information system**: hardware, software, data, people and procedures that produce information for day-to-day, short-range and long-range activities.

TPS

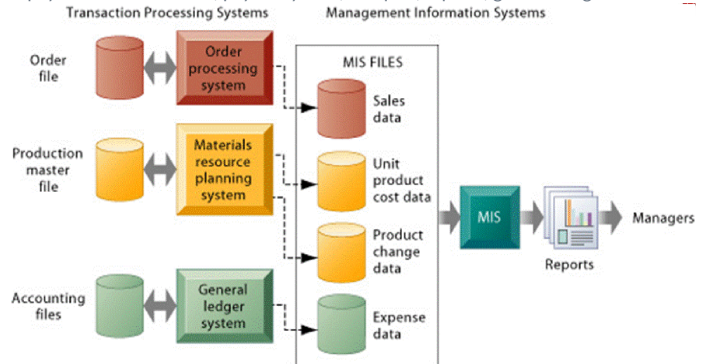
- Captures and processes data from **day-to-day transactions**: deposit, payment, order, reservation. Serves the operational level.
- Two modes:
 - **Online**: each transaction processed at once (deposits, reservations); most TPS today.
 - **Batch**: transactions grouped and run later (paychecks, invoices); efficient for bulk.

MIS: three kinds of report

- Accurate, timely, organized information to decide, solve problems, supervise, track progress; reports regularly, so also **management reporting system**.
- **Detailed**: confirms each transaction. Eg detailed order report.
- **Summary**: totals, tables, graphs. Eg inventory summary.
- **Exception**: only what is out of range. Eg items below reorder level.



A payroll TPS: master file, payroll system, cheques, reports, general ledger.



TPS files (orders, materials, ledger) feed MIS files and reports to managers.

Basis	TPS	MIS	DSS	ESS / EIS
Level served	Operational	Middle (management)	Middle, professionals	Top (strategic)
Inputs	Transactions, events	Summarized transaction data	Transaction data + models, external data	Aggregate internal + external data
Processing	Sort, list, update, batch or online	Simple models, routine reports	Interactive, what-if analysis	Interactive, graphics, drill-down
Outputs	Detailed reports, lists	Summary and exception reports	Decision analysis	Projections, dashboards
Decisions	Structured	Structured, recurring	Semi-structured	Unstructured
Example	Payroll, order entry, billing	Annual budgeting, sales by region	Contract cost analysis, production scheduling	5-year operating plan

DSS vs MIS (78 Bh, 75 Ash)

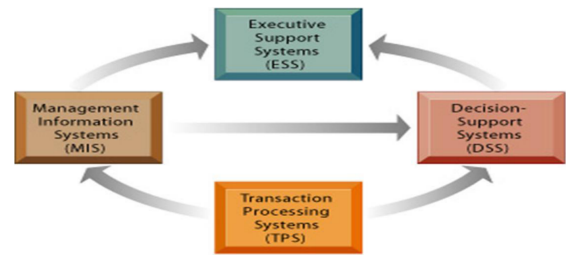
Basis	MIS	DSS
Purpose	Routine reporting, control	Support a specific decision
Problems	Structured	Semi-/unstructured
Output	Fixed, periodic reports	Ad hoc, interactive analysis
User role	Receives reports	Manipulates data and models
Data	Internal, past	Internal + external, future
Flexibility	Low	High (what-if)

How production managers use DSS in addition to MIS (82 Bh)

- MIS tells **what happened**: output, scrap, machine downtime reports.
- DSS asks **what if**: best production schedule and product mix (linear programming), make-or-buy, capacity expansion, inventory reorder levels, maintenance timing, with simulations of cost and delay.

Significance of EIS for top managers (79 Ba)

- One-screen dashboards of key indicators; drill-down to detail.
- External data: competitors, markets, economy.
- Fast, easy to use without technical skill.
- Supports strategic decisions and early warning; less dependence on staff reports.



TPS feeds MIS and DSS; all feed ESS.

Support to managers

- TPS: records every transaction; the data source for all others.
- MIS: regular detail, summary and exception reports → control (also called management reporting system).
- DSS: models + data to decide non-routine problems.
- ESS: strategic view for top managers.
- **Expert system**: knowledge base + inference rules imitate an expert (Eg medical or fault diagnosis).
- Modern software integrates them (ERP): firms consolidate TPS, MIS and DSS into one integrated system.

DSS: data from inside and outside

- **Internal**: sales, manufacturing, inventory, finance.
- **External**: interest rates, population trends, construction costs, raw material prices.
- Eg a voyage-estimating DSS runs models over ship, port, fuel and charter-cost files.

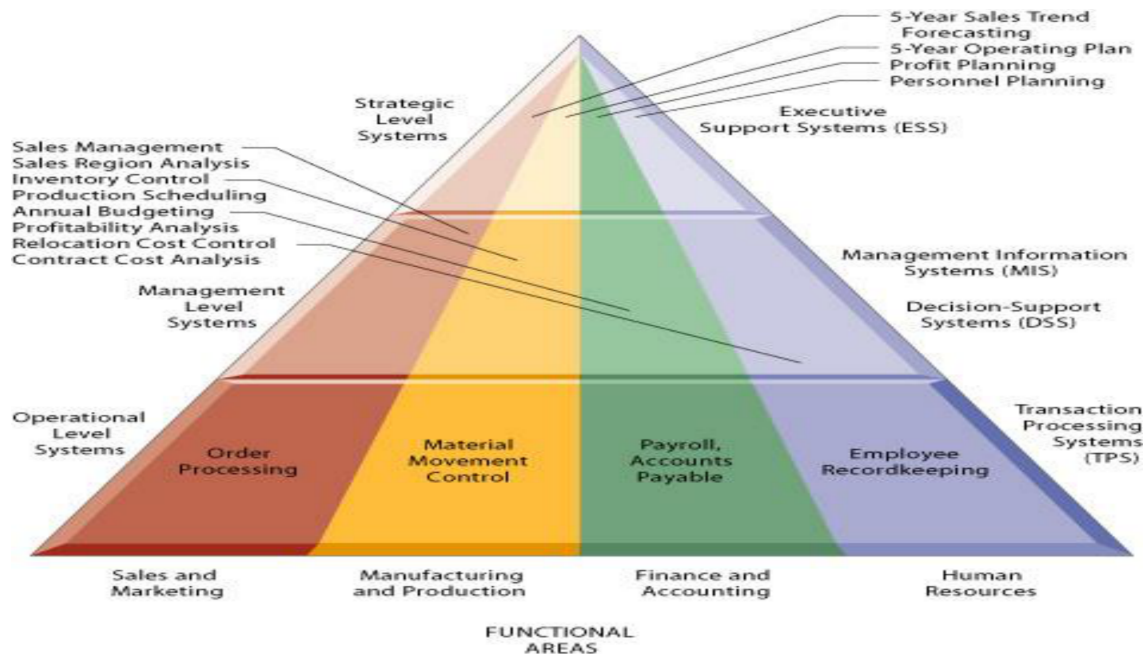
Expert system: how it reasons

1. User describes the situation.
2. Inference rules make logical judgements.
3. Knowledge base holds experts' knowledge and experience.
4. System gives advice or a decision.

Information support for functional areas of management; how MIS is used at different levels 4 5 6

8

HOT 5/29 74 Ch, 73 Shr, 72 Ka, 70 Ch, 70 Asa · 80 Ba's "MIS at different levels" is answered by the same figure



Four functional areas (columns) crossed with three levels (rows): each cell is the information system that supports it.

Sales and marketing

- TPS: order processing, invoicing.
- MIS: sales by region, salesperson, product; sales management.
- DSS: pricing, sales region analysis, promotion planning.
- ESS: 5-year sales trend forecast, market share.

Manufacturing and production

- TPS: material movement, machine control, purchase orders.
- MIS: inventory control, production scheduling reports, quality and scrap.
- DSS: production planning, relocation cost, capacity.
- ESS: 5-year operating plan, new plant decisions.

Finance and accounting

- TPS: payroll, accounts payable and receivable, general ledger.
- MIS: annual budgeting, cost reports.
- DSS: profitability, contract cost analysis, investment appraisal.
- ESS: profit planning, long-term financing.

Human resources

- TPS: employee record keeping, attendance.
- MIS: turnover, absenteeism, training reports, appraisal history.
- DSS: compensation analysis, manpower allocation.
- ESS: personnel (manpower) planning.

5.4 Computers, Databases, Networks and Organizing Information Systems

Computers and MIS; role of computers in MIS; what is a website 2 4

HOT 4/29 81 Bh, 76 Ash, 74 Ash, 68 Ba

- Computers are the **means** of modern MIS, not MIS itself: a computer alone does not make work efficient; identify the management problem first.
- **Before computers** information was: too late, incomplete, costlier than its worth, unfocused, irrelevant, rigid.
- **Role of computers:**
 - Fast, accurate processing of huge data.
 - Storage and quick retrieval.
 - Reports in any format, on demand.
 - Communication across sites.
 - Models and simulation for decisions.
- **Quick-response systems:**
 - **Online processing:** manager interacts directly with the computer.
 - **Real-time processing:** system runs alongside the activity and guides it (Eg ATM, airline booking).
 - **Time sharing:** many users share one computer at high speed.
- Two uses: **data (transaction) processing** with fixed reports, and **end-user computing** (managers, staff using spreadsheets, queries) with flexibility.
- **Relationship:** MIS defines what information is needed; the computer produces it quickly and cheaply. Computer = tool, MIS = purpose.
- **Website** (74 Ash): a set of linked web pages under one domain name, hosted on a server and reached through the internet. **verified/added**
 - Publishes information to customers.
 - Collects orders and feedback (data).
 - Gives staff remote access to reports.
- **Computer-aided advertising** (69 Ch, short note **LOW 1/29** 69 Ch): **verified/added**
 - Computer graphics and design for ads.
 - Targeted online, email and social media ads.
 - Analytics of clicks and conversions.

Database information system 4 5

HOT 4/29 81 Ba, 68 Ba, 66 Bh, 65 Shr · short note in 68 Ba

- **Database:** organized collection of related data serving many applications at once, appearing to be in one place.
- Kinds:
 - **Personal database:** an individual's contacts, schedules, notes.
 - **Corporate database:** large, shared, integrity critical.
- **DBMS:** software to create, maintain and query it (Oracle, MySQL); applications extract data without their own files.
- Components:
 - Data dictionary: describes the data.
 - Architecture: relations between data.
 - The objects described; rules for manipulation.
- Advantages: no redundancy, consistency, sharing, security, easy reporting.
- Examples:
 - A bank's customer database used by accounts, loans, ATM and mobile banking.
 - A college's student database used by admission, exams, library, fees.

Networking information system 4 5

PIN 2/29 67 Asa, 65 Shr · short note in 65 Shr

- Telecommunication network linking computers and terminals so users at different locations share data and resources.
- Components: computers, terminals, channels, communication processors, software; common rules = **protocols**; analog or digital signals.
- Types:
 - **By area:** LAN (a building), MAN (a city), WAN (country or world).
 - **By users:** internet (public), intranet (staff only), extranet (staff plus partners, suppliers).
- Needs a flexible, standard network architecture that grows with the business.
- Advantages: real-time sharing, distributed offices, e-commerce, email, lower cost.
- Eg bank branches and ATMs on one network; NEA billing linked across offices.

How information systems can be organized properly 4 6

PIN 3/29 81 Bh, 76 Ash, 71 Shr · short note in 71 Shr

- **Complete centralization:** one central IS department holds hardware, software, data and MIS personnel; serves all user departments.
 - + control, standards, economy of scale, security.
 - – slow response, users feel distant, bottleneck.
- **Complete decentralization:** each department has its own systems and MIS staff.
 - + responsive, users own their systems.
 - – duplication, incompatible data, higher cost.
- **Hybrid (distributed):** central database and standards, local processing and staff. Most common today.
 - + control of shared data with local responsiveness.
 - – needs clear rules on who owns what.

Other means of organizing it properly

- **Steering committee** of senior managers sets IS priorities.
- **CIO** (chief information officer) at top level.
- Clear **policies** for data ownership, access and security.
- **Standards** for hardware, software, data formats.
- **User training** and a help desk.
- Regular **audit and review** of the system.

Roles of an organized MIS department

- Hardware administration; database administration.
- Software development and support.
- End-user support and training.
- Computing standards; corporate data access.
- Advocacy: promotes IS use across the firm.

Evolution of MIS

syllabus, no PYQ yet

1. Manual ledgers and punch cards.
 2. Electronic data processing (1950s–60s): record keeping.
 3. MIS reports (1960s–70s).
 4. DSS (1970s): models, operations research.
 5. EIS / ESS (1980s).
 6. PCs and end-user computing: graphical, user-friendly interfaces.
 7. Networks, ERP: integrated systems.
 8. Internet, cloud and business intelligence.
- Driven by: growth of management theory, management science, new production methods and structures, and computers.

5.5 PYQ Mapping

Theory questions, covered in this document

- **Define MIS; need, functions, importance; MIS beyond task software** → §5.1. 17 papers; plus the second half of 73 Ch and 71 Ch, which the Detailed PYQ files under case studies.
- **Data, information, knowledge, wisdom; use in an office** 82 Ba 4, 78 Bh 4, 75 Ash 4 → §5.1.
- **Information for planning, decision making, control** → §5.2. 7 papers.
- **Hierarchy of information needs; MIS at levels** 80 Ba 8, 68 Ch 5, 66 Bh 5, 65 Shr 4; **information architecture** 68 Ba 3, 65 Shr 4 → §5.2.
- **Types of IS, DSS vs MIS, DSS for production, EIS** → §5.3. 7 papers.
- **Information support for functional areas** → §5.3. 74 Ch 8, 73 Shr 4, 72 Ka 8, 70 Ch 6, 70 Asa 5
- **Computers and MIS; website; computer-aided advertising** → §5.4. 81 Bh, 76 Ash, 74 Ash, 69 Ch, 68 Ba
- **Database IS** 81 Ba 5, 68 Ba 5, 66 Bh 5, 65 Shr 4; **networking IS** 67 Asa 5, 65 Shr 4; **organizing IS** 81 Bh 6, 76 Ash 6, 71 Shr 4 → §5.4.
- **Evolution of MIS:** syllabus, never set → §5.4.

Numerical content

- This chapter has no numerical component.

Sources and corrections

- Hierarchy of information needs, information architecture, planning and the quick-response systems come from S.K. Joshi, Ch. 3 (pp 99–135), read through `tools/ocr_page.py`; the lecture notes do not cover them.
- Adhikari p283 expands DSS as “design support system”; it is **decision** support system. Adhikari p296 heads the ESS slide but describes an **expert** system; both are given separately above.

- The Detailed PYQ's "hierarchy of needs" in 65 Shr sits in the MIS question and is read as hierarchy of information needs, as 66 Bh words it.

Figures

- Only diagrams an answer would draw are kept; no photos, clip art or decorative charts.
- Cropped: Adhikari's notes (pp277, 281, 291, 293, 299, 300) and S.K. Joshi Figs. 3.3, 3.6 (book photographs, rendered grayscale with auto-contrast).
- DIKW pyramid: Wikimedia Commons (Longlivetheux, CC BY-SA 4.0).

6. Last-Minute Revision

Built from the frequency analysis of all 29 papers (2065–2082 BS), ME 708. Nothing here is new: every line points back into a chapter.

R.1 Must Know (every TOP-tier topic: these carry the paper)

A topic is TOP tier when 8 or more of the 29 papers ask it. There are 27 of them. Every chapter has at least one; Chapter 1 has twelve.

Ch	Topic	Papers	The one thing to be able to write
2	Personnel management: definition, functions, role	17	Flippo's definition; managerial + 7 operative functions
5	MIS: definition, need, functions, importance	17	Need by manager type (production, marketing, personnel)
1	Organization: definition, necessity, importance in society	16	Mooney-Reiley; 8 reasons for necessity; role in society
3	Motivation: definition, human needs, types	16	Need → tension → drive → behaviour → satisfaction
1	Functions of management	15	POSDCoRB list; Fayol's five for "basic" functions
1	Marketing: definition, concept, importance	14	Stanton; 4 P's; selling vs marketing concept; digital era
2	Recruitment and selection process	13	11 steps from job analysis to induction; recruitment vs selection
2	Factors affecting wage / salary structure	12	10 factors with a Nepali example each
3	Leadership: definition, need, qualities	12	12 qualities; leader vs regular employee
1	Formal vs informal organization	11	10-row difference table; "do we need informal?"
1	Scientific management (Taylor)	11	5 principles, 7 techniques, pig-iron numbers
1	Behavioural approach (Mayo, Hawthorne)	11	4 experiments; 3-approach comparison table
3	Leadership styles; which style where	11	Autocratic, democratic, free-rein; situational answer
1	Managerial skills by level	10	Katz: technical low, human all, conceptual top; the figure
1	Joint stock company	10	Features, private vs public (101 / 7 min), OCR registration
1	Purchasing: 5 R's, functions	10	5 R's; 13 functions of purchasing department
2	Job analysis, description, specification	10	JD = the job, JS = the person; lecturer example
2	Incentives	10	Financial / non-financial; individual / group
3	Herzberg two-factor theory	10	Hygiene vs motivators lists; "motivation comes from the job"
1	Principles of organization	9	15 principles with an example each
1	Levels of management	9	Top / middle / lower duties; time-on-functions figure
1	Fayol's 14 principles	9	All 14 in order; applicable today
3	Techniques of motivation	9	9 techniques; attitude / group / executive
3	Maslow's hierarchy	9	5 levels + what the firm gives at each; pyramid
2	Manpower planning	8	7 steps; importance
3	Entrepreneurship and entrepreneur's characteristics	8	Schumpeter; 12 characteristics
4	Objectives of case study	8	7 objectives; value

R.2 Must Memorize (names, numbers and lists that cannot be improvised)

Ch 1: Introduction

- **Mooney and Reiley**: organization = form of every human association for a common purpose. **Barnard**: formal = consciously coordinated activities; informal = without conscious common purpose.
- **Katz**: technical, human, conceptual. **Mintzberg**: interpersonal, informational, decisional roles.
- **7 models**: hierarchical, task-oriented, allocational, transactional, team effort, knowledge-oriented, goal-concentrated.
- **Taylor** (1856–1915), *Principles of Scientific Management* 1911; pig iron 12.5 → 47.5 t/day, \$1.15 → \$1.85.
- **Fayol** (1841–1925), 1916; 14 principles; POCCC.
- **Mayo**, Hawthorne 1924–32: illumination, relay room, interviews (21,000), bank wiring (14 men).
- **Drucker** MBO 1954. **Graicunas**: $n(2^{n-1} + n - 1)$, 6 → 222.
- **Nepal law**: Companies Act 2063 (private 1–101, public min 7, Rs 1 crore); Cooperative Act 2074 (min 30); Labour Act 2074.
- **5 R's**: quality, quantity, price, time, source.

Ch 2: Personnel Management

- **Flippo**: procurement, development, compensation, integration, maintenance.
- Wage levels: minimum, fair, living. Nominal vs real wage.
- Job evaluation: ranking, classification, factor comparison, point rating; “rate the job, not the man”.
- Incentive plans: Taylor (two rates), Halsey (50% of time saved), Rowan (peaks at $T_a = T_s/2$), Emerson (bonus from 66.67%), Scanlon (labour / sales), Priestman (output / worker).

Ch 3: Motivation, Leadership, Entrepreneurship

- **Maslow** 1943: physiological, safety, social, esteem, self-actualization.
- **ERG** (Alderfer 1969): existence, relatedness, growth; progression + frustration-regression.
- **McClelland** 1961: achievement, power, affiliation.
- **McGregor** 1960: X (coerce) vs Y (involve).
- **Herzberg** 1959, 200 engineers and accountants: hygiene (policy, supervision, pay, security, conditions, relations) vs motivators (achievement, recognition, work itself, responsibility, advancement, growth).
- **Vroom** 1964: $M = E \times I \times V$.
- **Blake and Mouton** 1964: (1,1) impoverished, (1,9) country club, (9,1) authority, (5,5) middle, (9,9) team.
- Leadership approaches: trait, behavioural, contingency (Fiedler, path-goal, Hersey-Blanchard), integrated.
- Power: legitimate, reward, coercive, expert, referent.

Ch 4: Case Studies

- **5 phases**: study, collect facts, define problem, individual decisions, learn.
- **6 steps**: state problem, collect and analyse data, tentative solutions, recommend, write report, checklist.
- Types: illustrative, exploratory, cumulative, critical instance.

Ch 5: MIS

- Information qualities: accurate, timely, complete, concise, relevant.
- **TPS** (operational) → **MIS** (middle, reports) → **DSS** (what-if) → **ESS** (top, projections).
- Information needs: operational hourly/daily, tactical weekly/monthly, strategic quarterly/yearly.

R.3 Must Practice (opinion and applied questions: rehearse a structure, not a text)

O&M has no numerical in 29 papers. What it has instead is a steady supply of “in your opinion” and “prepare / recommend” questions, worth full marks to a student who takes a side and argues it in points.

Question	Papers	Take this side, in this order
Which management theory suits Nepal / is most practical today?	80 Ba, 74 Ash	Contingency; one reason per sector (factory, public enterprise, IT)
Structure for a (temporary) engineering project	73 Ch, 72 Ch	Project / matrix; why not line, functional, committee
CEO of a software company: ownership and structure	74 Ch	Private limited; matrix, draw it
Leadership style for an engineering project / industry / Nepal / start-ups	82 Ba, 80 Ba, 78 Bh, 74 Ash, 73 Ch	Democratic base, autocratic in crises, situational
Can Maslow explain Devkota?	69 Ch	Only partly; ERG and intrinsic motivation
Falling from grace while climbing Maslow's pyramid	78 Bh	Esteem without ethics, frustration-regression, remedies
Job description and specification of a lecturer	81 Ba	JD list, JS list
Which job analysis method suits Nepal?	82 Bh	Interview + observation; why not questionnaire
Salary and benefits you expect	82 Ba, 75 Ch	Market rate + allowances + PF / SSF + growth
HR manager vs talent poaching	79 Ba	Pay, ESOP, career path, culture, succession
Prepare a case study of NEA / Nepal Telecom	72 Ch	Report structure; planning horizon, leadership, motivation, HRD
Recommendations to your college	74 Ch	Academic, infrastructure, support, research, management, image
Would you start a startup after graduation?	74 Ch	Either, justified; then Nepal's risks
Why promote entrepreneurship in Nepal	82 Bh, 81 Ba, 75 Ash	Migration, remittance, trade deficit, unused resources

Figures to be able to draw in two minutes

- Line, functional, line & staff, matrix organization charts (§1.4, §1.5).
- Levels pyramid; skills-by-level bars (§1.2). Organization as an open system (§1.1).
- Maslow's pyramid; Vroom's E-I-V chain; managerial grid (§3.2, §3.3).
- Functional-areas pyramid of information systems; TPS-MIS-DSS-ESS cycle (§5.3).

R.4 Most Repeated PYQs (near-identical wording, set again and again)

1. Define personnel management and explain its functions. 17 papers. §2.1.
2. What are the factors affecting wage / salary structure? What is an incentive? 12 + 10 papers, often in one question (76 Ch, 78 Bh, 74 Ash, 72 Ka, 70 Asa). §2.7.
3. Define marketing and explain its importance in an organization. 14 papers. §1.6.
4. Explain the functions of the purchasing department. 10 papers. §1.6.
5. Principles of organization with a rider (necessity, formal / informal). 9 papers. §1.1.
6. Explain Herzberg's hygiene theory / Maslow's hierarchy / Vroom's expectancy theory, each after “define motivation”. §3.2.
7. Qualities of a good leader. 12 papers. §3.3.
8. Blake and Mouton's managerial grid. 6 papers, 3 of them a full 8 marks. §3.3.
9. Define entrepreneurship; its characteristics / steps for a small scale unit. §3.4.
10. Objectives of case study (or steps / phases), often as a 4-mark short note. §4.1, §4.2.
11. Define MIS. Explain information support for functional areas of management. §5.1, §5.3.

R.5 One-Day Revision Checklist

Morning (Ch 1, about 40% of the marks)

- ☐ Organization: definition, necessity, principles, formal vs informal.
- ☐ Management: functions, levels, skills (figure), models.
- ☐ Taylor, Fayol, Mayo + comparison table; contingency.
- ☐ Forms of ownership table; JSC formation in Nepal; cooperatives.
- ☐ Line / functional / line & staff / committee / matrix + charts.
- ☐ Purchasing 5 R's, functions, methods; marketing importance, functions.

Midday (Ch 2)

- ☐ Functions of personnel management; personnel policy contents.
- ☐ Manpower planning steps; recruitment and selection steps.
- ☐ Job analysis, JD, JS; job evaluation vs merit rating; appraisal methods.
- ☐ Wage factors; incentive types; one incentive formula.

Afternoon (Ch 3)

- ☐ Motivation definition, techniques.
- ☐ Maslow, ERG, Herzberg, X-Y, Vroom: one figure or table each.
- ☐ Leadership qualities, styles, grid, approaches, leader vs manager.
- ☐ Entrepreneur characteristics, small-scale steps, why promote in Nepal.

Evening (Ch 4, Ch 5)

- ☐ Case study objectives, 5 phases, 6 steps.
- ☐ MIS definition, need, importance; data vs information.
- ☐ TPS / MIS / DSS / ESS table; functional-areas pyramid.
- ☐ Computers and MIS; database; organizing IS.

Night before

- ☐ Read R.1 and R.2 only. Rehearse two opinion answers from R.3.
- ☐ Every answer: definition line, numbered points in bold, one Nepali example, one figure where it fits.

Abbreviations, chips and month codes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
JSC	joint stock company	MNC	multinational company
HR, HRM	human resource (management)	OB	organizational behaviour
MBO	management by objectives	CEO, MD	chief executive officer, managing director
AGM	annual general meeting	ESOP	employee share ownership plan
MoA, AoA	memorandum / articles of association	OCR	Office of the Company Registrar
SEBON	Securities Board of Nepal	NEPSE	Nepal Stock Exchange
IPO	initial public offering	PAN, VAT	permanent account no., value added tax
NEA	Nepal Electricity Authority	NOC	Nepal Oil Corporation
QC	quality control	R&D	research and development
OR	operations research	ROI	return on investment
CIM	computer integrated manufacturing	FMS	flexible manufacturing system
CAD, CAM	computer aided design / manufacturing	CAPP	computer aided process planning
CNC	computer numerical control	AGV	automated guided vehicle
MIS	management information system	IT	information technology
SaaS	software as a service	BPO	business process outsourcing
CSR	corporate social responsibility	POSDCoRB	plan, organize, staff, direct, coordinate, report, budget
OPC, PPC	ordinary Portland / Portland pozzolana cement	TPD	tonnes per day
DCS	distributed control system	ERP	enterprise resource planning
DG	diesel generator	KPI	key performance indicator
DPR	daily production report	SSF	Social Security Fund
NS	Nepal Standard (quality mark)	VED	vital, essential, desirable (spares)

Short	Means	Short	Means
no.	number	info	information
vs, Vs	versus	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	min, max	minimum, maximum
Ch.	chapter	§	section

Chip Means (29 papers, 2065–2082 BS)

TOP n/29	asked in 8 or more papers
HOT n/29	asked in 4 to 7 papers
PIN n/29	asked in 1 to 3 papers, inside an 8-mark question
LOW n/29	asked in 1 to 3 papers, as a 4-mark short note only
syllabus, no PYQ yet	in the syllabus, never examined
4+4	marks as printed; several chips = the question recurred at different marks
verified/added	researched and added beyond the lecture notes
80 Bh	bold year = Regular exam; plain = Back exam

Months: Ba = Baishakh, Jth = Jestha, Asa = Ashad, Shr = Shrawan, Bh = Bhadra, Ash = Ashwin, Ka = Kartik, Mng = Mangsir, Po = Poush, Ma = Magh, Ch = Chaitra. **Ash and Asa are different months.** **Sources:** AJ Sir = AJ Sir's lecture decks; Adhikari = Bikash Adhikari, *Organization and Management* slide notes; Joshi = S.K. Joshi, *Organization and Management*, A.K. Books.